MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3420 PRESS. OXDN.- AUTOCL. BLDG.
DESIGN OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
AUTOCL.	:	Autoclave
OXDN.	:	Oxidation
BLDG.	:	Building
PRESS.	:	Pressure

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1. INTENT

This document is to prepare design of foundation for **Pressure Oxidation Autoclave Building**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD PRO. For design of raft force values are directly taken from STAAD & designed in excel sheet. Column is designed in STAAD.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Doc. No. MD-216-3420-EG-ST-GAD-2230 attached in Attachment 1. Loads are also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Column	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 475 Kn/m^2 at 3.0 m below EGL. (Refer Table 5 - BH-9 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN CALCULATION OF FOOTING F1

Size of Columns at node	Along-X	Along-Z
2	2.00m	1.50m

Size of fdn. raft:	7.50m	6.00m
Thickness of raft:	1.00	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.50	m
U/S level of fdn:	-3.00	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

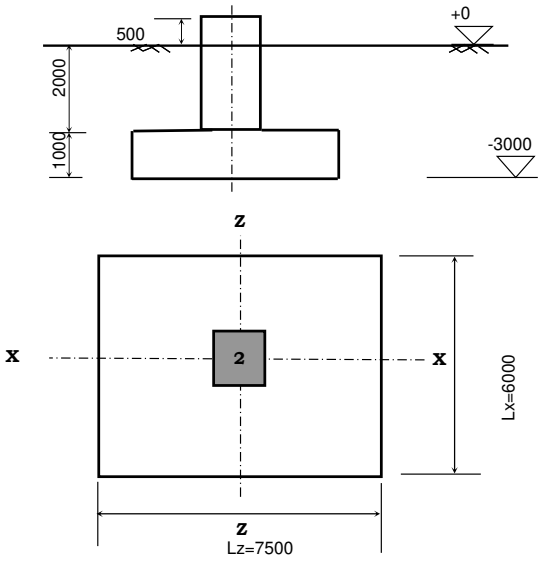
Weight of raft = (7.5*6*1*25) = 1125.00KN

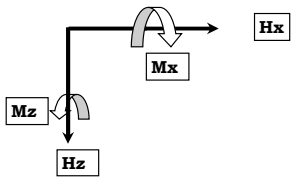
Weight of soil = ((7.5*6)-(2*1.5))*2*20 = 1680.00KN

Total = 2805.00KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	654.20	-11.90	130.25	0.00	0.00
202 DL+L+TL1	979.90	-31.20	341.70	0.00	0.00
203 DL+L+TL2	979.90	-29.00	286.20	0.00	0.00
204 DL+LR	798.70	-25.40	339.50	0.00	0.00
205 DL+0.75L+0.75TL1+0.75LR	1006.85	-36.50	445.77	0.00	0.00
206 DL+0.75L+0.75TL2+0.75LR	1006.85	-34.85	404.15	0.00	0.00
207 DL+0.625WL1	565.20	-207.96	2804.40	0.00	0.00
208 DL+0.625WL2	618.26	-129.71	2142.71	0.00	0.00
209 DL+0.625WL3	640.88	-227.15	3145.30	0.00	0.00
210 DL+0.625WL4	692.26	-147.34	2448.77	0.00	0.00
211 DL+0.625WL5	481.20	208.54	-3075.71	0.00	0.00
212 DL+0.625WL6	531.95	288.41	-3790.39	0.00	0.00
213 DL+0.625WL7	573.26	195.41	-2909.71	0.00	0.00
214 DL+0.625WL8	623.95	275.23	-3624.30	0.00	0.00
215 DL+0.625WL9	330.12	-187.09	1698.96	48.91	171.17
216 DL+0.625WL10	381.12	-107.28	995.31	48.91	171.17
217 DL+0.625WL11	839.77	-195.09	1823.83	-48.91	-171.17
218 DL+0.625WL12	890.77	-115.40	1120.87	-48.91	-171.17
219 DL+0.75WL1+0.75L+0.75LR	900.05	-270.95	3633.94	0.00	0.00
220 DL+0.75WL2+0.75L+0.75LR	963.72	-177.05	2839.92	0.00	0.00
221 DL+0.75WL3+0.75L+0.75LR	990.87	-293.98	4043.02	0.00	0.00
222 DL+0.75WL4+0.75L+0.75LR	1052.52	-198.20	3207.19	0.00	0.00
223 DL+0.75WL5+0.75L+0.75LR	799.25	228.85	-3422.19	0.00	0.00
224 DL+0.75WL67+0.75L+0.75LR	860.15	324.70	-4279.81	0.00	0.00
225 DL+0.75WL7+0.75L+0.75LR	909.72	213.10	-3222.99	0.00	0.00
226 DL+0.75WL8+0.75L+0.75LR	970.55	308.88	-4080.50	0.00	0.00
227 DL+0.75WL9+0.75L+0.75LR	617.96	-245.90	2307.41	58.69	205.40
228 DL+0.75WL10+0.75L+0.75LR	679.16	-150.13	1463.03	58.69	205.40
229 DL+0.75WL11+0.75L+0.75LR	1229.54	-255.50	2457.26	-58.69	-205.40
230 DL+0.75WL12+0.75L+0.75LR	1290.74	-159.88	1613.70	-58.69	-205.40
231 0.6DL+0.625WL1	303.52	-203.20	2752.30	0.00	0.00
232 0.6DL+0.625WL2	356.58	-124.95	2090.61	0.00	0.00
233 0.6DL+0.625WL3	379.21	-222.39	3093.20	0.00	0.00
234 0.6DL+0.625WL4	430.58	-142.58	2396.67	0.00	0.00
235 0.6DL+0.625WL5	219.52	213.30	-3127.81	0.00	0.00
236 0.6DL+0.625WL6	270.27	293.17	-3842.49	0.00	0.00
237 0.6DL+0.625WL7	311.58	200.17	-2961.81	0.00	0.00
238 0.6DL+0.625WL8	362.27	279.99	-3676.40	0.00	0.00
239 0.6DL+0.625WL9	68.44	-182.33	1646.86	48.91	171.17
240 0.6DL+0.625WL10	119.44	-102.52	943.21	48.91	171.17
241 0.6DL+0.625WL11	578.09	-190.33	1771.73	-48.91	-171.17
242 0.6DL+0.625WL12	629.09	-110.64	1068.77	-48.91	-171.17
243 0.6DL+0.7SL1	343.55	-24.57	291.33	10.70	37.44
244 0.6DL+0.7SL2	441.49	10.29	-135.03	-10.70	-37.44
245 DL+0.75WL1+0.75L+0.75LR+CR1	1040.55	-285.65	3900.28	0.00	0.00
246 DL+0.75WL2+0.75L+0.75LR+CR1	1104.22	-191.75	3106.26	0.00	0.00
247 DL+0.75WL3+0.75L+0.75LR+CR1	1131.37	-308.68	4309.37	0.00	0.00
248 DL+0.75WL4+0.75L+0.75LR+CR1	1193.02	-212.90	3473.54	0.00	0.00

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249 DL+0.75WL5+0.75L+0.75LR+CR1	939.75	214.15	-3155.84	0.00	0.00
250 DL+0.75WL67+0.75L+0.75LR+CR1	1000.65	310.00	-4013.46	0.00	0.00
251 DL+0.75WL7+0.75L+0.75LR+CR1	1050.22	198.40	-2956.64	0.00	0.00
252 DL+0.75WL8+0.75L+0.75LR+CR1	1111.05	294.18	-3814.15	0.00	0.00
253 DL+0.75WL9+0.75L+0.75LR+CR1	758.46	-260.60	2573.76	58.69	205.40
254 DL+0.75WL10+0.75L+0.75LR+CR1	819.66	-164.83	1729.38	58.69	205.40
255 DL+0.75WL11+0.75L+0.75LR+CR1	1370.04	-270.20	2723.61	-58.69	-205.40
256 DL+0.75WL12+0.75L+0.75LR+CR1	1431.24	-174.58	1880.05	-58.69	-205.40
257 DL+0.75WL1+0.75L+0.75LR+CR2	938.65	-285.75	3939.93	0.00	0.00
258 DL+0.75WL2+0.75L+0.75LR+CR2	1002.32	-191.85	3145.91	0.00	0.00
259 DL+0.75WL3+0.75L+0.75LR+CR2	1029.47	-308.78	4349.02	0.00	0.00
260 DL+0.75WL4+0.75L+0.75LR+CR2	1091.12	-213.00	3513.19	0.00	0.00
261 DL+0.75WL5+0.75L+0.75LR+CR2	837.85	214.05	-3116.19	0.00	0.00
262 DL+0.75WL67+0.75L+0.75LR+CR2	898.75	309.90	-3973.81	0.00	0.00
263 DL+0.75WL7+0.75L+0.75LR+CR2	948.32	198.30	-2916.99	0.00	0.00
264 DL+0.75WL8+0.75L+0.75LR+CR2	1009.15	294.08	-3774.50	0.00	0.00
265 DL+0.75WL9+0.75L+0.75LR+CR2	656.56	-260.70	2613.41	58.69	205.40
266 DL+0.75WL10+0.75L+0.75LR+CR2	717.76	-164.93	1769.03	58.69	205.40
267 DL+0.75WL11+0.75L+0.75LR+CR2	1268.14	-270.30	2763.26	-58.69	-205.40
268 DL+0.75WL12+0.75L+0.75LR+CR2	1329.34	-174.68	1919.70	-58.69	-205.40
269 DL+0.75WL1+0.75L+0.75LR+CR3	887.95	-262.55	3447.34	11.00	38.50
270 DL+0.75WL2+0.75L+0.75LR+CR3	951.62	-168.65	2653.32	11.00	38.50
271 DL+0.75WL3+0.75L+0.75LR+CR3	978.77	-285.58	3856.42	11.00	38.50
272 DL+0.75WL4+0.75L+0.75LR+CR3	1040.42	-189.80	3020.59	11.00	38.50
273 DL+0.75WL5+0.75L+0.75LR+CR3	787.15	237.25	-3608.79	11.00	38.50
274 DL+0.75WL67+0.75L+0.75LR+CR3	848.05	333.10	-4466.41	11.00	38.50
275 DL+0.75WL7+0.75L+0.75LR+CR3	897.62	221.50	-3409.59	11.00	38.50
276 DL+0.75WL8+0.75L+0.75LR+CR3	958.45	317.28	-4267.09	11.00	38.50
277 DL+0.75WL9+0.75L+0.75LR+CR3	605.86	-237.50	2120.81	69.69	243.90
278 DL+0.75WL10+0.75L+0.75LR+CR3	667.06	-141.73	1276.43	69.69	243.90
279 DL+0.75WL11+0.75L+0.75LR+CR3	1217.44	-247.10	2270.66	-47.69	-166.90
280 DL+0.75WL12+0.75L+0.75LR+CR3	1278.64	-151.48	1427.10	-47.69	-166.90
281 DL+0.75WL1+0.75L+0.75LR+CR4	1073.65	-262.55	3408.24	-11.00	-38.50
282 DL+0.75WL2+0.75L+0.75LR+CR4	1137.32	-168.65	2614.22	-11.00	-38.50
283 DL+0.75WL3+0.75L+0.75LR+CR4	1164.47	-285.58	3817.32	-11.00	-38.50
284 DL+0.75WL4+0.75L+0.75LR+CR4	1226.12	-189.80	2981.49	-11.00	-38.50
285 DL+0.75WL5+0.75L+0.75LR+CR4	972.85	237.25	-3647.89	-11.00	-38.50
286 DL+0.75WL67+0.75L+0.75LR+CR4	1033.75	333.10	-4505.51	-11.00	-38.50
287 DL+0.75WL7+0.75L+0.75LR+CR4	1083.32	221.50	-3448.69	-11.00	-38.50
288 DL+0.75WL8+0.75L+0.75LR+CR4	1144.15	317.28	-4306.19	-11.00	-38.50
289 DL+0.75WL9+0.75L+0.75LR+CR4	791.56	-237.50	2081.71	47.69	166.90
290 DL+0.75WL10+0.75L+0.75LR+CR4	852.76	-141.73	1237.33	47.69	166.90
291 DL+0.75WL11+0.75L+0.75LR+CR4	1403.14	-247.10	2231.56	-69.69	-243.90
292 DL+0.75WL12+0.75L+0.75LR+CR4	1464.34	-151.48	1388.00	-69.69	-243.90

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	3459.2	0.00	0.0	-11.9	130.2	99.6	NA	116.3	NA
202 DL+L+TL1	3784.9	0.00	0.0	-31.2	341.7	41.5	NA	48.5	NA
203 DL+L+TL2	3784.9	0.00	0.0	-29.0	286.2	49.6	NA	52.2	NA
204 DL+LR	3603.7	0.00	0.0	-25.4	339.5	39.8	NA	56.8	NA
205 DL+0.75L+0.75TL1+0.75LR	3811.8	0.00	0.0	-36.5	445.8	32.1	NA	41.8	NA
206 DL+0.75L+0.75TL2+0.75LR	3811.8	0.00	0.0	-34.9	404.1	35.4	NA	43.8	NA
207 DL+0.625WL1	3370.2	0.00	0.0	-208.0	2804.4	4.5	NA	6.5	NA
208 DL+0.625WL2	3423.3	0.00	0.0	-129.7	2142.7	6.0	NA	10.6	NA
209 DL+0.625WL3	3445.9	0.00	0.0	-227.2	3145.3	4.1	NA	6.1	NA
210 DL+0.625WL4	3497.3	0.00	0.0	-147.3	2448.8	5.4	NA	9.5	NA
211 DL+0.625WL5	3286.2	0.00	0.0	208.5	-3075.7	4.0	NA	6.3	NA
212 DL+0.625WL6	3336.9	0.00	0.0	288.4	-3790.4	3.3	NA	4.6	NA
213 DL+0.625WL7	3378.3	0.00	0.0	195.4	-2909.7	4.4	NA	6.9	NA
214 DL+0.625WL8	3428.9	0.00	0.0	275.2	-3624.3	3.5	NA	5.0	NA
215 DL+0.625WL9	3135.1	48.91	171.2	-187.1	1699.0	6.9	54.9	6.7	25.6
216 DL+0.625WL10	3186.1	48.91	171.2	-107.3	995.3	12.0	55.8	11.9	26.1
217 DL+0.625WL11	3644.8	-48.91	-171.2	-195.1	1823.8	7.5	63.9	7.5	29.8
218 DL+0.625WL12	3695.8	-48.91	-171.2	-115.4	1120.9	12.4	64.8	12.8	30.2
219 DL+0.75WL1+0.75L+0.75LR	3705.0	0.00	0.0	-271.0	3633.9	3.8	NA	5.5	NA
220 DL+0.75WL2+0.75L+0.75LR	3768.7	0.00	0.0	-177.1	2839.9	5.0	NA	8.5	NA
221 DL+0.75WL3+0.75L+0.75LR	3795.9	0.00	0.0	-294.0	4043.0	3.5	NA	5.2	NA
222 DL+0.75WL4+0.75L+0.75LR	3857.5	0.00	0.0	-198.2	3207.2	4.5	NA	7.8	NA
223 DL+0.75WL5+0.75L+0.75LR	3604.2	0.00	0.0	228.9	-3422.2	3.9	NA	6.3	NA
224 DL+0.75WL67+0.75L+0.75LR	3665.1	0.00	0.0	324.7	-4279.8	3.2	NA	4.5	NA
225 DL+0.75WL7+0.75L+0.75LR	3714.7	0.00	0.0	213.1	-3223.0	4.3	NA	7.0	NA
226 DL+0.75WL8+0.75L+0.75LR	3775.5	0.00	0.0	308.9	-4080.5	3.5	NA	4.9	NA
227 DL+0.75WL9+0.75L+0.75LR	3423.0	58.69	205.4	-245.9	2307.4	5.6	50.0	5.6	23.3
228 DL+0.75WL10+0.75L+0.75LR	3484.2	58.69	205.4	-150.1	1463.0	8.9	50.9	9.3	23.7
229 DL+0.75WL11+0.75L+0.75LR	4034.5	-58.69	-205.4	-255.5	2457.3	6.2	58.9	6.3	27.5
230 DL+0.75WL12+0.75L+0.75LR	4095.7	-58.69	-205.4	-159.9	1613.7	9.5	59.8	10.2	27.9
231 0.6DL+0.625WL1	3108.5	0.00	0.0	-203.2	2752.3	4.2	NA	6.1	NA
232 0.6DL+0.625WL2	3161.6	0.00	0.0	-125.0	2090.6	5.7	NA	10.1	NA
233 0.6DL+0.625WL3	3184.2	0.00	0.0	-222.4	3093.2	3.9	NA	5.7	NA
234 0.6DL+0.625WL4	3235.6	0.00	0.0	-142.6	2396.7	5.1	NA	9.1	NA
235 0.6DL+0.625WL5	3024.5	0.00	0.0	213.3	-3127.8	3.6	NA	5.7	NA
236 0.6DL+0.625WL6	3075.3	0.00	0.0	293.2	-3842.5	3.0	NA	4.2	NA
237 0.6DL+0.625WL7	3116.6	0.00	0.0	200.2	-2961.8	3.9	NA	6.2	NA
238 0.6DL+0.625WL8	3167.3	0.00	0.0	280.0	-3676.4	3.2	NA	4.5	NA
239 0.6DL+0.625WL9	2873.4	48.91	171.2	-182.3	1646.9	6.5	50.4	6.3	23.5
240 0.6DL+0.625WL10	2924.4	48.91	171.2	-102.5	943.2	11.6	51.3	11.4	23.9
241 0.6DL+0.625WL11	3383.1	-48.91	-171.2	-190.3	1771.7	7.2	59.3	7.1	27.7
242 0.6DL+0.625WL12	3434.1	-48.91	-171.2	-110.6	1068.8	12.0	60.2	12.4	28.1

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243 0.6DL+0.7SL1	3148.5	10.70	37.4	-24.6	291.3	40.5	252.3	51.3	117.7
244 0.6DL+0.7SL2	3246.5	-10.70	-37.4	10.3	-135.0	90.2	260.2	126.2	121.4
245 DL+0.75WL1+0.75L+0.75LR+CR1	3845.5	0.00	0.0	-285.7	3900.3	3.7	NA	5.4	NA
246 DL+0.75WL2+0.75L+0.75LR+CR1	3909.2	0.00	0.0	-191.8	3106.3	4.7	NA	8.2	NA
247 DL+0.75WL3+0.75L+0.75LR+CR1	3936.4	0.00	0.0	-308.7	4309.4	3.4	NA	5.1	NA
248 DL+0.75WL4+0.75L+0.75LR+CR1	3998.0	0.00	0.0	-212.9	3473.5	4.3	NA	7.5	NA
249 DL+0.75WL5+0.75L+0.75LR+CR1	3744.7	0.00	0.0	214.2	-3155.8	4.4	NA	7.0	NA
250 DL+0.75WL67+0.75L+0.75LR+CR1	3805.6	0.00	0.0	310.0	-4013.5	3.6	NA	4.9	NA
251 DL+0.75WL7+0.75L+0.75LR+CR1	3855.2	0.00	0.0	198.4	-2956.6	4.9	NA	7.8	NA
252 DL+0.75WL8+0.75L+0.75LR+CR1	3916.0	0.00	0.0	294.2	-3814.1	3.9	NA	5.3	NA
253 DL+0.75WL9+0.75L+0.75LR+CR1	3563.5	58.69	205.4	-260.6	2573.8	5.2	52.0	5.5	24.3
254 DL+0.75WL10+0.75L+0.75LR+CR1	3624.7	58.69	205.4	-164.8	1729.4	7.9	52.9	8.8	24.7
255 DL+0.75WL11+0.75L+0.75LR+CR1	4175.0	-58.69	-205.4	-270.2	2723.6	5.7	61.0	6.2	28.5
256 DL+0.75WL12+0.75L+0.75LR+CR1	4236.2	-58.69	-205.4	-174.6	1880.1	8.4	61.9	9.7	28.9
257 DL+0.75WL1+0.75L+0.75LR+CR2	3743.6	0.00	0.0	-285.8	3939.9	3.6	NA	5.2	NA
258 DL+0.75WL2+0.75L+0.75LR+CR2	3807.3	0.00	0.0	-191.9	3145.9	4.5	NA	7.9	NA
259 DL+0.75WL3+0.75L+0.75LR+CR2	3834.5	0.00	0.0	-308.8	4349.0	3.3	NA	5.0	NA
260 DL+0.75WL4+0.75L+0.75LR+CR2	3896.1	0.00	0.0	-213.0	3513.2	4.2	NA	7.3	NA
261 DL+0.75WL5+0.75L+0.75LR+CR2	3642.8	0.00	0.0	214.1	-3116.2	4.4	NA	6.8	NA
262 DL+0.75WL67+0.75L+0.75LR+CR2	3703.7	0.00	0.0	309.9	-3973.8	3.5	NA	4.8	NA
263 DL+0.75WL7+0.75L+0.75LR+CR2	3753.3	0.00	0.0	198.3	-2917.0	4.8	NA	7.6	NA
264 DL+0.75WL8+0.75L+0.75LR+CR2	3814.1	0.00	0.0	294.1	-3774.5	3.8	NA	5.2	NA
265 DL+0.75WL9+0.75L+0.75LR+CR2	3461.6	58.69	205.4	-260.7	2613.4	5.0	50.6	5.3	23.6
266 DL+0.75WL10+0.75L+0.75LR+CR2	3522.8	58.69	205.4	-164.9	1769.0	7.5	51.5	8.5	24.0
267 DL+0.75WL11+0.75L+0.75LR+CR2	4073.1	-58.69	-205.4	-270.3	2763.3	5.5	59.5	6.0	27.8
268 DL+0.75WL12+0.75L+0.75LR+CR2	4134.3	-58.69	-205.4	-174.7	1919.7	8.1	60.4	9.5	28.2
269 DL+0.75WL1+0.75L+0.75LR+CR3	3692.9	11.00	38.5	-262.6	3447.3	4.0	287.8	5.6	134.3
270 DL+0.75WL2+0.75L+0.75LR+CR3	3756.6	11.00	38.5	-168.7	2653.3	5.3	292.7	8.9	136.6
271 DL+0.75WL3+0.75L+0.75LR+CR3	3783.8	11.00	38.5	-285.6	3856.4	3.7	294.8	5.3	137.6
272 DL+0.75WL4+0.75L+0.75LR+CR3	3845.4	11.00	38.5	-189.8	3020.6	4.8	299.7	8.1	139.8
273 DL+0.75WL5+0.75L+0.75LR+CR3	3592.1	11.00	38.5	237.3	-3608.8	3.7	279.9	6.1	130.6
274 DL+0.75WL67+0.75L+0.75LR+CR3	3653.0	11.00	38.5	333.1	-4466.4	3.1	284.7	4.4	132.8
275 DL+0.75WL7+0.75L+0.75LR+CR3	3702.6	11.00	38.5	221.5	-3409.6	4.1	288.5	6.7	134.6
276 DL+0.75WL8+0.75L+0.75LR+CR3	3763.4	11.00	38.5	317.3	-4267.1	3.3	293.3	4.7	136.9
277 DL+0.75WL9+0.75L+0.75LR+CR3	3410.9	69.69	243.9	-237.5	2120.8	6.0	42.0	5.7	19.6
278 DL+0.75WL10+0.75L+0.75LR+CR3	3472.1	69.69	243.9	-141.7	1276.4	10.2	42.7	9.8	19.9
279 DL+0.75WL11+0.75L+0.75LR+CR3	4022.4	-47.69	-166.9	-247.1	2270.7	6.6	72.3	6.5	33.7
280 DL+0.75WL12+0.75L+0.75LR+CR3	4083.6	-47.69	-166.9	-151.5	1427.1	10.7	73.4	10.8	34.3
281 DL+0.75WL1+0.75L+0.75LR+CR4	3878.6	-11.00	-38.5	-262.6	3408.2	4.3	302.2	5.9	141.0
282 DL+0.75WL2+0.75L+0.75LR+CR4	3942.3	-11.00	-38.5	-168.7	2614.2	5.7	307.2	9.4	143.4
283 DL+0.75WL3+0.75L+0.75LR+CR4	3969.5	-11.00	-38.5	-285.6	3817.3	3.9	309.3	5.6	144.3
284 DL+0.75WL4+0.75L+0.75LR+CR4	4031.1	-11.00	-38.5	-189.8	2981.5	5.1	314.1	8.5	146.6
285 DL+0.75WL5+0.75L+0.75LR+CR4	3777.8	-11.00	-38.5	237.3	-3647.9	3.9	294.4	6.4	137.4
286 DL+0.75WL67+0.75L+0.75LR+CR4	3838.7	-11.00	-38.5	333.1	-4505.5	3.2	299.1	4.6	139.6
287 DL+0.75WL7+0.75L+0.75LR+CR4	3888.3	-11.00	-38.5	221.5	-3448.7	4.2	303.0	7.0	141.4
288 DL+0.75WL8+0.75L+0.75LR+CR4	3949.1	-11.00	-38.5	317.3	-4306.2	3.4	307.7	5.0	143.6
289 DL+0.75WL9+0.75L+0.75LR+CR4	3596.6	47.69	166.9	-237.5	2081.7	6.5	64.6	6.1	30.2
290 DL+0.75WL10+0.75L+0.75LR+CR4	3657.8	47.69	166.9	-141.7	1237.3	11.1	65.7	10.3	30.7
291 DL+0.75WL11+0.75L+0.75LR+CR4	4208.1	-69.69	-243.9	-247.1	2231.6	7.1	51.8	6.8	24.2
292 DL+0.75WL12+0.75L+0.75LR+CR4	4269.3	-69.69	-243.9	-151.5	1388.0	11.5	52.5	11.3	24.5

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	128.16	0
Min Px	2	201	0	128.16	0
Max Py	827	286	0	198.15	0
Min Py	793	236	0	1.39	0
Max Pz	2	201	0	128.16	0
Min Pz	2	201	0	128.16	0

Gross Maximum Base pressure, p_{max} = 198.2 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2
Depth of soil above footing = 2.00 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 515 KN/m^2
> 198.2 KN/m^2
Hence OK

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Design of foundation raft F1

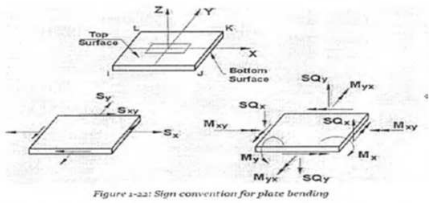
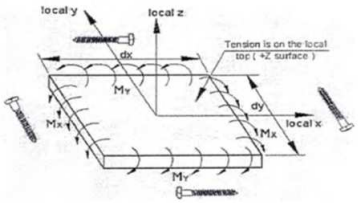


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	338	0.86	-0.134	0	0	0	-615.98	-373.95	5.086	-621.063	-379.03	135:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
Min Qx	341	-0.909	-0.141	0	0	0	-657.35	-383	-8.442	-665.789	-391.45	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Qy	393	-0.018	0.766	0	0	0	-492.86	-226.5	118.095	-610.955	-344.59	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Min Qy	363	-0.028	-0.773	0	0	0	-496.45	-233.6	-120.01	-616.459	-353.61	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Sx	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Min Sx	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Max Sy	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Min Sy	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Max Sxy	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Min Sxy	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Max Mx	338	-0.728	0.125	0	0	0	569.44	231.025	-16.888	586.324	247.913	124:0.9DL+1.0WL6
Min Mx	341	-0.909	-0.141	0	0	0	-657.35	-383	-8.442	-665.789	-391.45	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max My	338	-0.728	0.125	0	0	0	569.44	231.025	-16.888	586.324	247.913	124:0.9DL+1.0WL6
Min My	341	-0.894	-0.137	0	0	0	-640.94	-386.1	-6.919	-647.861	-393.02	176:1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
Max Mxy	397	0.413	-0.402	0	0	0	129.01	66.2	224.019	353.03	290.219	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Min Mxy	367	0.42	0.392	0	0	0	125.89	57.762	-222.69	348.585	280.456	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	529	0.38	-0.035	0	0	0	-219.41	-255.5	-9.06	-228.465	-264.56	135:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
Min Qx	543	-0.399	-0.036	0	0	0	-233.94	-263.12	8.103	-242.041	-271.22	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Qy	548	-0.208	0.181	0	0	0	-206.65	-126.42	-44.083	-250.735	-170.51	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Min Qy	582	0.034	-0.181	0	0	0	-246.75	-5.776	-52.553	-299.302	-58.329	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Sx	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Min Sx	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Max Sy	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Min Sy	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Max Sxy	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Min Sxy	1091	0	-0.001	0	0	0	-0.417	-0.465	0.284	-0.701	-0.749	101:1.4DL
Max Mx	529	-0.31	0.031	0	0	0	201.87	168.813	4.366	206.238	173.179	124:0.9DL+1.0WL6
Min Mx	582	0.027	-0.18	0	0	0	-250.2	-7.693	-46.911	-297.112	-54.604	176:1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
Max My	529	-0.31	0.031	0	0	0	201.87	168.813	4.366	206.238	173.179	124:0.9DL+1.0WL6
Min My	543	-0.394	-0.035	0	0	0	-228.2	-263.99	8.318	-236.521	-272.31	176:1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
Max Mxy	560	0.186	0.026	0	0	0	-73.039	-9.462	130.388	-203.427	-139.85	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Min Mxy	576	0.187	-0.03	0	0	0	-75.862	-11.36	-130.42	-206.284	-141.78	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		1000	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

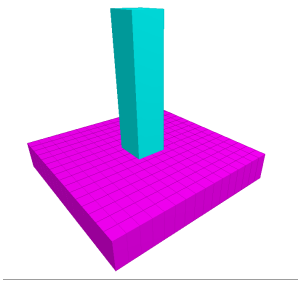
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
124:0.9D	586.32	top face	50	938	586.32	0.89	0.0022	2016.0	1687.5	25	243.58	150	3273.9	46.2	57.7	0.046	tension control
174:1.2D	-665.8	bottom face	50	938	665.79	1.01	0.0024	2294.2		25	214.05	150	3273.9	46.2	57.7	0.046	tension control
174:1.2D	353.03	top face	50	938	353.03	0.54	0.0013	1206.4		25	291.01	150	3273.9	46.2	57.7	0.046	tension control
174:1.2D	348.59	top face	50	938	348.59	0.53	0.0013	1191.1		25	291.01	150	3273.9	46.2	57.7	0.046	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u			Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	0.38	1	380	228.47	0.003	0.9375	974.198	943.068	1608.43	943.07	3453	> 380	ok	707.3	φVc>Vu	not required	
	Min Qx	0.399	1	399	242.04			973.426			943.07	3453	> 399	ok	707.3	φVc>Vu	not required	

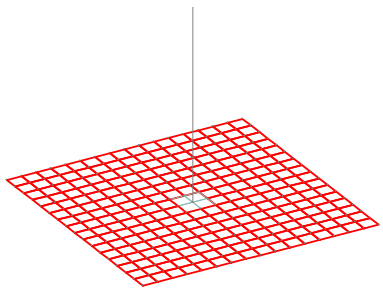
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
124:0.9D	247.91	top face	50	940	247.91	0.37	0.0009	842.6	1692	20	186.24	150	2095.3	29.5	36.93	0.073	tension control
176:1.2D	-393.02	bottom face	50	940	393.02	0.59	0.0014	1340.8		20	186.24	150	2095.3	29.5	36.93	0.073	tension control
174:1.2D	290.22	top face	50	940	290.22	0.44	0.0011	987.5		20	186.24	150	2095.3	29.5	36.93	0.073	tension control
174:1.2D	280.46	top face	50	940	280.46	0.42	0.0010	954.0		20	186.24	150	2095.3	29.5	36.93	0.073	tension control

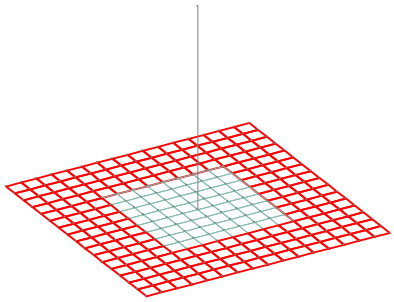
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u			Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.181	1	181	170.51	0.002	0.94	942.801	943.068	fc'1608	942.8	982	> 181	ok	707.1	φVc>Vu	not required	
	Min Qy	0.181	1	181	58.329			1049.32			943.07	982	> 181	ok	707.3	φVc>Vu	not required	



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**25ϕT**	**150**	c/c
At Top	**25ϕT**	**150**	c/c
Along Z Direction :			
At Bottom	**20ϕT**	**150**	c/c
At Top	**20ϕT**	**150**	c/c

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DESIGN CALCULATION OF FOOTING F2

Size of Columns at node	Along-X	Along-Z
2	2.00m	1.50m

Size of fdn. raft:	7.25m	6.00m
Thickness of raft:	1.20	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.50	m
U/S level of fdn:	-3.00	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

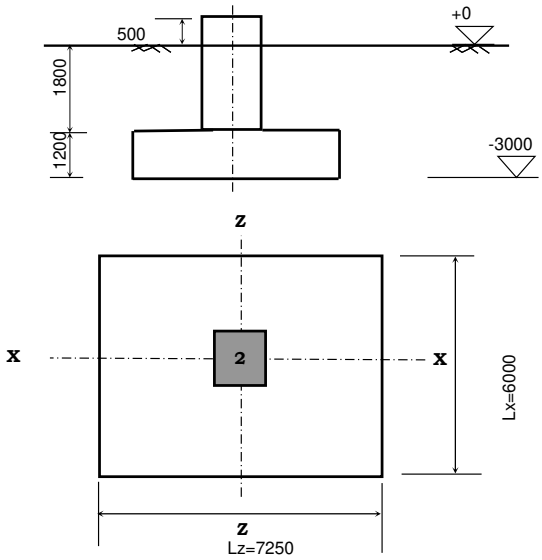
Weight of raft = (7.25*6*1.2*25) = 1305.00KN

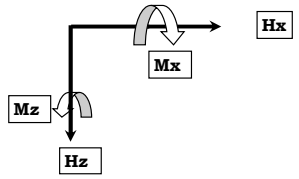
Weight of soil = ((7.25*6)-(2*1.5))*1.8*20 = 1458.00KN

Total = 2763.00KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F2

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	654.20	-11.90	130.25	0.00	0.00
202 DL+L+TL1	979.90	-31.20	341.70	0.00	0.00
203 DL+L+TL2	979.90	-29.00	286.20	0.00	0.00
204 DL+LR	798.70	-25.40	339.50	0.00	0.00
205 DL+0.75L+0.75TL1+0.75LR	1006.85	-36.50	445.77	0.00	0.00
206 DL+0.75L+0.75TL2+0.75LR	1006.85	-34.85	404.15	0.00	0.00
207 DL+0.625WL1	565.20	-207.96	2804.40	0.00	0.00
208 DL+0.625WL2	618.26	-129.71	2142.71	0.00	0.00
209 DL+0.625WL3	640.88	-227.15	3145.30	0.00	0.00
210 DL+0.625WL4	692.26	-147.34	2448.77	0.00	0.00
211 DL+0.625WL5	481.20	208.54	-3075.71	0.00	0.00
212 DL+0.625WL6	531.95	288.41	-3790.39	0.00	0.00
213 DL+0.625WL7	573.26	195.41	-2909.71	0.00	0.00
214 DL+0.625WL8	623.95	275.23	-3624.30	0.00	0.00
215 DL+0.625WL9	330.12	-187.09	1698.96	48.91	171.17
216 DL+0.625WL10	381.12	-107.28	995.31	48.91	171.17
217 DL+0.625WL11	839.77	-195.09	1823.83	-48.91	-171.17
218 DL+0.625WL12	890.77	-115.40	1120.87	-48.91	-171.17
219 DL+0.75WL1+0.75L+0.75LR	900.05	-270.95	3633.94	0.00	0.00
220 DL+0.75WL2+0.75L+0.75LR	963.72	-177.05	2839.92	0.00	0.00
221 DL+0.75WL3+0.75L+0.75LR	990.87	-293.98	4043.02	0.00	0.00
222 DL+0.75WL4+0.75L+0.75LR	1052.52	-198.20	3207.19	0.00	0.00
223 DL+0.75WL5+0.75L+0.75LR	799.25	228.85	-3422.19	0.00	0.00
224 DL+0.75WL67+0.75L+0.75LR	860.15	324.70	-4279.81	0.00	0.00
225 DL+0.75WL7+0.75L+0.75LR	909.72	213.10	-3222.99	0.00	0.00
226 DL+0.75WL8+0.75L+0.75LR	970.55	308.88	-4080.50	0.00	0.00
227 DL+0.75WL9+0.75L+0.75LR	617.96	-245.90	2307.41	58.69	205.40
228 DL+0.75WL10+0.75L+0.75LR	679.16	-150.13	1463.03	58.69	205.40
229 DL+0.75WL11+0.75L+0.75LR	1229.54	-255.50	2457.26	-58.69	-205.40
230 DL+0.75WL12+0.75L+0.75LR	1290.74	-159.88	1613.70	-58.69	-205.40
231 0.6DL+0.625WL1	303.52	-203.20	2752.30	0.00	0.00
232 0.6DL+0.625WL2	356.58	-124.95	2090.61	0.00	0.00
233 0.6DL+0.625WL3	379.21	-222.39	3093.20	0.00	0.00
234 0.6DL+0.625WL4	430.58	-142.58	2396.67	0.00	0.00
235 0.6DL+0.625WL5	219.52	213.30	-3127.81	0.00	0.00
236 0.6DL+0.625WL6	270.27	293.17	-3842.49	0.00	0.00
237 0.6DL+0.625WL7	311.58	200.17	-2961.81	0.00	0.00
238 0.6DL+0.625WL8	362.27	279.99	-3676.40	0.00	0.00
239 0.6DL+0.625WL9	68.44	-182.33	1646.86	48.91	171.17
240 0.6DL+0.625WL10	119.44	-102.52	943.21	48.91	171.17
241 0.6DL+0.625WL11	578.09	-190.33	1771.73	-48.91	-171.17
242 0.6DL+0.625WL12	629.09	-110.64	1068.77	-48.91	-171.17
243 0.6DL+0.7SL1	343.55	-24.57	291.33	10.70	37.44
244 0.6DL+0.7SL2	441.49	10.29	-135.03	-10.70	-37.44
245 DL+0.75WL1+0.75L+0.75LR+CR1	1040.55	-285.65	3900.28	0.00	0.00
246 DL+0.75WL2+0.75L+0.75LR+CR1	1104.22	-191.75	3106.26	0.00	0.00
247 DL+0.75WL3+0.75L+0.75LR+CR1	1131.37	-308.68	4309.37	0.00	0.00
248 DL+0.75WL4+0.75L+0.75LR+CR1	1193.02	-212.90	3473.54	0.00	0.00

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249 DL+0.75WL5+0.75L+0.75LR+CR1	939.75	214.15	-3155.84	0.00	0.00
250 DL+0.75WL67+0.75L+0.75LR+CR1	1000.65	310.00	-4013.46	0.00	0.00
251 DL+0.75WL7+0.75L+0.75LR+CR1	1050.22	198.40	-2956.64	0.00	0.00
252 DL+0.75WL8+0.75L+0.75LR+CR1	1111.05	294.18	-3814.15	0.00	0.00
253 DL+0.75WL9+0.75L+0.75LR+CR1	758.46	-260.60	2573.76	58.69	205.40
254 DL+0.75WL10+0.75L+0.75LR+CR1	819.66	-164.83	1729.38	58.69	205.40
255 DL+0.75WL11+0.75L+0.75LR+CR1	1370.04	-270.20	2723.61	-58.69	-205.40
256 DL+0.75WL12+0.75L+0.75LR+CR1	1431.24	-174.58	1880.05	-58.69	-205.40
257 DL+0.75WL1+0.75L+0.75LR+CR2	938.65	-285.75	3939.93	0.00	0.00
258 DL+0.75WL2+0.75L+0.75LR+CR2	1002.32	-191.85	3145.91	0.00	0.00
259 DL+0.75WL3+0.75L+0.75LR+CR2	1029.47	-308.78	4349.02	0.00	0.00
260 DL+0.75WL4+0.75L+0.75LR+CR2	1091.12	-213.00	3513.19	0.00	0.00
261 DL+0.75WL5+0.75L+0.75LR+CR2	837.85	214.05	-3116.19	0.00	0.00
262 DL+0.75WL67+0.75L+0.75LR+CR2	898.75	309.90	-3973.81	0.00	0.00
263 DL+0.75WL7+0.75L+0.75LR+CR2	948.32	198.30	-2916.99	0.00	0.00
264 DL+0.75WL8+0.75L+0.75LR+CR2	1009.15	294.08	-3774.50	0.00	0.00
265 DL+0.75WL9+0.75L+0.75LR+CR2	656.56	-260.70	2613.41	58.69	205.40
266 DL+0.75WL10+0.75L+0.75LR+CR2	717.76	-164.93	1769.03	58.69	205.40
267 DL+0.75WL11+0.75L+0.75LR+CR2	1268.14	-270.30	2763.26	-58.69	-205.40
268 DL+0.75WL12+0.75L+0.75LR+CR2	1329.34	-174.68	1919.70	-58.69	-205.40
269 DL+0.75WL1+0.75L+0.75LR+CR3	887.95	-262.55	3447.34	11.00	38.50
270 DL+0.75WL2+0.75L+0.75LR+CR3	951.62	-168.65	2653.32	11.00	38.50
271 DL+0.75WL3+0.75L+0.75LR+CR3	978.77	-285.58	3856.42	11.00	38.50
272 DL+0.75WL4+0.75L+0.75LR+CR3	1040.42	-189.80	3020.59	11.00	38.50
273 DL+0.75WL5+0.75L+0.75LR+CR3	787.15	237.25	-3608.79	11.00	38.50
274 DL+0.75WL67+0.75L+0.75LR+CR3	848.05	333.10	-4466.41	11.00	38.50
275 DL+0.75WL7+0.75L+0.75LR+CR3	897.62	221.50	-3409.59	11.00	38.50
276 DL+0.75WL8+0.75L+0.75LR+CR3	958.45	317.28	-4267.09	11.00	38.50
277 DL+0.75WL9+0.75L+0.75LR+CR3	605.86	-237.50	2120.81	69.69	243.90
278 DL+0.75WL10+0.75L+0.75LR+CR3	667.06	-141.73	1276.43	69.69	243.90
279 DL+0.75WL11+0.75L+0.75LR+CR3	1217.44	-247.10	2270.66	-47.69	-166.90
280 DL+0.75WL12+0.75L+0.75LR+CR3	1278.64	-151.48	1427.10	-47.69	-166.90
281 DL+0.75WL1+0.75L+0.75LR+CR4	1073.65	-262.55	3408.24	-11.00	-38.50
282 DL+0.75WL2+0.75L+0.75LR+CR4	1137.32	-168.65	2614.22	-11.00	-38.50
283 DL+0.75WL3+0.75L+0.75LR+CR4	1164.47	-285.58	3817.32	-11.00	-38.50
284 DL+0.75WL4+0.75L+0.75LR+CR4	1226.12	-189.80	2981.49	-11.00	-38.50
285 DL+0.75WL5+0.75L+0.75LR+CR4	972.85	237.25	-3647.89	-11.00	-38.50
286 DL+0.75WL67+0.75L+0.75LR+CR4	1033.75	333.10	-4505.51	-11.00	-38.50
287 DL+0.75WL7+0.75L+0.75LR+CR4	1083.32	221.50	-3448.69	-11.00	-38.50
288 DL+0.75WL8+0.75L+0.75LR+CR4	1144.15	317.28	-4306.19	-11.00	-38.50
289 DL+0.75WL9+0.75L+0.75LR+CR4	791.56	-237.50	2081.71	47.69	166.90
290 DL+0.75WL10+0.75L+0.75LR+CR4	852.76	-141.73	1237.33	47.69	166.90
291 DL+0.75WL11+0.75L+0.75LR+CR4	1403.14	-247.10	2231.56	-69.69	-243.90
292 DL+0.75WL12+0.75L+0.75LR+CR4	1464.34	-151.48	1388.00	-69.69	-243.90

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	3417.2	0.00	0.0	-11.9	130.2	95.1	NA	114.9	NA
202 DL+L+TL1	3742.9	0.00	0.0	-31.2	341.7	39.7	NA	48.0	NA
203 DL+L+TL2	3742.9	0.00	0.0	-29.0	286.2	47.4	NA	51.6	NA
204 DL+LR	3561.7	0.00	0.0	-25.4	339.5	38.0	NA	56.1	NA
205 DL+0.75L+0.75TL1+0.75LR	3769.8	0.00	0.0	-36.5	445.8	30.7	NA	41.3	NA
206 DL+0.75L+0.75TL2+0.75LR	3769.8	0.00	0.0	-34.9	404.1	33.8	NA	43.3	NA
207 DL+0.625WL1	3328.2	0.00	0.0	-208.0	2804.4	4.3	NA	6.4	NA
208 DL+0.625WL2	3381.3	0.00	0.0	-129.7	2142.7	5.7	NA	10.4	NA
209 DL+0.625WL3	3403.9	0.00	0.0	-227.2	3145.3	3.9	NA	6.0	NA
210 DL+0.625WL4	3455.3	0.00	0.0	-147.3	2448.8	5.1	NA	9.4	NA
211 DL+0.625WL5	3244.2	0.00	0.0	208.5	-3075.7	3.8	NA	6.2	NA
212 DL+0.625WL6	3294.9	0.00	0.0	288.4	-3790.4	3.2	NA	4.6	NA
213 DL+0.625WL7	3336.3	0.00	0.0	195.4	-2909.7	4.2	NA	6.8	NA
214 DL+0.625WL8	3386.9	0.00	0.0	275.2	-3624.3	3.4	NA	4.9	NA
215 DL+0.625WL9	3093.1	48.91	171.2	-187.1	1699.0	6.6	54.2	6.6	25.3
216 DL+0.625WL10	3144.1	48.91	171.2	-107.3	995.3	11.5	55.1	11.7	25.7
217 DL+0.625WL11	3602.8	-48.91	-171.2	-195.1	1823.8	7.2	63.1	7.4	29.5
218 DL+0.625WL12	3653.8	-48.91	-171.2	-115.4	1120.9	11.8	64.0	12.7	29.9
219 DL+0.75WL1+0.75L+0.75LR	3663.0	0.00	0.0	-271.0	3633.9	3.7	NA	5.4	NA
220 DL+0.75WL2+0.75L+0.75LR	3726.7	0.00	0.0	-177.1	2839.9	4.8	NA	8.4	NA
221 DL+0.75WL3+0.75L+0.75LR	3753.9	0.00	0.0	-294.0	4043.0	3.4	NA	5.1	NA
222 DL+0.75WL4+0.75L+0.75LR	3815.5	0.00	0.0	-198.2	3207.2	4.3	NA	7.7	NA
223 DL+0.75WL5+0.75L+0.75LR	3562.2	0.00	0.0	228.9	-3422.2	3.8	NA	6.2	NA
224 DL+0.75WL67+0.75L+0.75LR	3623.1	0.00	0.0	324.7	-4279.8	3.1	NA	4.5	NA
225 DL+0.75WL7+0.75L+0.75LR	3672.7	0.00	0.0	213.1	-3223.0	4.1	NA	6.9	NA
226 DL+0.75WL8+0.75L+0.75LR	3733.5	0.00	0.0	308.9	-4080.5	3.3	NA	4.8	NA
227 DL+0.75WL9+0.75L+0.75LR	3381.0	58.69	205.4	-245.9	2307.4	5.3	49.4	5.5	23.0
228 DL+0.75WL10+0.75L+0.75LR	3442.2	58.69	205.4	-150.1	1463.0	8.5	50.3	9.2	23.5
229 DL+0.75WL11+0.75L+0.75LR	3992.5	-58.69	-205.4	-255.5	2457.3	5.9	58.3	6.3	27.2
230 DL+0.75WL12+0.75L+0.75LR	4053.7	-58.69	-205.4	-159.9	1613.7	9.1	59.2	10.1	27.6
231 0.6DL+0.625WL1	3066.5	0.00	0.0	-203.2	2752.3	4.0	NA	6.0	NA
232 0.6DL+0.625WL2	3119.6	0.00	0.0	-125.0	2090.6	5.4	NA	10.0	NA
233 0.6DL+0.625WL3	3142.2	0.00	0.0	-222.4	3093.2	3.7	NA	5.7	NA
234 0.6DL+0.625WL4	3193.6	0.00	0.0	-142.6	2396.7	4.8	NA	9.0	NA
235 0.6DL+0.625WL5	2982.5	0.00	0.0	213.3	-3127.8	3.5	NA	5.6	NA
236 0.6DL+0.625WL6	3033.3	0.00	0.0	293.2	-3842.5	2.9	NA	4.1	NA
237 0.6DL+0.625WL7	3074.6	0.00	0.0	200.2	-2961.8	3.8	NA	6.1	NA
238 0.6DL+0.625WL8	3125.3	0.00	0.0	280.0	-3676.4	3.1	NA	4.5	NA
239 0.6DL+0.625WL9	2831.4	48.91	171.2	-182.3	1646.9	6.2	49.6	6.2	23.2
240 0.6DL+0.625WL10	2882.4	48.91	171.2	-102.5	943.2	11.1	50.5	11.2	23.6
241 0.6DL+0.625WL11	3341.1	-48.91	-171.2	-190.3	1771.7	6.8	58.6	7.0	27.3
242 0.6DL+0.625WL12	3392.1	-48.91	-171.2	-110.6	1068.8	11.5	59.5	12.3	27.7

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243 0.6DL+0.7SL1	3106.5	10.70	37.4	-24.6	291.3	38.7	249.0	50.6	116.2
244 0.6DL+0.7SL2	3204.5	-10.70	-37.4	10.3	-135.0	86.0	256.8	124.6	119.8
245 DL+0.75WL1+0.75L+0.75LR+CR1	3803.5	0.00	0.0	-285.7	3900.3	3.5	NA	5.3	NA
246 DL+0.75WL2+0.75L+0.75LR+CR1	3867.2	0.00	0.0	-191.8	3106.3	4.5	NA	8.1	NA
247 DL+0.75WL3+0.75L+0.75LR+CR1	3894.4	0.00	0.0	-308.7	4309.4	3.3	NA	5.0	NA
248 DL+0.75WL4+0.75L+0.75LR+CR1	3956.0	0.00	0.0	-212.9	3473.5	4.1	NA	7.4	NA
249 DL+0.75WL5+0.75L+0.75LR+CR1	3702.7	0.00	0.0	214.2	-3155.8	4.3	NA	6.9	NA
250 DL+0.75WL67+0.75L+0.75LR+CR1	3763.6	0.00	0.0	310.0	-4013.5	3.4	NA	4.9	NA
251 DL+0.75WL7+0.75L+0.75LR+CR1	3813.2	0.00	0.0	198.4	-2956.6	4.7	NA	7.7	NA
252 DL+0.75WL8+0.75L+0.75LR+CR1	3874.0	0.00	0.0	294.2	-3814.1	3.7	NA	5.3	NA
253 DL+0.75WL9+0.75L+0.75LR+CR1	3521.5	58.69	205.4	-260.6	2573.8	5.0	51.4	5.4	24.0
254 DL+0.75WL10+0.75L+0.75LR+CR1	3582.7	58.69	205.4	-164.8	1729.4	7.5	52.3	8.7	24.4
255 DL+0.75WL11+0.75L+0.75LR+CR1	4133.0	-58.69	-205.4	-270.2	2723.6	5.5	60.4	6.1	28.2
256 DL+0.75WL12+0.75L+0.75LR+CR1	4194.2	-58.69	-205.4	-174.6	1880.1	8.1	61.3	9.6	28.6
257 DL+0.75WL1+0.75L+0.75LR+CR2	3701.6	0.00	0.0	-285.8	3939.9	3.4	NA	5.2	NA
258 DL+0.75WL2+0.75L+0.75LR+CR2	3765.3	0.00	0.0	-191.9	3145.9	4.3	NA	7.9	NA
259 DL+0.75WL3+0.75L+0.75LR+CR2	3792.5	0.00	0.0	-308.8	4349.0	3.2	NA	4.9	NA
260 DL+0.75WL4+0.75L+0.75LR+CR2	3854.1	0.00	0.0	-213.0	3513.2	4.0	NA	7.2	NA
261 DL+0.75WL5+0.75L+0.75LR+CR2	3600.8	0.00	0.0	214.1	-3116.2	4.2	NA	6.7	NA
262 DL+0.75WL67+0.75L+0.75LR+CR2	3661.7	0.00	0.0	309.9	-3973.8	3.3	NA	4.7	NA
263 DL+0.75WL7+0.75L+0.75LR+CR2	3711.3	0.00	0.0	198.3	-2917.0	4.6	NA	7.5	NA
264 DL+0.75WL8+0.75L+0.75LR+CR2	3772.1	0.00	0.0	294.1	-3774.5	3.6	NA	5.1	NA
265 DL+0.75WL9+0.75L+0.75LR+CR2	3419.6	58.69	205.4	-260.7	2613.4	4.7	49.9	5.2	23.3
266 DL+0.75WL10+0.75L+0.75LR+CR2	3480.8	58.69	205.4	-164.9	1769.0	7.1	50.8	8.4	23.7
267 DL+0.75WL11+0.75L+0.75LR+CR2	4031.1	-58.69	-205.4	-270.3	2763.3	5.3	58.9	6.0	27.5
268 DL+0.75WL12+0.75L+0.75LR+CR2	4092.3	-58.69	-205.4	-174.7	1919.7	7.7	59.8	9.4	27.9
269 DL+0.75WL1+0.75L+0.75LR+CR3	3650.9	11.00	38.5	-262.6	3447.3	3.8	284.5	5.6	132.8
270 DL+0.75WL2+0.75L+0.75LR+CR3	3714.6	11.00	38.5	-168.7	2653.3	5.1	289.5	8.8	135.1
271 DL+0.75WL3+0.75L+0.75LR+CR3	3741.8	11.00	38.5	-285.6	3856.4	3.5	291.6	5.2	136.1
272 DL+0.75WL4+0.75L+0.75LR+CR3	3803.4	11.00	38.5	-189.8	3020.6	4.6	296.4	8.0	138.3
273 DL+0.75WL5+0.75L+0.75LR+CR3	3550.1	11.00	38.5	237.3	-3608.8	3.6	276.6	6.0	129.1
274 DL+0.75WL67+0.75L+0.75LR+CR3	3611.0	11.00	38.5	333.1	-4466.4	2.9	281.4	4.3	131.3
275 DL+0.75WL7+0.75L+0.75LR+CR3	3660.6	11.00	38.5	221.5	-3409.6	3.9	285.3	6.6	133.1
276 DL+0.75WL8+0.75L+0.75LR+CR3	3721.4	11.00	38.5	317.3	-4267.1	3.2	290.0	4.7	135.3
277 DL+0.75WL9+0.75L+0.75LR+CR3	3368.9	69.69	243.9	-237.5	2120.8	5.8	41.4	5.7	19.3
278 DL+0.75WL10+0.75L+0.75LR+CR3	3430.1	69.69	243.9	-141.7	1276.4	9.7	42.2	9.7	19.7
279 DL+0.75WL11+0.75L+0.75LR+CR3	3980.4	-47.69	-166.9	-247.1	2270.7	6.4	71.5	6.4	33.4
280 DL+0.75WL12+0.75L+0.75LR+CR3	4041.6	-47.69	-166.9	-151.5	1427.1	10.3	72.6	10.7	33.9
281 DL+0.75WL1+0.75L+0.75LR+CR4	3836.6	-11.00	-38.5	-262.6	3408.2	4.1	299.0	5.8	139.5
282 DL+0.75WL2+0.75L+0.75LR+CR4	3900.3	-11.00	-38.5	-168.7	2614.2	5.4	303.9	9.3	141.8
283 DL+0.75WL3+0.75L+0.75LR+CR4	3927.5	-11.00	-38.5	-285.6	3817.3	3.7	306.0	5.5	142.8
284 DL+0.75WL4+0.75L+0.75LR+CR4	3989.1	-11.00	-38.5	-189.8	2981.5	4.9	310.8	8.4	145.1
285 DL+0.75WL5+0.75L+0.75LR+CR4	3735.8	-11.00	-38.5	237.3	-3647.9	3.7	291.1	6.3	135.8
286 DL+0.75WL67+0.75L+0.75LR+CR4	3796.7	-11.00	-38.5	333.1	-4505.5	3.1	295.9	4.6	138.1
287 DL+0.75WL7+0.75L+0.75LR+CR4	3846.3	-11.00	-38.5	221.5	-3448.7	4.0	299.7	6.9	139.9
288 DL+0.75WL8+0.75L+0.75LR+CR4	3907.1	-11.00	-38.5	317.3	-4306.2	3.3	304.5	4.9	142.1
289 DL+0.75WL9+0.75L+0.75LR+CR4	3554.6	47.69	166.9	-237.5	2081.7	6.2	63.9	6.0	29.8
290 DL+0.75WL10+0.75L+0.75LR+CR4	3615.8	47.69	166.9	-141.7	1237.3	10.6	65.0	10.2	30.3
291 DL+0.75WL11+0.75L+0.75LR+CR4	4166.1	-69.69	-243.9	-247.1	2231.6	6.8	51.2	6.7	23.9
292 DL+0.75WL12+0.75L+0.75LR+CR4	4227.3	-69.69	-243.9	-151.5	1388.0	11.0	52.0	11.2	24.3

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Base pressure check of footing F2

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	128.51	0
Min Px	2	201	0	128.51	0
Max Py	695	286	0	242.29	0
Min Py	1235	236	0	0.65	0
Max Pz	2	201	0	128.51	0
Min Pz	2	201	0	128.51	0

Gross Maximum Base pressure, p_{max} = 242.3 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2
Depth of soil above footing = 1.80 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 511 KN/m^2
> 242.3 KN/m^2
Hence OK

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Design of foundation raft F2

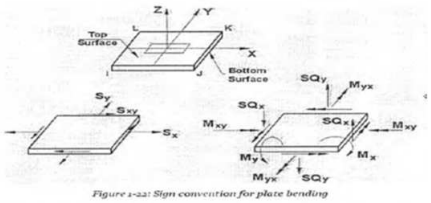
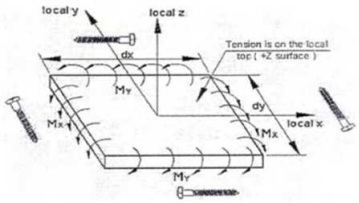


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F2_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	338	0.706	-0.103	0	0	0	-800	-324.72	17.015	-816.72	-341.73	135:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
Min Qx	341	-0.698	-0.109	0	0	0	-534.19	-362.48	-23.26	-557.447	-385.74	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Qy	393	-0.056	0.585	0	0	0	-282.09	-272.92	156.396	-438.489	-429.32	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Min Qy	363	-0.063	-0.591	0	0	0	-285.59	-280.2	-158.35	-443.948	-438.55	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Sx	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Min Sx	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Max Sy	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Min Sy	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Max Sxy	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Min Sxy	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Max Mx	338	-0.601	0.094	0	0	0	783.47	171.465	-30.746	814.22	202.211	124:0.9DL+1.0WL6
Min Mx	338	0.706	-0.103	0	0	0	-799.71	-324.72	17.015	-816.72	-341.73	135:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
Max My	430	-0.601	-0.095	0	0	0	783.27	171.485	30.921	814.191	202.406	124:0.9DL+1.0WL6
Min My	341	-0.687	-0.106	0	0	0	-519.59	-368.11	-21.148	-540.741	-389.26	176:1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
Max Mxy	397	0.237	-0.276	0	0	0	276.95	88.041	239.871	516.825	327.912	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Min Mxy	367	0.242	0.268	0	0	0	274.43	79.023	-237.29	511.721	316.311	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4

PLATE CENTER STRESS : FDN_F2_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	529	0.342	-0.028	0	0	0	-357.51	-232.79	-6.496	-364.002	-239.29	135:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
Min Qx	529	-0.29	0.025	0	0	0	357.19	141.744	0.93	358.12	142.674	124:0.9DL+1.0WL6
Max Qy	510	0.037	0.215	0	0	0	-94.902	-54.394	66.886	-161.788	-121.28	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Min Qy	524	0.034	-0.218	0	0	0	-96.77	-56.978	-66.405	-163.175	-123.38	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Sx	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Min Sx	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Max Sy	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Min Sy	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Max Sxy	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Min Sxy	1229	0	-0.001	0	0	0	-0.233	-0.333	-0.168	-0.401	-0.501	101:1.4DL
Max Mx	529	-0.29	0.025	0	0	0	357.19	141.744	0.93	358.12	142.674	124:0.9DL+1.0WL6
Min Mx	521	-0.003	-0.204	0	0	0	-401.12	-45.967	40.411	-441.533	-86.378	135:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
Max My	529	-0.29	0.025	0	0	0	357.19	141.744	0.93	358.12	142.674	124:0.9DL+1.0WL6
Min My	543	-0.254	-0.031	0	0	0	-106.79	-302.3	2.602	-109.389	-304.91	176:1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
Max Mxy	504	0.185	0.027	0	0	0	100.13	-16.361	142.419	242.549	-158.78	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Min Mxy	518	0.186	-0.032	0	0	0	97.436	-19.683	-141.69	239.127	-161.37	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		1200	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

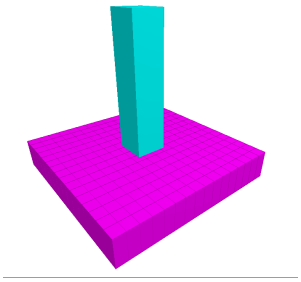
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
124:0.9D	814.22	top face	50	1138	814.22	0.84	0.0020	2305.4	2047.5	25	213.01	150	3273.9	46.2	57.7	0.056	tension control
135:1.2D	-816.7	bottom face	50	1138	816.72	0.84	0.0020	2312.5		25	212.35	150	3273.9	46.2	57.7	0.056	tension control
174:1.2D	516.83	top face	50	1138	516.83	0.53	0.0013	1455.5		25	239.84	150	3273.9	46.2	57.7	0.056	tension control
174:1.2D	511.72	top face	50	1138	511.72	0.53	0.0013	1441.0		25	239.84	150	3273.9	46.2	57.7	0.056	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.342	1	410.4	364	0.003	1.1375	1148.11	1132.38	1951.57	1132.4	4180	> 410.4	ok	849.3	φVc>Vu	not required
	Min Qx	0.29	1	348	358.12			1138.25			1132.4	4180	> 348	ok	849.3	φVc>Vu	not required

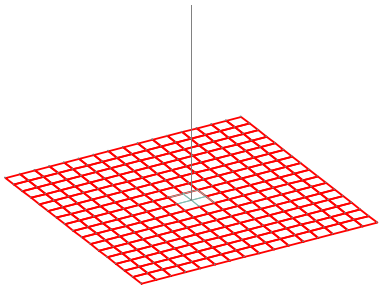
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
124:0.9D	202.41	top face	50	1140	202.41	0.21	0.0005	565.6	2052	20	153.5	150	2095.3	29.5	36.93	0.09	tension control
176:1.2D	-389.26	bottom face	50	1140	389.26	0.40	0.0010	1091.4		20	153.5	150	2095.3	29.5	36.93	0.09	tension control
174:1.2D	327.91	top face	50	1140	327.91	0.34	0.0008	918.4		20	153.5	150	2095.3	29.5	36.93	0.09	tension control
174:1.2D	316.31	top face	50	1140	316.31	0.32	0.0008	885.7		20	153.5	150	2095.3	29.5	36.93	0.09	tension control

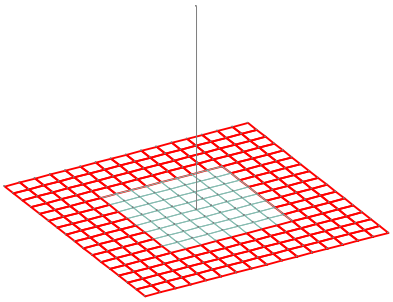
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.215	1	258	121.28	0.002	1.14	1211.4	1132.38	fc'1952	1132.4	1182	> 258	ok	849.3	φVc>Vu	not required
	Min Qy	0.218	1	261.6	123.38			1210.96			1132.4	1182	> 261.6	ok	849.3	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**25ϕT**	**150**	c/c
At Top	**25ϕT**	**150**	c/c
Along Z Direction :			
At Bottom	**20ϕT**	**150**	c/c
At Top	**20ϕT**	**150**	c/c

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DESIGN CALCULATION OF FOOTING F3

Size of Columns at node	Along-X	Along-Z
2	2.00m	1.50m
469	0.70m	0.70m
Size of fdn. raft:	7.50m	6.00m
Thickness of raft:	1.00	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.50	m
U/S level of fdn:	-3.00	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

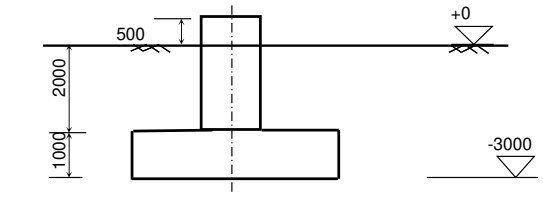
Weight of raft = (7.5*6*1*25) = 1125.00KN

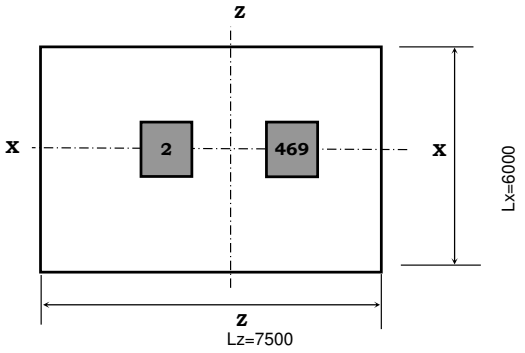
Weight of soil = {(7.5*6)-(2*1.5)-(0.7*0.7)}*2*20 = 1660.40KN

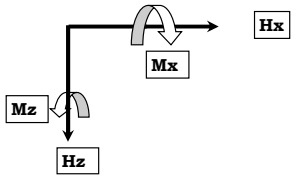
Total = 2785.40KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F3

Load at bottom of footing	Node 2					Node 469				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	654.20	-11.90	130.25	0.00	0.00	124.9	-7.7	28.5	0.0	0.0
202 DL+L+TL1	979.90	-31.20	341.70	0.00	0.00	323.6	-30.6	113.2	0.0	0.0
203 DL+L+TL2	979.90	-29.00	286.20	0.00	0.00	323.6	-30.6	113.2	0.0	0.0
204 DL+LR	798.70	-25.40	339.50	0.00	0.00	124.9	-7.7	28.5	0.0	0.0
205 DL+0.75L+0.75TL1+0.75LR	1006.85	-36.50	445.77	0.00	0.00	274.0	-24.9	92.0	0.0	0.0
206 DL+0.75L+0.75TL2+0.75LR	1006.85	-34.85	404.15	0.00	0.00	274.0	-24.9	92.0	0.0	0.0
207 DL+0.625WL1	565.20	-207.96	2804.40	0.00	0.00	100.9	-50.0	185.0	0.0	0.0
208 DL+0.625WL2	618.26	-129.71	2142.71	0.00	0.00	100.9	-50.0	185.0	0.0	0.0
209 DL+0.625WL3	640.88	-227.15	3145.30	0.00	0.00	100.9	-50.0	185.0	0.0	0.0
210 DL+0.625WL4	692.26	-147.34	2448.77	0.00	0.00	100.9	-50.0	185.0	0.0	0.0
211 DL+0.625WL5	481.20	208.54	-3075.71	0.00	0.00	148.9	34.6	-128.1	0.0	0.0
212 DL+0.625WL6	531.95	288.41	-3790.39	0.00	0.00	148.9	34.6	-128.1	0.0	0.0
213 DL+0.625WL7	573.26	195.41	-2909.71	0.00	0.00	148.9	34.6	-128.1	0.0	0.0
214 DL+0.625WL8	623.95	275.23	-3624.30	0.00	0.00	148.9	34.6	-128.1	0.0	0.0
215 DL+0.625WL9	330.12	-187.09	1698.96	48.91	171.17	124.6	-11.7	43.3	1.3	4.6
216 DL+0.625WL10	381.12	-107.28	995.31	48.91	171.17	124.6	-11.7	43.3	1.3	4.6
217 DL+0.625WL11	839.77	-195.09	1823.83	-48.91	-171.17	125.2	-11.7	43.3	-1.3	-4.6
218 DL+0.625WL12	890.77	-115.40	1120.87	-48.91	-171.17	125.2	-11.7	43.3	-1.3	-4.6
219 DL+0.75WL1+0.75L+0.75LR	900.05	-270.95	3633.94	0.00	0.00	245.2	-75.7	279.9	0.0	0.0
220 DL+0.75WL2+0.75L+0.75LR	963.72	-177.05	2839.92	0.00	0.00	245.2	-75.7	279.9	0.0	0.0
221 DL+0.75WL3+0.75L+0.75LR	990.87	-293.98	4043.02	0.00	0.00	245.2	-75.7	279.9	0.0	0.0
222 DL+0.75WL4+0.75L+0.75LR	1052.52	-198.20	3207.19	0.00	0.00	245.2	-75.7	279.9	0.0	0.0
223 DL+0.75WL5+0.75L+0.75LR	799.25	228.85	-3422.19	0.00	0.00	302.8	25.9	-95.8	0.0	0.0
224 DL+0.75WL67+0.75L+0.75LR	860.15	324.70	-4279.81	0.00	0.00	302.8	25.9	-95.8	0.0	0.0
225 DL+0.75WL7+0.75L+0.75LR	909.72	213.10	-3222.99	0.00	0.00	302.8	25.9	-95.8	0.0	0.0
226 DL+0.75WL8+0.75L+0.75LR	970.55	308.88	-4080.50	0.00	0.00	302.8	25.9	-95.8	0.0	0.0
227 DL+0.75WL9+0.75L+0.75LR	617.96	-245.90	2307.41	58.69	205.40	273.6	-29.7	109.8	1.5	5.6
228 DL+0.75WL10+0.75L+0.75LR	679.16	-150.13	1463.03	58.69	205.40	273.6	-29.7	109.8	1.5	5.6
229 DL+0.75WL11+0.75L+0.75LR	1229.54	-255.50	2457.26	-58.69	-205.40	274.3	-29.7	109.8	-1.5	-5.6
230 DL+0.75WL12+0.75L+0.75LR	1290.74	-159.88	1613.70	-58.69	-205.40	274.3	-29.7	109.8	-1.5	-5.6
231 0.6DL+0.625WL1	303.52	-203.20	2752.30	0.00	0.00	50.9	-46.9	173.6	0.0	0.0
232 0.6DL+0.625WL2	356.58	-124.95	2090.61	0.00	0.00	50.9	-46.9	173.6	0.0	0.0
233 0.6DL+0.625WL3	379.21	-222.39	3093.20	0.00	0.00	50.9	-46.9	173.6	0.0	0.0
234 0.6DL+0.625WL4	430.58	-142.58	2396.67	0.00	0.00	50.9	-46.9	173.6	0.0	0.0
235 0.6DL+0.625WL5	219.52	213.30	-3127.81	0.00	0.00	98.9	37.7	-139.5	0.0	0.0
236 0.6DL+0.625WL6	270.27	293.17	-3842.49	0.00	0.00	98.9	37.7	-139.5	0.0	0.0
237 0.6DL+0.625WL7	311.58	200.17	-2961.81	0.00	0.00	98.9	37.7	-139.5	0.0	0.0
238 0.6DL+0.625WL8	362.27	279.99	-3676.40	0.00	0.00	98.9	37.7	-139.5	0.0	0.0
239 0.6DL+0.625WL9	68.44	-182.33	1646.86	48.91	171.17	74.6	-8.6	31.9	1.3	4.6
240 0.6DL+0.625WL10	119.44	-102.52	943.21	48.91	171.17	74.6	-8.6	31.9	1.3	4.6
241 0.6DL+0.625WL11	578.09	-190.33	1771.73	-48.91	-171.17	75.2	-8.6	31.9	-1.3	-4.6
242 0.6DL+0.625WL12	629.09	-110.64	1068.77	-48.91	-171.17	75.2	-8.6	31.9	-1.3	-4.6
243 0.6DL+0.7SL1	343.55	-24.57	291.33	10.70	37.44	74.9	-4.6	17.1	0.0	0.0
244 0.6DL+0.7SL2	441.49	10.29	-135.03	-10.70	-37.44	74.9	-4.6	17.1	0.0	0.0
245 DL+0.75WL1+0.75L+0.75LR+CR1	1040.55	-285.65	3900.28	0.00	0.00	245.2	-75.7	279.9	0.0	0.0
246 DL+0.75WL2+0.75L+0.75LR+CR1	1104.22	-191.75	3106.26	0.00	0.00	245.2	-75.7	279.9	0.0	0.0
247 DL+0.75WL3+0.75L+0.75LR+CR1	1131.37	-308.68	4309.37	0.00	0.00	245.2	-75.7	279.9	0.0	0.0
248 DL+0.75WL4+0.75L+0.75LR+CR1	1193.02	-212.90	3473.54	0.00	0.00	245.2	-75.7	279.9	0.0	0.0

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249 DL+0.75WL5+0.75L+0.75LR+CR1	939.75	214.15	-3155.84	0.00	0.00
250 DL+0.75WL67+0.75L+0.75LR+CR1	1000.65	310.00	-4013.46	0.00	0.00
251 DL+0.75WL7+0.75L+0.75LR+CR1	1050.22	198.40	-2956.64	0.00	0.00
252 DL+0.75WL8+0.75L+0.75LR+CR1	1111.05	294.18	-3814.15	0.00	0.00
253 DL+0.75WL9+0.75L+0.75LR+CR1	758.46	-260.60	2573.76	58.69	205.40
254 DL+0.75WL10+0.75L+0.75LR+CR1	819.66	-164.83	1729.38	58.69	205.40
255 DL+0.75WL11+0.75L+0.75LR+CR1	1370.04	-270.20	2723.61	-58.69	-205.40
256 DL+0.75WL12+0.75L+0.75LR+CR1	1431.24	-174.58	1880.05	-58.69	-205.40
257 DL+0.75WL1+0.75L+0.75LR+CR2	938.65	-285.75	3939.93	0.00	0.00
258 DL+0.75WL2+0.75L+0.75LR+CR2	1002.32	-191.85	3145.91	0.00	0.00
259 DL+0.75WL3+0.75L+0.75LR+CR2	1029.47	-308.78	4349.02	0.00	0.00
260 DL+0.75WL4+0.75L+0.75LR+CR2	1091.12	-213.00	3513.19	0.00	0.00
261 DL+0.75WL5+0.75L+0.75LR+CR2	837.85	214.05	-3116.19	0.00	0.00
262 DL+0.75WL67+0.75L+0.75LR+CR2	898.75	309.90	-3973.81	0.00	0.00
263 DL+0.75WL7+0.75L+0.75LR+CR2	948.32	198.30	-2916.99	0.00	0.00
264 DL+0.75WL8+0.75L+0.75LR+CR2	1009.15	294.08	-3774.50	0.00	0.00
265 DL+0.75WL9+0.75L+0.75LR+CR2	656.56	-260.70	2613.41	58.69	205.40
266 DL+0.75WL10+0.75L+0.75LR+CR2	717.76	-164.93	1769.03	58.69	205.40
267 DL+0.75WL11+0.75L+0.75LR+CR2	1268.14	-270.30	2763.26	-58.69	-205.40
268 DL+0.75WL12+0.75L+0.75LR+CR2	1329.34	-174.68	1919.70	-58.69	-205.40
269 DL+0.75WL1+0.75L+0.75LR+CR3	887.95	-262.55	3447.34	11.00	38.50
270 DL+0.75WL2+0.75L+0.75LR+CR3	951.62	-168.65	2653.32	11.00	38.50
271 DL+0.75WL3+0.75L+0.75LR+CR3	978.77	-285.58	3856.42	11.00	38.50
272 DL+0.75WL4+0.75L+0.75LR+CR3	1040.42	-189.80	3020.59	11.00	38.50
273 DL+0.75WL5+0.75L+0.75LR+CR3	787.15	237.25	-3608.79	11.00	38.50
274 DL+0.75WL67+0.75L+0.75LR+CR3	848.05	333.10	-4466.41	11.00	38.50
275 DL+0.75WL7+0.75L+0.75LR+CR3	897.62	221.50	-3409.59	11.00	38.50
276 DL+0.75WL8+0.75L+0.75LR+CR3	958.45	317.28	-4267.09	11.00	38.50
277 DL+0.75WL9+0.75L+0.75LR+CR3	605.86	-237.50	2120.81	69.69	243.90
278 DL+0.75WL10+0.75L+0.75LR+CR3	667.06	-141.73	1276.43	69.69	243.90
279 DL+0.75WL11+0.75L+0.75LR+CR3	1217.44	-247.10	2270.66	-47.69	-166.90
280 DL+0.75WL12+0.75L+0.75LR+CR3	1278.64	-151.48	1427.10	-47.69	-166.90
281 DL+0.75WL1+0.75L+0.75LR+CR4	1073.65	-262.55	3408.24	-11.00	-38.50
282 DL+0.75WL2+0.75L+0.75LR+CR4	1137.32	-168.65	2614.22	-11.00	-38.50
283 DL+0.75WL3+0.75L+0.75LR+CR4	1164.47	-285.58	3817.32	-11.00	-38.50
284 DL+0.75WL4+0.75L+0.75LR+CR4	1226.12	-189.80	2981.49	-11.00	-38.50
285 DL+0.75WL5+0.75L+0.75LR+CR4	972.85	237.25	-3647.89	-11.00	-38.50
286 DL+0.75WL67+0.75L+0.75LR+CR4	1033.75	333.10	-4505.51	-11.00	-38.50
287 DL+0.75WL7+0.75L+0.75LR+CR4	1083.32	221.50	-3448.69	-11.00	-38.50
288 DL+0.75WL8+0.75L+0.75LR+CR4	1144.15	317.28	-4306.19	-11.00	-38.50
289 DL+0.75WL9+0.75L+0.75LR+CR4	791.56	-237.50	2081.71	47.69	166.90
290 DL+0.75WL10+0.75L+0.75LR+CR4	852.76	-141.73	1237.33	47.69	166.90
291 DL+0.75WL11+0.75L+0.75LR+CR4	1403.14	-247.10	2231.56	-69.69	-243.90
292 DL+0.75WL12+0.75L+0.75LR+CR4	1464.34	-151.48	1388.00	-69.69	-243.90

302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
273.6	-29.7	109.8	1.5	5.6
273.6	-29.7	109.8	1.5	5.6
274.3	-29.7	109.8	-1.5	-5.6
274.3	-29.7	109.8	-1.5	-5.6
245.2	-75.7	279.9	0.0	0.0
245.2	-75.7	279.9	0.0	0.0
245.2	-75.7	279.9	0.0	0.0
245.2	-75.7	279.9	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
273.6	-29.7	109.8	1.5	5.6
273.6	-29.7	109.8	1.5	5.6
274.3	-29.7	109.8	-1.5	-5.6
274.3	-29.7	109.8	-1.5	-5.6
245.2	-75.7	279.9	0.0	0.0
245.2	-75.7	279.9	0.0	0.0
245.2	-75.7	279.9	0.0	0.0
245.2	-75.7	279.9	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
273.6	-29.7	109.8	1.5	5.6
273.6	-29.7	109.8	1.5	5.6
274.3	-29.7	109.8	-1.5	-5.6
274.3	-29.7	109.8	-1.5	-5.6
245.2	-75.7	279.9	0.0	0.0
245.2	-75.7	279.9	0.0	0.0
245.2	-75.7	279.9	0.0	0.0
245.2	-75.7	279.9	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
302.8	25.9	-95.8	0.0	0.0
273.6	-29.7	109.8	1.5	5.6
273.6	-29.7	109.8	1.5	5.6
274.3	-29.7	109.8	-1.5	-5.6
274.3	-29.7	109.8	-1.5	-5.6

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	3564.5	0.00	0.0	-19.6	158.7	84.2	NA	72.7	NA
202 DL+L+TL1	4088.9	0.00	0.0	-61.8	454.9	33.7	NA	26.5	NA
203 DL+L+TL2	4088.9	0.00	0.0	-59.6	399.4	38.4	NA	27.4	NA
204 DL+LR	3709.0	0.00	0.0	-33.1	368.0	37.8	NA	44.8	NA
205 DL+0.75L+0.75TL1+0.75LR	4066.2	0.00	0.0	-61.4	537.8	28.4	NA	26.5	NA
206 DL+0.75L+0.75TL2+0.75LR	4066.2	0.00	0.0	-59.7	496.2	30.7	NA	27.2	NA
207 DL+0.625WL1	3451.5	0.00	0.0	-258.0	2989.4	4.3	NA	5.4	NA
208 DL+0.625WL2	3504.6	0.00	0.0	-179.7	2327.8	5.6	NA	7.8	NA
209 DL+0.625WL3	3527.2	0.00	0.0	-277.2	3330.3	4.0	NA	5.1	NA
210 DL+0.625WL4	3578.6	0.00	0.0	-197.4	2633.8	5.1	NA	7.3	NA
211 DL+0.625WL5	3415.5	0.00	0.0	243.2	-3203.8	4.0	NA	5.6	NA
212 DL+0.625WL6	3466.2	0.00	0.0	323.0	-3918.5	3.3	NA	4.3	NA
213 DL+0.625WL7	3507.6	0.00	0.0	230.0	-3037.8	4.3	NA	6.1	NA
214 DL+0.625WL8	3558.2	0.00	0.0	309.8	-3752.4	3.6	NA	4.6	NA
215 DL+0.625WL9	3240.1	50.16	175.8	-198.8	1742.2	7.0	55.3	6.5	25.8
216 DL+0.625WL10	3291.1	50.16	175.8	-119.0	1038.6	11.9	56.2	11.1	26.2
217 DL+0.625WL11	3750.4	-50.16	-175.8	-206.8	1867.1	7.5	64.0	7.3	29.9
218 DL+0.625WL12	3801.4	-50.16	-175.8	-127.1	1164.2	12.2	64.9	12.0	30.3
219 DL+0.75WL1+0.75L+0.75LR	3930.6	0.00	0.0	-346.6	3913.8	3.8	NA	4.5	NA
220 DL+0.75WL2+0.75L+0.75LR	3994.3	0.00	0.0	-252.7	3119.8	4.8	NA	6.3	NA
221 DL+0.75WL3+0.75L+0.75LR	4021.4	0.00	0.0	-369.6	4322.9	3.5	NA	4.4	NA
222 DL+0.75WL4+0.75L+0.75LR	4083.1	0.00	0.0	-273.9	3487.1	4.4	NA	6.0	NA
223 DL+0.75WL5+0.75L+0.75LR	3887.4	0.00	0.0	254.8	-3518.0	4.1	NA	6.1	NA
224 DL+0.75WL67+0.75L+0.75LR	3948.3	0.00	0.0	350.6	-4375.6	3.4	NA	4.5	NA
225 DL+0.75WL7+0.75L+0.75LR	3997.9	0.00	0.0	239.0	-3318.8	4.5	NA	6.7	NA
226 DL+0.75WL8+0.75L+0.75LR	4058.7	0.00	0.0	334.8	-4176.3	3.6	NA	4.8	NA
227 DL+0.75WL9+0.75L+0.75LR	3676.9	60.19	211.0	-275.6	2417.2	5.7	52.3	5.3	24.4
228 DL+0.75WL10+0.75L+0.75LR	3738.1	60.19	211.0	-179.8	1572.8	8.9	53.2	8.3	24.8
229 DL+0.75WL11+0.75L+0.75LR	4289.3	-60.19	-211.0	-285.2	2567.1	6.3	61.0	6.0	28.5
230 DL+0.75WL12+0.75L+0.75LR	4350.5	-60.19	-211.0	-189.6	1723.5	9.5	61.9	9.2	28.9
231 0.6DL+0.625WL1	3139.9	0.00	0.0	-250.1	2925.9	4.0	NA	5.0	NA
232 0.6DL+0.625WL2	3192.9	0.00	0.0	-171.9	2264.3	5.3	NA	7.4	NA
233 0.6DL+0.625WL3	3215.5	0.00	0.0	-269.3	3266.8	3.7	NA	4.8	NA
234 0.6DL+0.625WL4	3266.9	0.00	0.0	-189.5	2570.3	4.8	NA	6.9	NA
235 0.6DL+0.625WL5	3103.9	0.00	0.0	251.0	-3267.3	3.6	NA	4.9	NA
236 0.6DL+0.625WL6	3154.6	0.00	0.0	330.9	-3982.0	3.0	NA	3.8	NA
237 0.6DL+0.625WL7	3195.9	0.00	0.0	237.9	-3101.3	3.9	NA	5.4	NA
238 0.6DL+0.625WL8	3246.6	0.00	0.0	317.7	-3815.9	3.2	NA	4.1	NA
239 0.6DL+0.625WL9	2928.5	50.16	175.8	-190.9	1678.8	6.5	50.0	6.1	23.4
240 0.6DL+0.625WL10	2979.5	50.16	175.8	-111.1	975.1	11.5	50.8	10.7	23.8
241 0.6DL+0.625WL11	3438.7	-50.16	-175.8	-198.9	1803.6	7.1	58.7	6.9	27.4
242 0.6DL+0.625WL12	3489.7	-50.16	-175.8	-119.3	1100.7	11.9	59.6	11.7	27.8

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243 0.6DL+0.7SL1	3203.9	10.70	37.4	-29.2	308.4	39.0	256.8	43.9	119.8
244 0.6DL+0.7SL2	3301.8	-10.70	-37.4	5.7	-117.9	105.0	264.6	232.9	123.5
245 DL+0.75WL1+0.75L+0.75LR+CR1	4071.1	0.00	0.0	-361.3	4180.2	3.7	NA	4.5	NA
246 DL+0.75WL2+0.75L+0.75LR+CR1	4134.8	0.00	0.0	-267.4	3386.2	4.6	NA	6.2	NA
247 DL+0.75WL3+0.75L+0.75LR+CR1	4219.5	0.00	0.0	-282.8	4213.5	3.8	NA	6.0	NA
248 DL+0.75WL4+0.75L+0.75LR+CR1	4281.2	0.00	0.0	-187.0	3377.7	4.8	NA	9.2	NA
249 DL+0.75WL5+0.75L+0.75LR+CR1	4027.9	0.00	0.0	240.1	-3251.7	4.6	NA	6.7	NA
250 DL+0.75WL67+0.75L+0.75LR+CR1	4088.8	0.00	0.0	335.9	-4109.3	3.7	NA	4.9	NA
251 DL+0.75WL7+0.75L+0.75LR+CR1	4109.2	1.50	5.6	168.7	-2846.8	5.4	2221.2	9.7	1095.8
252 DL+0.75WL8+0.75L+0.75LR+CR1	4170.0	1.50	5.6	264.5	-3704.3	4.2	2254.1	6.3	1112.0
253 DL+0.75WL9+0.75L+0.75LR+CR1	3818.2	57.19	199.9	-290.3	2683.6	5.3	57.3	5.3	26.7
254 DL+0.75WL10+0.75L+0.75LR+CR1	3879.4	57.19	199.9	-194.5	1839.2	7.9	58.2	8.0	27.1
255 DL+0.75WL11+0.75L+0.75LR+CR1	4400.6	-58.69	-205.4	-345.9	3003.5	5.5	64.3	5.1	30.0
256 DL+0.75WL12+0.75L+0.75LR+CR1	4461.8	-58.69	-205.4	-250.2	2160.0	7.7	65.2	7.1	30.4
257 DL+0.75WL1+0.75L+0.75LR+CR2	3969.2	0.00	0.0	-361.4	4219.8	3.5	NA	4.4	NA
258 DL+0.75WL2+0.75L+0.75LR+CR2	4032.9	0.00	0.0	-267.5	3425.8	4.4	NA	6.0	NA
259 DL+0.75WL3+0.75L+0.75LR+CR2	4117.6	0.00	0.0	-282.9	4253.2	3.6	NA	5.8	NA
260 DL+0.75WL4+0.75L+0.75LR+CR2	4179.3	0.00	0.0	-187.1	3417.4	4.6	NA	8.9	NA
261 DL+0.75WL5+0.75L+0.75LR+CR2	3926.0	0.00	0.0	240.0	-3212.0	4.6	NA	6.5	NA
262 DL+0.75WL67+0.75L+0.75LR+CR2	3986.9	0.00	0.0	335.8	-4069.6	3.7	NA	4.7	NA
263 DL+0.75WL7+0.75L+0.75LR+CR2	4007.3	1.50	5.6	168.6	-2807.2	5.4	2166.1	9.5	1068.6
264 DL+0.75WL8+0.75L+0.75LR+CR2	4068.1	1.50	5.6	264.4	-3664.7	4.2	2199.0	6.2	1084.8
265 DL+0.75WL9+0.75L+0.75LR+CR2	3716.3	57.19	199.9	-290.4	2723.2	5.1	55.8	5.1	26.0
266 DL+0.75WL10+0.75L+0.75LR+CR2	3777.5	57.19	199.9	-194.6	1878.8	7.5	56.7	7.8	26.4
267 DL+0.75WL11+0.75L+0.75LR+CR2	4298.7	-58.69	-205.4	-346.0	3043.2	5.3	62.8	5.0	29.3
268 DL+0.75WL12+0.75L+0.75LR+CR2	4359.9	-58.69	-205.4	-250.3	2199.6	7.4	63.7	7.0	29.7
269 DL+0.75WL1+0.75L+0.75LR+CR3	3918.5	11.00	38.5	-338.2	3727.2	3.9	305.3	4.6	142.5
270 DL+0.75WL2+0.75L+0.75LR+CR3	3982.2	11.00	38.5	-244.3	2933.2	5.1	310.3	6.5	144.8
271 DL+0.75WL3+0.75L+0.75LR+CR3	4066.9	11.00	38.5	-259.7	3760.6	4.1	316.9	6.3	147.9
272 DL+0.75WL4+0.75L+0.75LR+CR3	4128.6	11.00	38.5	-163.9	2924.8	5.3	321.7	10.1	150.1
273 DL+0.75WL5+0.75L+0.75LR+CR3	3875.3	11.00	38.5	263.2	-3704.6	3.9	302.0	5.9	140.9
274 DL+0.75WL67+0.75L+0.75LR+CR3	3936.2	11.00	38.5	359.0	-4562.2	3.2	306.7	4.4	143.1
275 DL+0.75WL7+0.75L+0.75LR+CR3	3956.6	12.50	44.0	191.8	-3299.8	4.5	269.5	8.3	126.6
276 DL+0.75WL8+0.75L+0.75LR+CR3	4017.4	12.50	44.0	287.6	-4157.3	3.6	273.6	5.6	128.6
277 DL+0.75WL9+0.75L+0.75LR+CR3	3665.6	68.19	238.4	-267.2	2230.6	6.2	46.1	5.5	21.5
278 DL+0.75WL10+0.75L+0.75LR+CR3	3726.8	68.19	238.4	-171.4	1386.2	10.1	46.9	8.7	21.9
279 DL+0.75WL11+0.75L+0.75LR+CR3	4248.0	-47.69	-166.9	-322.8	2550.6	6.2	76.4	5.3	35.6
280 DL+0.75WL12+0.75L+0.75LR+CR3	4309.2	-47.69	-166.9	-227.1	1707.0	9.5	77.5	7.6	36.1
281 DL+0.75WL1+0.75L+0.75LR+CR4	4104.2	-11.00	-38.5	-338.2	3688.1	4.2	319.8	4.9	149.2
282 DL+0.75WL2+0.75L+0.75LR+CR4	4167.9	-11.00	-38.5	-244.3	2894.1	5.4	324.8	6.8	151.6
283 DL+0.75WL3+0.75L+0.75LR+CR4	4252.6	-11.00	-38.5	-259.7	3721.5	4.3	331.4	6.6	154.6
284 DL+0.75WL4+0.75L+0.75LR+CR4	4314.3	-11.00	-38.5	-163.9	2885.7	5.6	336.2	10.5	156.9
285 DL+0.75WL5+0.75L+0.75LR+CR4	4061.0	-11.00	-38.5	263.2	-3743.7	4.1	316.5	6.2	147.7
286 DL+0.75WL67+0.75L+0.75LR+CR4	4121.9	-11.00	-38.5	359.0	-4601.3	3.4	321.2	4.6	149.9
287 DL+0.75WL7+0.75L+0.75LR+CR4	4142.3	-9.50	-32.9	191.8	-3338.9	4.7	377.2	8.6	174.4
288 DL+0.75WL8+0.75L+0.75LR+CR4	4203.1	-9.50	-32.9	287.6	-4196.4	3.8	382.7	5.8	177.0
289 DL+0.75WL9+0.75L+0.75LR+CR4	3851.3	46.19	161.4	-267.2	2191.5	6.6	71.6	5.8	33.4
290 DL+0.75WL10+0.75L+0.75LR+CR4	3912.5	46.19	161.4	-171.4	1347.1	10.9	72.7	9.1	33.9
291 DL+0.75WL11+0.75L+0.75LR+CR4	4188.5	-69.69	-243.9	-247.1	2231.6	7.0	51.5	6.8	24.0
292 DL+0.75WL12+0.75L+0.75LR+CR4	4249.7	-69.69	-243.9	-151.5	1388.0	11.5	52.3	11.2	24.4

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Base pressure check of footing F3

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	131.17	0
Min Px	2	201	0	131.17	0
Max Py	793	247	0	217.30	0
Min Py	793	236	0	3.03	0
Max Pz	2	201	0	131.17	0
Min Pz	2	201	0	131.17	0

Gross Maximum Base pressure, p_{max} = 217.3 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2
Depth of soil above footing = 2.00 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 515 KN/m^2
> 217.3 KN/m^2
Hence OK

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Design of foundation raft F3

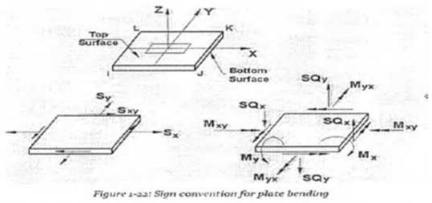
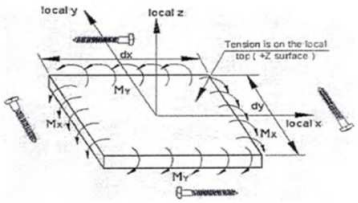


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F3_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	338	0.891	-0.124	0	0	0	-541.49	-405.09	14.99	-556.483	-420.08	135:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
Min Qx	341	-0.919	-0.141	0	0	0	-655.81	-398.17	-7.672	-663.483	-405.84	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Qy	393	-0.031	0.772	0	0	0	-495.58	-241.85	111.698	-607.277	-353.55	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Min Qy	363	-0.041	-0.779	0	0	0	-499.17	-248.96	-113.62	-612.787	-362.58	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Sx	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Min Sx	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Max Sy	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Min Sy	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Max Sxy	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Min Sxy	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Max Mx	338	-0.817	0.106	0	0	0	502.39	200.781	-19.464	521.851	220.245	124:0.9DL+1.0WL6
Min Mx	341	-0.919	-0.141	0	0	0	-655.81	-398.17	-7.672	-663.483	-405.84	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max My	338	-0.817	0.106	0	0	0	502.39	200.781	-19.464	521.851	220.245	124:0.9DL+1.0WL6
Min My	338	0.891	-0.124	0	0	0	-541.49	-405.09	14.99	-556.483	-420.08	135:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
Max Mxy	397	0.386	-0.382	0	0	0	111.44	42.271	213.64	325.075	255.911	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Min Mxy	394	-0.391	-0.331	0	0	0	109.03	34.976	-213.1	322.128	248.078	135:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1

PLATE CENTER STRESS : FDN_F3_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	543	0.24	0.024	0	0	0	162.28	120.033	-2.741	165.02	122.774	121:0.9DL+1.0WL3
Min Qx	543	-0.403	-0.036	0	0	0	-231.11	-273.81	8.564	-239.672	-282.37	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Qy	540	0.136	0.184	0	0	0	-182.11	-97.399	28.291	-210.396	-125.69	135:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
Min Qy	582	0.028	-0.186	0	0	0	-250.24	-11.127	-45.167	-295.407	-56.294	174:1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
Max Sx	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Min Sx	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Max Sy	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Min Sy	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Max Sxy	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Min Sxy	1091	0	-0.001	0	0	0	-0.41	-0.515	0.322	-0.732	-0.837	101:1.4DL
Max Mx	543	0.237	0.025	0	0	0	174.37	95.128	-1.133	175.5	96.261	147:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR2
Min Mx	582	0.021	-0.185	0	0	0	-253.69	-13.045	-39.525	-293.217	-52.57	176:1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
Max My	543	0.24	0.024	0	0	0	162.28	120.033	-2.741	165.02	122.774	121:0.9DL+1.0WL3
Min My	543	-0.398	-0.035	0	0	0	-225.37	-274.68	8.779	-234.153	-283.46	176:1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
Max Mxy	575	-0.17	-0.032	0	0	0	-58.929	-14.559	128.172	-187.101	-142.73	147:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR2
Min Mxy	559	-0.17	0.032	0	0	0	-58.921	-14.55	-128.18	-187.102	-142.73	147:1.2DL+1.0WL3+1.0L+0.5LR+1.0CR2

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		1000	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

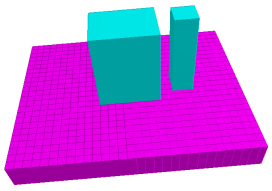
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
124:0.9D	521.85	top face	50	938	521.85	0.79	0.0019	1791.3	1687.5	25	274.15	150	3273.9	46.2	57.7	0.046	tension control
174:1.2D	-663.5	bottom face	50	938	663.48	1.01	0.0024	2286.1		25	214.81	150	3273.9	46.2	57.7	0.046	tension control
174:1.2D	325.08	top face	50	938	325.08	0.49	0.0012	1110.1		25	291.01	150	3273.9	46.2	57.7	0.046	tension control
135:1.2D	322.13	top face	50	938	322.13	0.49	0.0012	1099.9		25	291.01	150	3273.9	46.2	57.7	0.046	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.24	1	240	165.02	0.003	0.9375	963.298	943.068	1608.43	943.07	3453	> 240	ok	707.3	φVc>Vu	not required
	Min Qx	0.403	1	403	239.67			975.147			943.07	3453	> 403	ok	707.3	φVc>Vu	not required

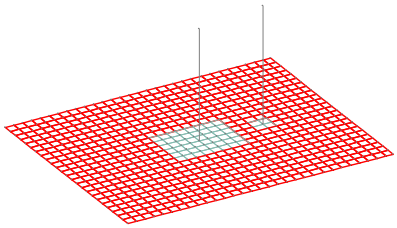
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
124:0.9D	220.25	top face	50	940	220.25	0.33	0.0008	748.0	1692	20	186.24	150	2095.3	29.5	36.93	0.073	tension control
135:1.2D	-420.08	bottom face	50	940	420.08	0.63	0.0015	1434.2		20	186.24	150	2095.3	29.5	36.93	0.073	tension control
174:1.2D	255.91	top face	50	940	255.91	0.39	0.0009	870.0		20	186.24	150	2095.3	29.5	36.93	0.073	tension control
135:1.2D	248.08	top face	50	940	248.08	0.37	0.0009	843.2		20	186.24	150	2095.3	29.5	36.93	0.073	tension control

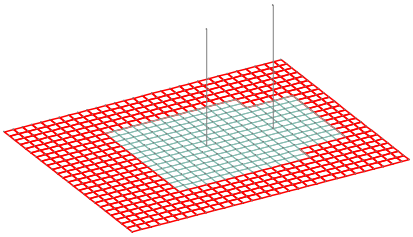
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.184	1	184	125.69	0.002	0.94	963.796	943.068	fc'1608	943.07	982	> 184	ok	707.3	φVc>Vu	not required
	Min Qy	0.186	1	186	56.294			1059.81			943.07	982	> 186	ok	707.3	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**25φT**	**150**	c/c
At Top	**25φT**	**150**	c/c
Along Z Direction :			
At Bottom	**20φT**	**150**	c/c
At Top	**20φT**	**150**	c/c

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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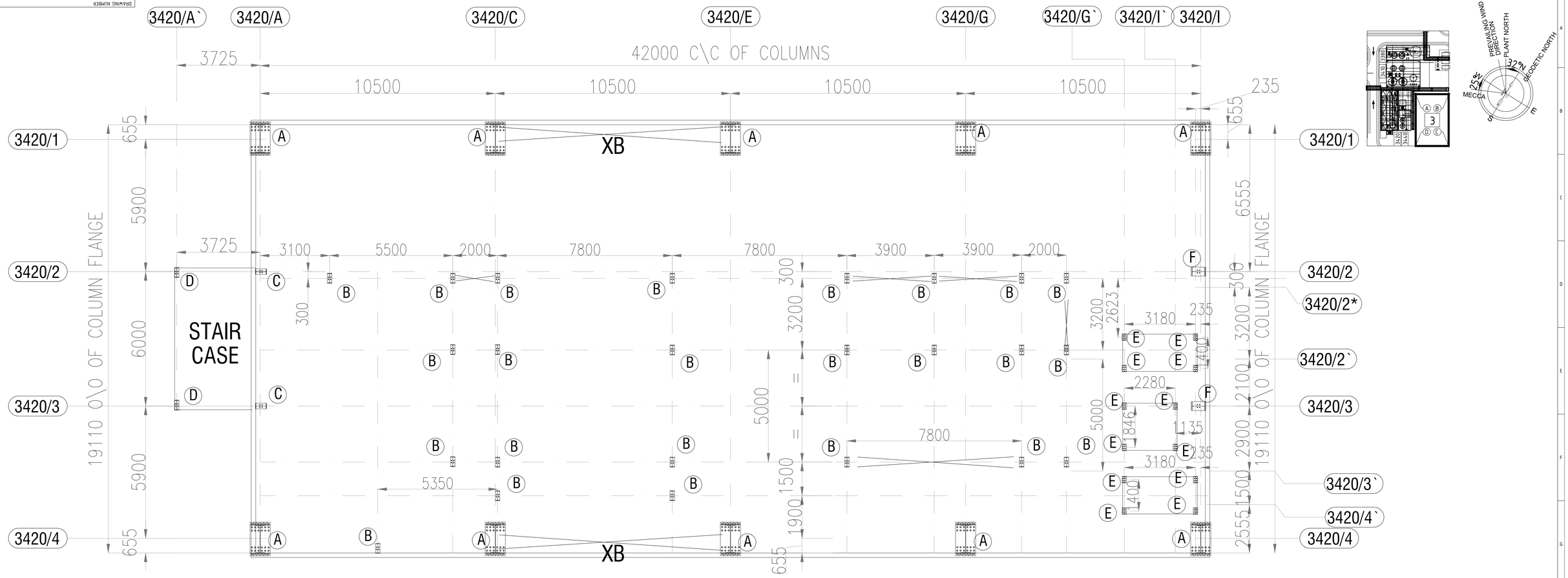
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ATTACHMENT 1 INPUT DRAWINGS

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DRAWING NUMBER MD-216-3420-EG-ST-GAD-2229 REV A01

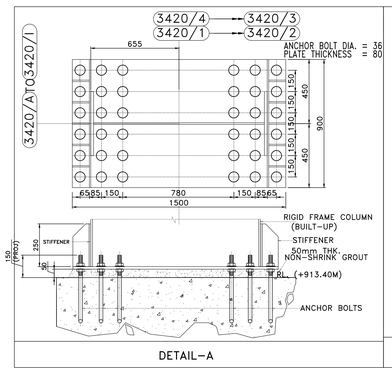


ANCHOR BOLT PLAN

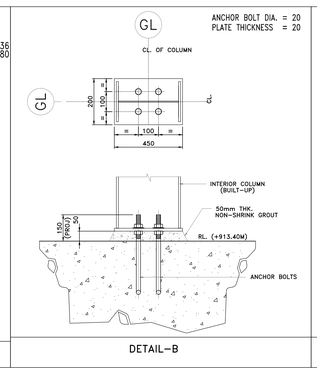
FFL IS AT LEVEL +913.40

I DENOTES TO PLATFORM COLUMNS

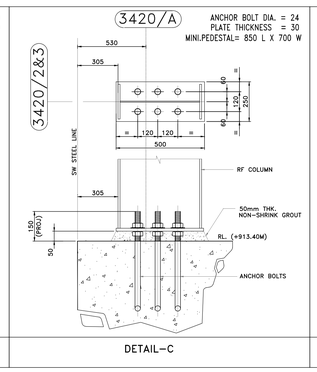
XB DENOTE TO X-BRACING.



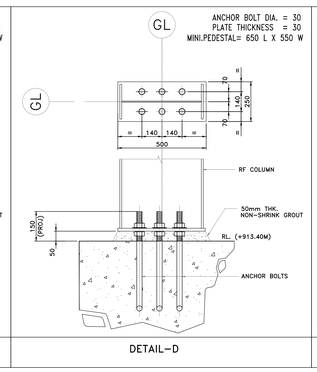
DETAIL-A



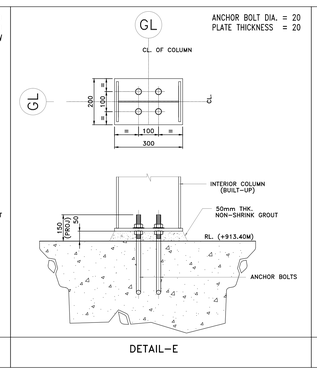
DETAIL-B



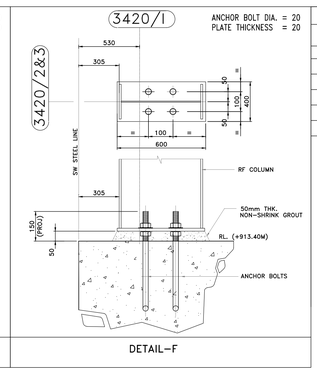
DETAIL-C



DETAIL-D



DETAIL-E



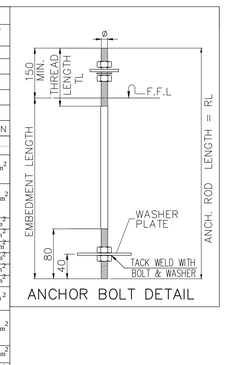
DETAIL-F

NOTE: ANCHOR BOLT SPECIFICATION (AS PER GR. 8.8)

QTY.	DIA. OF BOLT	DIA. OF HOLE	THREAD LENGTH	ANCHOR ROD LENGTH
	Ø	D	TL	RL
--	M20	26	150	450
--	M24	30	150	500
--	M30	36	200	700
	M36	42	200	1000
	>M36	Ø + 8	200	AS PER DESIGN

MATERIAL SPECIFICATIONS

NO.	MATERIALS		GRADE & SPECIFICATION	STRENGTH
01	ANCHOR BOLTS	J-type	ASTM F1554 Grade A36M, Hot dip galvanized to ASTM A153M, Heavy Hex Nuts A563M-10S Washers-ASTMF436	Fu = 400 N/mm^2
		STRAIGHT -type	BS EN ISO 898-1 GRDE 8.8, Hot dip galvanized to BS EN ISO1461 BS EN ISO 898-2 10 STYLE2 WASHER-BS4320	Fu = 800 N/mm^2
02	BUILT-UP SECTIONS		ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm^2
03	HOT ROLLED SECTIONS	Tubes	E10025 S355HR or equivalent	Fy = 345 N/mm^2
		I-Sections	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm^2
		Channels	E10025 S355JR or equivalent	Fy = 355 N/mm^2
		Angles	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm^2
		Pipes	ASTM A53 Grade B or equivalent	Fy = 240 N/mm^2
04	Primary Connection bolts		ASTM A325M TYPE-1; Fully threaded hot dip galvanized	Ft = 827 N/mm^2
05	DIAGONAL "X" BRACING	HR Angles	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm^2
06	Cold formed secondary members		ASTM A653M SS Grade 340 Class 1: Coating Z275 (Grade 90)	Fy = 344 N/mm^2
07	Secondary Connection bolts (for purlins,girts and flange braces)		Grade 8.8 Fully threaded hot dip galvanized	Ft = 400 N/mm^2
08	Sag Rods		ASTM Grade A36M Electro galvanized, Fully threaded with 4 nuts Coating Z275 (Grade 90)	Fy = 245 N/mm^2
09	Mezzanine Deck Panels		ASTM A653M SS Grade 340 Class 1: Coating Z275 (Grade 90)	Fy = 344 N/mm^2



ANCHOR BOLT DETAIL

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3420-EG-ME-GAD-1015	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL929.300								
MD-216-3420-EG-ME-GAD-1014	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL920.100								
MD-216-3420-EG-ME-GAD-1013	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL912.900								
MD-216-3420-EG-ME-GAD-1012	LOAD PLAN, FEED STORAGE,PLAN AT LEVEL929.30								
MD-216-3420-EG-ME-GAD-1011	LOAD PLAN, FEED STORAGE,PLAN AT LEVEL912.900								
MD-216-3420-EG-ME-GAD-1001	GENERAL ARRANGEMENT DRAWING PLAN	A01	02/07/2020	ISSUED FOR REFERENCE	--	SYC	--	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

P.O.Box : 41017 Jeddah 21521,K.S.A
Tel. : +966 2 591 0999
Fax : +966 2 591 0099
E-mail : aic@aicsteel.com
Site : www.aicsteel.com

DRAWING NUMBER					REVISION
MD-P20131-B03-A001					03
DATE	29/06/2020	DATE	29/06/2020	DATE	29/06/2020
DRAWN	7014	CHECKED	7013	APPROVED	7001

LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_450_S_G_26043_001	0
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707 _ GAU01 _ HH12 _ MTL01 _ 34211	OU100668474 / 0

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DRAWN	--	--	
CHECKED	SYC	02/07/2020	
ENGINEER	--	--	
APPROVED	PRI	02/07/2020	
EMPLOYER			
SCALE	NTS	A0	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRESSURE OXIDATION-AUTOCLAVE BUILDING
ANCHOR BOLT PLAN AND DETAILS

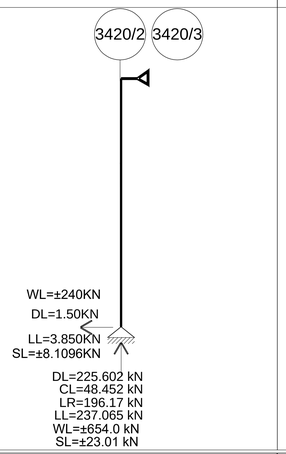
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DRAWING NUMBER MD-216-3420-EG-ST-GAD-2230 | REV A01

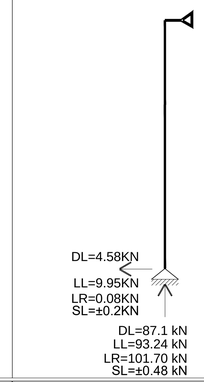
LOADS LEGEND	
Wind_Left1	Denotes wind load from left to right with internal pressure
Wind_Right1	Denotes wind load from right to left with internal pressure
Wind_Left2	Denotes wind load from left to right with internal suction
Wind_Right2	Denotes wind load from right to left with internal suction
Wind_Left2	Denotes wind load from left to right with internal pressure
Wind_Right2	Denotes wind load from right to left with internal pressure
Wind_Left3	Denotes wind load from left to right with internal suction
Wind_Right4	Denotes wind load from right to left with internal suction
Wind_Long1	Denotes wind load parallel to ridge from end wall with internal pressure
Wind_Long2	Denotes wind load parallel to ridge from end wall with internal suction
Seismic_L	Denotes seismic load from left
Seismic_R	Denotes seismic load from right

IMPORTANT NOTES :
(1) BEARING CAPACITY OF CONCRETE UNDER BASE PLATE SHOULD NOT BE LESS THAN 1.05 KN/CM2.
(2) GIVEN REACTIONS ARE FOR UNFACTORED LOADS.
(3) BRACING REACTIONS TO BE ADDED WITH MAIN FRAME COLUMN REACTIONS FOR THE DESIGN OF FOUNDATION.
(4) LOADS IN THE SAME CATEGORY , NOT TO BE COMBINED IN THE SAME LOAD COMBINATION .

ENDWALL COLUMN REACTION AT GL I



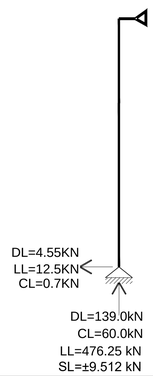
PLATFORM COLUMN REACTION AT GL 3420/2



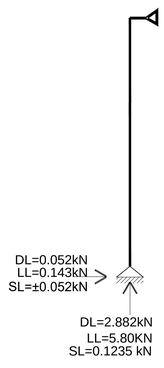
PLATFORM COLUMN REACTION AT GL 3420/2`

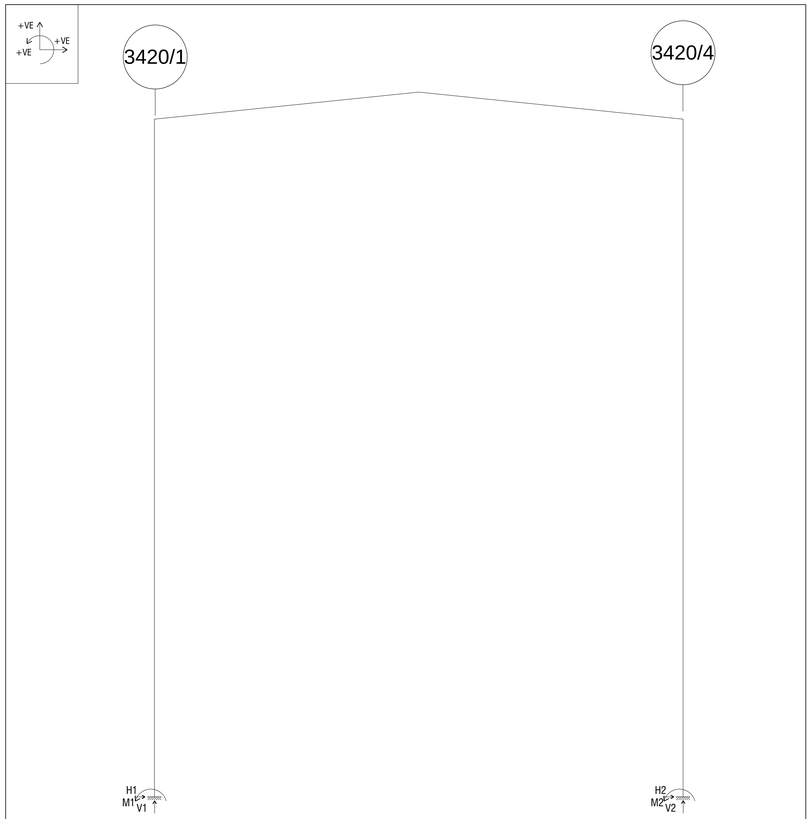


PLATFORM COLUMN REACTION AT GL 3420/3`



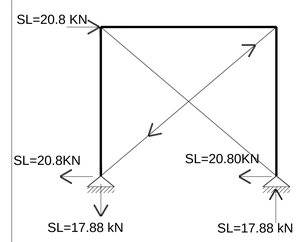
TYP.PLATFORM COLUMN REACTION AT GL 3420/G`&I` (separate platform)



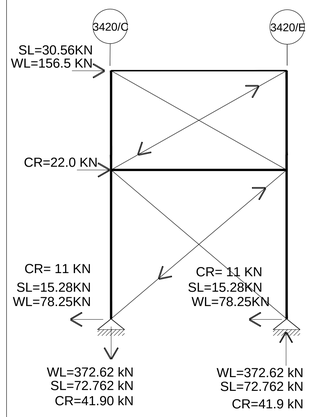


TYP. FRAME @ G.L 3420/A TO GL 3420/I							
CATEGORY	LOAD CASE	LEFT COL.@ GL 3420/1			RIGHT COL.@ GL 3420/4		
		H1 (KN)	V1 (KN)	M1(KN.M)	H2 (KN)	V2 (KN)	M2 (KN.M)
Dead	Dead	4.9	94.8	-55.6	-12.1	224.4	108.3
Collateral	Collateral	-0.2	24.8	-14.4	0.2	167.3	-19.7
Roof Live	Live	13.5	72.6	-160.4	-13.5	144.5	162.0
FLOOR Live	Floor Live	-1.0	52.2	19.8	-18.2	325.7	120.0
Crane	Crane1	-8.0	29.8	148.2	-14.7	140.5	214.9
	Crane2	-7.9	131.7	184.6	-14.8	38.6	254.2
	Crane3	14.3	140.5	-205.7	8.4	29.8	-157.2
	Crane4	14.3	38.6	-242.2	8.4	131.7	-196.3
Wind	Wind_Left1	-338.8	-281.6	3672.1	-313.7	-142.4	3180.7
	Wind_Left2	-473.9	-196.1	4432.0	-188.5	-57.5	2560.2
	Wind_Left3	-324.9	-129.6	3543.1	-344.4	-21.3	3618.7
	Wind_Left4	-452.6	-48.7	4237.0	-216.7	60.9	2951.2
	Wind_Right1	309.8	-140.0	-3077.7	352.7	-276.8	-3895.1
	Wind_Right2	182.0	-58.0	-2410.4	480.5	-195.6	-4591.3
	Wind_Right3	337.6	-21.3	-3460.1	331.7	-129.5	-3703.0
	Wind_Right4	209.8	60.6	-2794.0	459.4	-48.4	-4399.4
	Wind_Long1	280.3	-145.4	-1523.8	-280.3	-145.9	1528.9
	Wind_Long2	152.6	-64.1	-847.1	-152.6	-64.3	850.0
	Wind_Long3	293.1	-75.1	-1678.0	-293.1	-75.7	1683.9
	Wind_Long4	165.6	6.3	-1001.7	-165.6	5.9	1005.4
Seismic	Seismic_L	-5.1	-2.8	98.3	-24.9	2.8	217.4
	Seismic_R	5.1	2.8	-98.3	24.9	-2.8	-217.4
Temperature	25Degree	1.1	0.0	-23.7	-1.1	0.0	23.9
	-25Degree	-1.1	0.0	23.7	1.1	0.0	-23.9

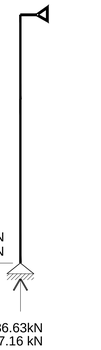
SIDEWALL (XB) BRACING REACTION AT GL 3420/2&3420/3`



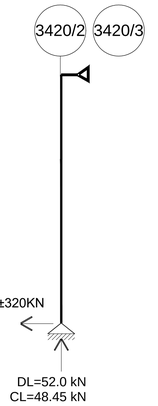
SIDEWALL (XB) BRACING REACTION AT GL 3420/1& 3420/4



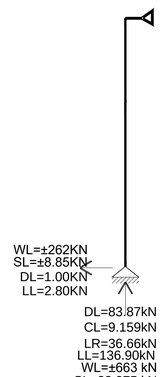
PLATFORM COLUMN REACTION AT GL 3420/4



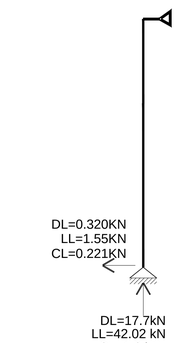
ENDWALL COLUMN REACTION AT GL 3240/A



STAIR COLUMN REACTION



PLATFORM COLUMN REACTION AT GL 3420/4`



WIND IS CALCULATED BASED ON SBC-301(2018)
ALL HORIZONTAL REACTION UNDER PLATFORM COLUMNS ARE IN TRANSVERSE DIRECTION.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3420-EG-ME-GAD-1015	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL929.300								
MD-216-3420-EG-ME-GAD-1014	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL920.100								
MD-216-3420-EG-ME-GAD-1013	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL912.900								
MD-216-3420-EG-ME-GAD-1012	LOAD PLAN, FEED STORAGE,PLAN AT LEVEL929.30								
MD-216-3420-EG-ME-GAD-1011	LOAD PLAN, FEED STORAGE,PLAN AT LEVEL912.900								
MD-216-3420-EG-ME-GAD-1001	GENERAL ARRANGEMENT DRAWING PLAN	A01	02/07/2020	ISSUED FOR REFERENCE	--	SYC	--	PRI	

P.O.Box : 41017 Jeddah 21521,K.S.A
Tel. : +966 2 591 0999
Fax : +966 2 591 0099
E-mail : aic@aicsteel.com
Site : www.aicsteel.com

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DRAWING NUMBER	MD-P20131-B03-A002
REVISION	03
DATE / DRAWN	29/06/2020 / 7014
DATE / CHECKED	29/06/2020 / 7013
DATE / APPROVED	29/06/2020 / 7001
LARSEN & TOUBRO DRAWING NUMBER	L19045_450_S_G_26043_002
REVISION	0
OUTOTEC DRAWING NUMBER	900707 __ GAU01 __ HH12 __ MTL01 __ 34212
DOCUMENT ID	OU100668475
REVISION	0
DRAWN	-- / --
CHECKED	SYC / 02/07/2020
ENGINEER	-- / --
APPROVED	PRI / 02/07/2020
EMPLOYER	
SCALE	NTS / A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRESSURE OXIDATION-AUTOCLAVE BUILDING
COLUMN REACTIONS

DRAWING NUMBER MD-216-3420-EG-ST-GAD-2230 | REV A01

ATTACHMENT 2 STAAD INPUT – FOUNDATION F1

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28/05/2020
JOB NAME FDN. MKD. F1
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
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START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091
_RAFT_MOM 338 341 350 351 360 361 363 TO 402 410 411 420 421 430 431 -
433 TO 794 809 811 813 TO 826 841 843 845 TO 901 903 905 907 909 TO 965 967 -
969 971 TO 1091
_RAFT_SHEAR 529 TO 794 809 811 813 TO 826 841 843 845 TO 901 903 905 907 909 -
910 TO 965 967 969 971 TO 1091
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 THICKNESS 1
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 2 ZD 1.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1090 -
1091 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
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LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 PR GY -40
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 PR GY -40
JOINT LOAD
128 FX 11.9 FY -391.7 MZ -88.6
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FX 13.5 FY -144.5 MZ -162
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX 18.2 FY -325.7 MZ -120
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE CRANE LOAD  ( CR1 )
JOINT LOAD
128 FX 14.7 FY -140.5 MZ -214.9
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE CRANE LOAD  ( CR2 )
JOINT LOAD
128 FX 14.8 FY -38.6 MZ -254.2
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE CRANE LOAD  ( CR3 )
JOINT LOAD
128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE CRANE LOAD  ( CR4 )
JOINT LOAD
128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FX 313.7 FY 142.4 MZ -3180.7
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FX 188.5 FY 57.5 MZ -2560.2
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FX 344.4 FY 21.3 MZ -3618.7
*********************************************************************************************
LOAD 11 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FX 216.7 FY -60.9 MZ -2951.2
*********************************************************************************************
LOAD 12 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FX -352.7 FY 276.8 MZ 3895.1
*********************************************************************************************
LOAD 13 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FX -480.5 FY 195.6 MZ 4591.3
*********************************************************************************************
LOAD 14 LOADTYPE None  TITLE WIND LOAD  ( WL7 )
JOINT LOAD
128 FX -331.7 FY 129.5 MZ 3703
*********************************************************************************************
LOAD 15 LOADTYPE None  TITLE WIND LOAD  ( WL8 )
JOINT LOAD
128 FX -459.4 FY 48.4 MZ 4399.4
*********************************************************************************************
LOAD 16 LOADTYPE None  TITLE WIND LOAD  ( WL9 )
JOINT LOAD
128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
*********************************************************************************************
LOAD 17 LOADTYPE None  TITLE WIND LOAD  ( WL10 )
JOINT LOAD
128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
*********************************************************************************************
LOAD 18 LOADTYPE None  TITLE WIND LOAD  ( WL11 )
JOINT LOAD
128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
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128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
*********************************************************************************************
LOAD 19 LOADTYPE None  TITLE WIND LOAD  ( WL12 )
JOINT LOAD
128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
*********************************************************************************************
LOAD 20 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
*********************************************************************************************
LOAD 21 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
*********************************************************************************************
LOAD 22 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FX 1.1 FY 0 MZ -23.9
*********************************************************************************************
LOAD 23 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FX -1.1 FY 0 MZ 23.9
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L+TL1
1 1.0 3 1.0 22 1.0
LOAD COMB 203 DL+L+TL2
1 1.0 3 1.0 23 1.0
LOAD COMB 204 DL+LR
1 1.0 2 1.0
LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
1 1.0 2 0.75 3 0.75 22 0.75
LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
1 1.0 2 0.75 3 0.75 23 0.75
LOAD COMB 207 DL+0.625WL1
1 1.0 8 0.625
LOAD COMB 208 DL+0.625WL2
1 1.0 9 0.625
LOAD COMB 209 DL+0.625WL3
1 1.0 10 0.625
LOAD COMB 210 DL+0.625WL4
1 1.0 11 0.625
LOAD COMB 211 DL+0.625WL5
1 1.0 12 0.625
LOAD COMB 212 DL+0.625WL6
1 1.0 13 0.625
LOAD COMB 213 DL+0.625WL7
1 1.0 14 0.625
LOAD COMB 214 DL+0.625WL8
1 1.0 15 0.625
LOAD COMB 215 DL+0.625WL9
1 1.0 16 0.625
LOAD COMB 216 DL+0.625WL10
1 1.0 17 0.625
LOAD COMB 217 DL+0.625WL11
1 1.0 18 0.625
LOAD COMB 218 DL+0.625WL12
1 1.0 19 0.625
LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 8 0.75
LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 9 0.75
LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 10 0.75
LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 11 0.75
LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 12 0.75
LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 13 0.75
LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 14 0.75
LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 15 0.75
LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 16 0.75
LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 17 0.75
```



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1 1.0 2 0.75 3 0.75 17 0.75
LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 18 0.75
LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 19 0.75
LOAD COMB 231 0.6DL+0.625WL1
1 0.6 8 0.625
LOAD COMB 232 0.6DL+0.625WL2
1 0.6 9 0.625
LOAD COMB 233 0.6DL+0.625WL3
1 0.6 10 0.625
LOAD COMB 234 0.6DL+0.625WL4
1 0.6 11 0.625
LOAD COMB 235 0.6DL+0.625WL5
1 0.6 12 0.625
LOAD COMB 236 0.6DL+0.625WL6
1 0.6 13 0.625
LOAD COMB 237 0.6DL+0.625WL7
1 0.6 14 0.625
LOAD COMB 238 0.6DL+0.625WL8
1 0.6 15 0.625
LOAD COMB 239 0.6DL+0.625WL9
1 0.6 16 0.625
LOAD COMB 240 0.6DL+0.625WL10
1 0.6 17 0.625
LOAD COMB 241 0.6DL+0.625WL11
1 0.6 18 0.625
LOAD COMB 242 0.6DL+0.625WL12
1 0.6 19 0.625
LOAD COMB 243 0.6DL+0.7SL1
1 0.6 20 0.7
LOAD COMB 244 0.6DL+0.7SL2
1 0.6 21 0.7
LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
```

```
1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
LOAD COMB 275 DL+0.75WL7+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
LOAD COMB 287 DL+0.75WL7+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.4TL1+1.7L+0.5LR
1 1.4 22 1.4 3 1.7 2 0.5
LOAD COMB 103 1.4DL+1.4TL2+1.7L+0.5LR
1 1.4 23 1.4 3 1.7 2 0.5
LOAD COMB 104 1.2DL+1.0LR+1.0L
1 1.2 2 1.0 3 1.0
LOAD COMB 105 1.2DL+1.0WL1+1.0L+0.5LR
1 1.2 8 1.0 2 0.5 3 1.0
LOAD COMB 106 1.2DL+1.0WL2+0.5L+0.5LR
1 1.2 9 1.0 2 0.5 3 1.0
LOAD COMB 107 1.2DL+1.0WL3+0.5L+0.5LR
1 1.2 10 1.0 2 0.5 3 1.0
LOAD COMB 108 1.2DL+1.0WL4+0.5L+0.5LR
1 1.2 11 1.0 2 0.5 3 1.0
LOAD COMB 109 1.2DL+1.0WL5+0.5L+0.5LR
1 1.2 12 1.0 2 0.5 3 1.0
LOAD COMB 110 1.2DL+1.0WL6+0.5L+0.5LR
1 1.2 13 1.0 2 0.5 3 1.0
LOAD COMB 111 1.2DL+1.0WL7+0.5L+0.5LR
1 1.2 14 1.0 2 0.5 3 1.0
LOAD COMB 112 1.2DL+1.0WL8+0.5L+0.5LR
1 1.2 15 1.0 2 0.5 3 1.0
LOAD COMB 113 1.2DL+1.0WL9+0.5L+0.5LR
1 1.2 16 1.0 2 0.5 3 1.0
```



```
1 1.2 16 1.0 2 0.5 3 1.0
LOAD COMB 114 1.2DL+1.0WL10+0.5L+0.5LR
1 1.2 17 1.0 2 0.5 3 1.0
LOAD COMB 115 1.2DL+1.0WL11+0.5L+0.5LR
1 1.2 18 1.0 2 0.5 3 1.0
LOAD COMB 116 1.2DL+1.0WL12+0.5L+0.5LR
1 1.2 19 1.0 2 0.5 3 1.0
LOAD COMB 117 1.2DL+1.0SL1+1.0L
1 1.2 20 1.0 3 1.0
LOAD COMB 118 1.2DL+1.0SL2+1.0L
1 1.2 21 1.0 3 1.0
LOAD COMB 119 0.9DL+1.0WL1
1 0.9 8 1.0
LOAD COMB 120 0.9DL+1.0WL2
1 0.9 9 1.0
LOAD COMB 121 0.9DL+1.0WL3
1 0.9 10 1.0
LOAD COMB 122 0.9DL+1.0WL4
1 0.9 11 1.0
LOAD COMB 123 0.9DL+1.0WL5
1 0.9 12 1.0
LOAD COMB 124 0.9DL+1.0WL6
1 0.9 13 1.0
LOAD COMB 125 0.9DL+1.0WL7
1 0.9 14 1.0
LOAD COMB 126 0.9DL+1.0WL8
1 0.9 15 1.0
LOAD COMB 127 0.9DL+1.0WL9
1 0.9 16 1.0
LOAD COMB 128 0.9DL+1.0WL10
1 0.9 17 1.0
LOAD COMB 129 0.9DL+1.0WL11
1 0.9 18 1.0
LOAD COMB 130 0.9DL+1.0WL12
1 0.9 19 1.0
LOAD COMB 131 0.9DL+1.0SL1
1 0.9 20 1.0
LOAD COMB 132 0.9DL+1.0SL2
1 0.9 21 1.0
LOAD COMB 133 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR1
1 1.2 8 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 134 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR1
1 1.2 9 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 135 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
1 1.2 10 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 136 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR1
1 1.2 11 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 137 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR1
1 1.2 12 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 138 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR1
1 1.2 13 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 139 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR1
1 1.2 14 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 140 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR1
1 1.2 15 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 141 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR1
1 1.2 16 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 142 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR1
1 1.2 17 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 143 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR1
1 1.2 18 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 144 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR1
1 1.2 19 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 145 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR2
1 1.2 8 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 146 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR2
1 1.2 9 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 147 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR2
1 1.2 10 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 148 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR2
1 1.2 11 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 149 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR2
1 1.2 12 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 150 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR2
1 1.2 13 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 151 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR2
1 1.2 14 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 152 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR2
1 1.2 15 1.0 2 0.5 3 1.0 5 1.0
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1 1.2 15 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 153 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR2
1 1.2 16 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 154 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR2
1 1.2 17 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 155 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR2
1 1.2 18 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 156 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR2
1 1.2 19 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 157 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR3
1 1.2 8 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 158 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR3
1 1.2 9 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 159 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR3
1 1.2 10 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 160 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR3
1 1.2 11 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 161 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR3
1 1.2 12 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 162 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR3
1 1.2 13 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 163 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR3
1 1.2 14 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 164 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR3
1 1.2 15 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 165 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR3
1 1.2 16 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 166 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR3
1 1.2 17 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 167 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR3
1 1.2 18 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 168 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR3
1 1.2 19 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 169 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR4
1 1.2 8 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 170 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR4
1 1.2 9 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 171 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR4
1 1.2 10 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 172 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR4
1 1.2 11 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 173 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR4
1 1.2 12 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 174 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
1 1.2 13 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 175 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR4
1 1.2 14 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 176 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
1 1.2 15 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 177 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR4
1 1.2 16 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 178 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR4
1 1.2 17 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 179 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR4
1 1.2 18 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 180 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR4
1 1.2 19 1.0 2 0.5 3 1.0 7 1.0
********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 180
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28/05/2020
JOB NAME FDN. MKD. F1_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 913.4 3.175;
MEMBER INCIDENCES
185 2 128;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 2 ZD 1.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 FIXED
************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FX 11.9 FY -391.7 MZ -88.6
************************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FX 13.5 FY -144.5 MZ -162
************************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX 18.2 FY -325.7 MZ -120
************************************************************************************************
LOAD 4 LOADTYPE None  TITLE CRANE LOAD  ( CR1 )
JOINT LOAD
128 FX 14.7 FY -140.5 MZ -214.9
************************************************************************************************
LOAD 5 LOADTYPE None  TITLE CRANE LOAD  ( CR2 )
JOINT LOAD
128 FX 14.8 FY -38.6 MZ -254.2
************************************************************************************************
LOAD 6 LOADTYPE None  TITLE CRANE LOAD  ( CR3 )
JOINT LOAD
128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
************************************************************************************************
LOAD 7 LOADTYPE None  TITLE CRANE LOAD  ( CR4 )
JOINT LOAD
128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
************************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FX 313.7 FY 142.4 MZ -3180.7
************************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FX 188.5 FY 57.5 MZ -2560.2
************************************************************************************************
LOAD 10 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FX 344.4 FY 21.3 MZ -3618.7
************************************************************************************************
LOAD 11 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FX 216.7 FY -60.9 MZ -2951.2
************************************************************************************************
LOAD 12 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
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JOINT LOAD
128 FX -352.7 FY 276.8 MZ 3895.1
************************************************************************************************
LOAD 13 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FX -480.5 FY 195.6 MZ 4591.3
************************************************************************************************
LOAD 14 LOADTYPE None  TITLE WIND LOAD  ( WL7 )
JOINT LOAD
128 FX -331.7 FY 129.5 MZ 3703
************************************************************************************************
LOAD 15 LOADTYPE None  TITLE WIND LOAD  ( WL8 )
JOINT LOAD
128 FX -459.4 FY 48.4 MZ 4399.4
************************************************************************************************
LOAD 16 LOADTYPE None  TITLE WIND LOAD  ( WL9 )
JOINT LOAD
128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
************************************************************************************************
LOAD 17 LOADTYPE None  TITLE WIND LOAD  ( WL10 )
JOINT LOAD
128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
************************************************************************************************
LOAD 18 LOADTYPE None  TITLE WIND LOAD  ( WL11 )
JOINT LOAD
128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
************************************************************************************************
LOAD 19 LOADTYPE None  TITLE WIND LOAD  ( WL12 )
JOINT LOAD
128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
************************************************************************************************
LOAD 20 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
************************************************************************************************
LOAD 21 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
************************************************************************************************
LOAD 22 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FX 1.1 FY 0 MZ -23.9
************************************************************************************************
LOAD 23 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FX -1.1 FY 0 MZ 23.9
************************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L+TL1
1 1.0 3 1.0 22 1.0
LOAD COMB 203 DL+L+TL2
1 1.0 3 1.0 23 1.0
LOAD COMB 204 DL+LR
1 1.0 2 1.0
LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
1 1.0 2 0.75 3 0.75 22 0.75
LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
1 1.0 2 0.75 3 0.75 23 0.75
LOAD COMB 207 DL+0.625WL1
1 1.0 8 0.625
LOAD COMB 208 DL+0.625WL2
1 1.0 9 0.625
LOAD COMB 209 DL+0.625WL3
1 1.0 10 0.625
LOAD COMB 210 DL+0.625WL4
1 1.0 11 0.625
LOAD COMB 211 DL+0.625WL5
1 1.0 12 0.625
LOAD COMB 212 DL+0.625WL6
1 1.0 13 0.625
LOAD COMB 213 DL+0.625WL7
1 1.0 14 0.625
LOAD COMB 214 DL+0.625WL8
1 1.0 15 0.625
LOAD COMB 215 DL+0.625WL9
1 1.0 16 0.625
LOAD COMB 216 DL+0.625WL10
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LOAD COMB 216 DL+0.625WL10
1 1.0 17 0.625
LOAD COMB 217 DL+0.625WL11
1 1.0 18 0.625
LOAD COMB 218 DL+0.625WL12
1 1.0 19 0.625
LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 8 0.75
LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 9 0.75
LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 10 0.75
LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 11 0.75
LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 12 0.75
LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 13 0.75
LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 14 0.75
LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 15 0.75
LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 16 0.75
LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 17 0.75
LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 18 0.75
LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 19 0.75
LOAD COMB 231 0.6DL+0.625WL1
1 0.6 8 0.625
LOAD COMB 232 0.6DL+0.625WL2
1 0.6 9 0.625
LOAD COMB 233 0.6DL+0.625WL3
1 0.6 10 0.625
LOAD COMB 234 0.6DL+0.625WL4
1 0.6 11 0.625
LOAD COMB 235 0.6DL+0.625WL5
1 0.6 12 0.625
LOAD COMB 236 0.6DL+0.625WL6
1 0.6 13 0.625
LOAD COMB 237 0.6DL+0.625WL7
1 0.6 14 0.625
LOAD COMB 238 0.6DL+0.625WL8
1 0.6 15 0.625
LOAD COMB 239 0.6DL+0.625WL9
1 0.6 16 0.625
LOAD COMB 240 0.6DL+0.625WL10
1 0.6 17 0.625
LOAD COMB 241 0.6DL+0.625WL11
1 0.6 18 0.625
LOAD COMB 242 0.6DL+0.625WL12
1 0.6 19 0.625
LOAD COMB 243 0.6DL+0.7SL1
1 0.6 20 0.7
LOAD COMB 244 0.6DL+0.7SL2
1 0.6 21 0.7
LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
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LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
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LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
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LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
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LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
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LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
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LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
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LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
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LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
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LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
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LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
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LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
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LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
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LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
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LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
********************************************************************************************************
PERFORM ANALYSIS
FINISH
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FINISH

ATTACHMENT 3 STAAD INPUT – FOUNDATION F2

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 10/06/2020
JOB NAME FDN. MKD. F2
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
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35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175;
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694 702 543 542 699; 695 703 704 701 700; 696 704 702 699 701;
697 705 544 543 702; 698 706 707 704 703; 699 707 705 702 704;
700 708 545 546 709; 701 710 711 712 713; 702 711 708 709 712;
703 714 547 545 708; 704 715 716 711 710; 705 716 714 708 711;
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709 720 549 548 717; 710 721 722 719 718; 711 722 720 717 719;
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913 930 931 576 575; 914 931 932 577 576; 915 932 933 578 577;
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961 980 978 972 974; 965 965 984 824 689; 967 971 986 984 965;
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981 886 878 992 1000; 982 885 888 1002 999; 983 889 887 1001 1003;
984 888 890 1004 1002; 985 891 889 1003 1005; 986 890 892 1006 1004;
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993 996 997 1014 1013; 994 997 998 1015 1014; 995 998 999 1016 1015;
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999 1003 1001 1018 1020; 1000 1002 1004 1021 1019; 1001 1005 1003 1020 1022;
1002 1004 1006 1023 1021; 1003 1007 1005 1022 1024; 1004 892 900 1025 1006;
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1029 1030 1049 1050 1036; 1032 1053 1054 944 943; 1033 1054 1055 945 944;
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1073 1095 1096 1090 1089; 1074 1096 1097 1091 1090; 1075 1097 1098 1092 1091;

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1091 794 1115 1113 792; 1092 795 1116 1115 794; 1093 796 1117 1116 795;
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1097 801 1122 1121 800; 1098 802 1123 1122 801; 1099 793 1114 1123 802;
1100 1113 1124 1125 1114; 1101 1115 1126 1124 1113; 1102 1116 1127 1126 1115;
1103 1117 1128 1127 1116; 1104 1118 1129 1128 1117; 1105 1119 1130 1131 1120;
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1109 1114 1125 1134 1123; 1110 803 1135 1118 797; 1111 1135 1136 1129 1118;
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1127 1137 1138 1153 1152; 1129 806 1155 1135 803; 1130 1155 1156 1136 1135;
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1195 1218 1219 1223 1222; 1200 1111 1228 1212 986; 1201 1228 1229 1213 1212;
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1207 1126 1236 1234 1124; 1208 1127 1237 1236 1126; 1209 1128 1238 1237 1127;
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1225 1213 1254 1253 1209; 1226 1203 1252 1255 1219; 1227 1219 1255 1256 1223;
1228 1229 1257 1254 1213; 1229 1233 1258 1257 1229;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 TO 731 -
745 TO 747 750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 845 TO 891 -
895 TO 897 901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 TO 1024 -
1028 1029 1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 -
1147 1150 1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 -
1200 1201 1204 TO 1229
_RAFT_MOM 338 341 350 351 360 361 363 TO 402 410 411 420 421 430 431 -
433 TO 714 725 TO 727 729 TO 731 745 TO 747 750 TO 752 769 TO 771 -
775 TO 777 781 TO 794 809 811 845 TO 891 895 TO 897 901 903 909 TO 955 959 -
960 TO 961 965 967 972 TO 1018 1022 TO 1024 1028 1029 1032 TO 1078 -
1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 1147 1150 1151 1170 1171 -
1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 1200 1201 1204 TO 1229
_RAFT_SHEAR 501 TO 714 725 TO 727 729 TO 731 745 TO 747 750 TO 752 -
769 TO 771 775 TO 777 781 TO 794 809 811 845 TO 891 895 TO 897 901 903 909 -
910 TO 955 959 TO 961 965 967 972 TO 1018 1022 TO 1024 1028 1029 1032 TO 1078 -
1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 1147 1150 1151 1170 1171 -
1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 1200 1201 1204 TO 1229
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 TO 731 745 TO 747 -
750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 845 TO 891 895 TO 897 -
901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 TO 1024 1028 1029 -
1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 1147 1150 -
1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 1200 1201 -
1204 TO 1229 THICKNESS 1.2
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 2 ZD 1.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 TO 731 745 TO 747 -
750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 845 TO 891 895 TO 897 -
901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 TO 1024 1028 1029 -
1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 1147 1150 -
1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 1200 1201 -
1204 TO 1229 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
*******************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
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LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 TO 731 745 TO 747 -
750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 845 TO 891 895 TO 897 -
901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 TO 1024 1028 1029 -
1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 1147 1150 -
1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 1200 1201 -
1204 TO 1229 PR GY -36
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 TO 731 745 TO 747 -
750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 845 TO 891 895 TO 897 -
901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 TO 1024 1028 1029 -
1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 1147 1150 -
1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 1200 1201 -
1204 TO 1229 PR GY -40
JOINT LOAD
128 FX 11.9 FY -391.7 MZ -88.6
*******************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FX 13.5 FY -144.5 MZ -162
*******************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX 18.2 FY -325.7 MZ -120
*******************************************************************************
LOAD 4 LOADTYPE None  TITLE CRANE LOAD  ( CR1 )
JOINT LOAD
128 FX 14.7 FY -140.5 MZ -214.9
*******************************************************************************
LOAD 5 LOADTYPE None  TITLE CRANE LOAD  ( CR2 )
JOINT LOAD
128 FX 14.8 FY -38.6 MZ -254.2
*******************************************************************************
LOAD 6 LOADTYPE None  TITLE CRANE LOAD  ( CR3 )
JOINT LOAD
128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
*******************************************************************************
LOAD 7 LOADTYPE None  TITLE CRANE LOAD  ( CR4 )
JOINT LOAD
128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
*******************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FX 313.7 FY 142.4 MZ -3180.7
*******************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FX 188.5 FY 57.5 MZ -2560.2
*******************************************************************************
LOAD 10 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FX 344.4 FY 21.3 MZ -3618.7
*******************************************************************************
LOAD 11 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FX 216.7 FY -60.9 MZ -2951.2
*******************************************************************************
LOAD 12 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FX -352.7 FY 276.8 MZ 3895.1
*******************************************************************************
LOAD 13 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FX -480.5 FY 195.6 MZ 4591.3
*******************************************************************************
LOAD 14 LOADTYPE None  TITLE WIND LOAD  ( WL7 )
JOINT LOAD
128 FX -331.7 FY 129.5 MZ 3703
*******************************************************************************
LOAD 15 LOADTYPE None  TITLE WIND LOAD  ( WL8 )
JOINT LOAD
128 FX -459.4 FY 48.4 MZ 4399.4
*******************************************************************************
LOAD 16 LOADTYPE None  TITLE WIND LOAD  ( WL9 )
JOINT LOAD
128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
```

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128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
*********************************************************************************************
LOAD 17 LOADTYPE None  TITLE WIND LOAD  ( WL10 )
JOINT LOAD
128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
*********************************************************************************************
LOAD 18 LOADTYPE None  TITLE WIND LOAD  ( WL11 )
JOINT LOAD
128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
*********************************************************************************************
LOAD 19 LOADTYPE None  TITLE WIND LOAD  ( WL12 )
JOINT LOAD
128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
*********************************************************************************************
LOAD 20 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
*********************************************************************************************
LOAD 21 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
*********************************************************************************************
LOAD 22 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FX 1.1 FY 0 MZ -23.9
*********************************************************************************************
LOAD 23 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FX -1.1 FY 0 MZ 23.9
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L+TL1
1 1.0 3 1.0 22 1.0
LOAD COMB 203 DL+L+TL2
1 1.0 3 1.0 23 1.0
LOAD COMB 204 DL+LR
1 1.0 2 1.0
LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
1 1.0 2 0.75 3 0.75 22 0.75
LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
1 1.0 2 0.75 3 0.75 23 0.75
LOAD COMB 207 DL+0.625WL1
1 1.0 8 0.625
LOAD COMB 208 DL+0.625WL2
1 1.0 9 0.625
LOAD COMB 209 DL+0.625WL3
1 1.0 10 0.625
LOAD COMB 210 DL+0.625WL4
1 1.0 11 0.625
LOAD COMB 211 DL+0.625WL5
1 1.0 12 0.625
LOAD COMB 212 DL+0.625WL6
1 1.0 13 0.625
LOAD COMB 213 DL+0.625WL7
1 1.0 14 0.625
LOAD COMB 214 DL+0.625WL8
1 1.0 15 0.625
LOAD COMB 215 DL+0.625WL9
1 1.0 16 0.625
LOAD COMB 216 DL+0.625WL10
1 1.0 17 0.625
LOAD COMB 217 DL+0.625WL11
1 1.0 18 0.625
LOAD COMB 218 DL+0.625WL12
1 1.0 19 0.625
LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 8 0.75
LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 9 0.75
LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 10 0.75
LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 11 0.75
LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 12 0.75
LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 13 0.75
```



```
1 1.0 2 0.75 3 0.75 13 0.75
LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 14 0.75
LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 15 0.75
LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 16 0.75
LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 17 0.75
LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 18 0.75
LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 19 0.75
LOAD COMB 231 0.6DL+0.625WL1
1 0.6 8 0.625
LOAD COMB 232 0.6DL+0.625WL2
1 0.6 9 0.625
LOAD COMB 233 0.6DL+0.625WL3
1 0.6 10 0.625
LOAD COMB 234 0.6DL+0.625WL4
1 0.6 11 0.625
LOAD COMB 235 0.6DL+0.625WL5
1 0.6 12 0.625
LOAD COMB 236 0.6DL+0.625WL6
1 0.6 13 0.625
LOAD COMB 237 0.6DL+0.625WL7
1 0.6 14 0.625
LOAD COMB 238 0.6DL+0.625WL8
1 0.6 15 0.625
LOAD COMB 239 0.6DL+0.625WL9
1 0.6 16 0.625
LOAD COMB 240 0.6DL+0.625WL10
1 0.6 17 0.625
LOAD COMB 241 0.6DL+0.625WL11
1 0.6 18 0.625
LOAD COMB 242 0.6DL+0.625WL12
1 0.6 19 0.625
LOAD COMB 243 0.6DL+0.7SL1
1 0.6 20 0.7
LOAD COMB 244 0.6DL+0.7SL2
1 0.6 21 0.7
LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
```

```
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
LOAD COMB 275 DL+0.75WL7+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
LOAD COMB 287 DL+0.75WL7+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.4TL1+1.7L+0.5LR
1 1.4 22 1.4 3 1.7 2 0.5
LOAD COMB 103 1.4DL+1.4TL2+1.7L+0.5LR
1 1.4 23 1.4 3 1.7 2 0.5
LOAD COMB 104 1.2DL+1.0LR+1.0L
1 1.2 2 1.0 3 1.0
LOAD COMB 105 1.2DL+1.0WL1+1.0L+0.5LR
1 1.2 8 1.0 2 0.5 3 1.0
LOAD COMB 106 1.2DL+1.0WL2+0.5L+0.5LR
1 1.2 9 1.0 2 0.5 3 1.0
LOAD COMB 107 1.2DL+1.0WL3+0.5L+0.5LR
1 1.2 10 1.0 2 0.5 3 1.0
LOAD COMB 108 1.2DL+1.0WL4+0.5L+0.5LR
1 1.2 11 1.0 2 0.5 3 1.0
LOAD COMB 109 1.2DL+1.0WL5+0.5L+0.5LR
1 1.2 12 1.0 2 0.5 3 1.0
```



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1 1.2 12 1.0 2 0.5 3 1.0
LOAD COMB 110 1.2DL+1.0WL6+0.5L+0.5LR
1 1.2 13 1.0 2 0.5 3 1.0
LOAD COMB 111 1.2DL+1.0WL7+0.5L+0.5LR
1 1.2 14 1.0 2 0.5 3 1.0
LOAD COMB 112 1.2DL+1.0WL8+0.5L+0.5LR
1 1.2 15 1.0 2 0.5 3 1.0
LOAD COMB 113 1.2DL+1.0WL9+0.5L+0.5LR
1 1.2 16 1.0 2 0.5 3 1.0
LOAD COMB 114 1.2DL+1.0WL10+0.5L+0.5LR
1 1.2 17 1.0 2 0.5 3 1.0
LOAD COMB 115 1.2DL+1.0WL11+0.5L+0.5LR
1 1.2 18 1.0 2 0.5 3 1.0
LOAD COMB 116 1.2DL+1.0WL12+0.5L+0.5LR
1 1.2 19 1.0 2 0.5 3 1.0
LOAD COMB 117 1.2DL+1.0SL1+1.0L
1 1.2 20 1.0 3 1.0
LOAD COMB 118 1.2DL+1.0SL2+1.0L
1 1.2 21 1.0 3 1.0
LOAD COMB 119 0.9DL+1.0WL1
1 0.9 8 1.0
LOAD COMB 120 0.9DL+1.0WL2
1 0.9 9 1.0
LOAD COMB 121 0.9DL+1.0WL3
1 0.9 10 1.0
LOAD COMB 122 0.9DL+1.0WL4
1 0.9 11 1.0
LOAD COMB 123 0.9DL+1.0WL5
1 0.9 12 1.0
LOAD COMB 124 0.9DL+1.0WL6
1 0.9 13 1.0
LOAD COMB 125 0.9DL+1.0WL7
1 0.9 14 1.0
LOAD COMB 126 0.9DL+1.0WL8
1 0.9 15 1.0
LOAD COMB 127 0.9DL+1.0WL9
1 0.9 16 1.0
LOAD COMB 128 0.9DL+1.0WL10
1 0.9 17 1.0
LOAD COMB 129 0.9DL+1.0WL11
1 0.9 18 1.0
LOAD COMB 130 0.9DL+1.0WL12
1 0.9 19 1.0
LOAD COMB 131 0.9DL+1.0SL1
1 0.9 20 1.0
LOAD COMB 132 0.9DL+1.0SL2
1 0.9 21 1.0
LOAD COMB 133 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR1
1 1.2 8 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 134 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR1
1 1.2 9 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 135 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
1 1.2 10 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 136 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR1
1 1.2 11 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 137 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR1
1 1.2 12 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 138 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR1
1 1.2 13 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 139 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR1
1 1.2 14 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 140 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR1
1 1.2 15 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 141 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR1
1 1.2 16 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 142 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR1
1 1.2 17 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 143 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR1
1 1.2 18 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 144 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR1
1 1.2 19 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 145 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR2
1 1.2 8 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 146 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR2
1 1.2 9 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 147 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR2
1 1.2 10 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 148 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR2
1 1.2 11 1.0 2 0.5 3 1.0 5 1.0
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1 1.2 11 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 149 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR2
1 1.2 12 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 150 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR2
1 1.2 13 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 151 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR2
1 1.2 14 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 152 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR2
1 1.2 15 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 153 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR2
1 1.2 16 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 154 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR2
1 1.2 17 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 155 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR2
1 1.2 18 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 156 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR2
1 1.2 19 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 157 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR3
1 1.2 8 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 158 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR3
1 1.2 9 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 159 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR3
1 1.2 10 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 160 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR3
1 1.2 11 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 161 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR3
1 1.2 12 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 162 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR3
1 1.2 13 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 163 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR3
1 1.2 14 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 164 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR3
1 1.2 15 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 165 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR3
1 1.2 16 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 166 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR3
1 1.2 17 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 167 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR3
1 1.2 18 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 168 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR3
1 1.2 19 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 169 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR4
1 1.2 8 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 170 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR4
1 1.2 9 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 171 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR4
1 1.2 10 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 172 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR4
1 1.2 11 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 173 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR4
1 1.2 12 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 174 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
1 1.2 13 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 175 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR4
1 1.2 14 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 176 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
1 1.2 15 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 177 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR4
1 1.2 16 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 178 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR4
1 1.2 17 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 179 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR4
1 1.2 18 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 180 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR4
1 1.2 19 1.0 2 0.5 3 1.0 7 1.0
********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 180
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 -
730 TO 731 745 TO 747 750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 -
845 TO 891 895 TO 897 901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 -
1023 TO 1024 1028 1029 1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 -
1130 1146 1147 1150 1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 -
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1130 1146 1147 1150 1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 -
1194 1195 1200 1201 1204 1205
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 10/06/2020
JOB NAME FDN. MKD. F2_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 913.4 3.175;
MEMBER INCIDENCES
185 2 128;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 2 ZD 1.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FX 11.9 FY -391.7 MZ -88.6
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FX 13.5 FY -144.5 MZ -162
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX 18.2 FY -325.7 MZ -120
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE CRANE LOAD  ( CR1 )
JOINT LOAD
128 FX 14.7 FY -140.5 MZ -214.9
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE CRANE LOAD  ( CR2 )
JOINT LOAD
128 FX 14.8 FY -38.6 MZ -254.2
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE CRANE LOAD  ( CR3 )
JOINT LOAD
128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE CRANE LOAD  ( CR4 )
JOINT LOAD
128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FX 313.7 FY 142.4 MZ -3180.7
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FX 188.5 FY 57.5 MZ -2560.2
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FX 344.4 FY 21.3 MZ -3618.7
*********************************************************************************************
LOAD 11 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FX 216.7 FY -60.9 MZ -2951.2
*********************************************************************************************
LOAD 12 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
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JOINT LOAD
128 FX -352.7 FY 276.8 MZ 3895.1
*********************************************************************************************
LOAD 13 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FX -480.5 FY 195.6 MZ 4591.3
*********************************************************************************************
LOAD 14 LOADTYPE None  TITLE WIND LOAD  ( WL7 )
JOINT LOAD
128 FX -331.7 FY 129.5 MZ 3703
*********************************************************************************************
LOAD 15 LOADTYPE None  TITLE WIND LOAD  ( WL8 )
JOINT LOAD
128 FX -459.4 FY 48.4 MZ 4399.4
*********************************************************************************************
LOAD 16 LOADTYPE None  TITLE WIND LOAD  ( WL9 )
JOINT LOAD
128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
*********************************************************************************************
LOAD 17 LOADTYPE None  TITLE WIND LOAD  ( WL10 )
JOINT LOAD
128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
*********************************************************************************************
LOAD 18 LOADTYPE None  TITLE WIND LOAD  ( WL11 )
JOINT LOAD
128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
*********************************************************************************************
LOAD 19 LOADTYPE None  TITLE WIND LOAD  ( WL12 )
JOINT LOAD
128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
*********************************************************************************************
LOAD 20 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
*********************************************************************************************
LOAD 21 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
*********************************************************************************************
LOAD 22 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FX 1.1 FY 0 MZ -23.9
*********************************************************************************************
LOAD 23 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FX -1.1 FY 0 MZ 23.9
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L+TL1
1 1.0 3 1.0 22 1.0
LOAD COMB 203 DL+L+TL2
1 1.0 3 1.0 23 1.0
LOAD COMB 204 DL+LR
1 1.0 2 1.0
LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
1 1.0 2 0.75 3 0.75 22 0.75
LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
1 1.0 2 0.75 3 0.75 23 0.75
LOAD COMB 207 DL+0.625WL1
1 1.0 8 0.625
LOAD COMB 208 DL+0.625WL2
1 1.0 9 0.625
LOAD COMB 209 DL+0.625WL3
1 1.0 10 0.625
LOAD COMB 210 DL+0.625WL4
1 1.0 11 0.625
LOAD COMB 211 DL+0.625WL5
1 1.0 12 0.625
LOAD COMB 212 DL+0.625WL6
1 1.0 13 0.625
LOAD COMB 213 DL+0.625WL7
1 1.0 14 0.625
LOAD COMB 214 DL+0.625WL8
1 1.0 15 0.625
LOAD COMB 215 DL+0.625WL9
1 1.0 16 0.625
LOAD COMB 216 DL+0.625WL10
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LOAD COMB 216 DL+0.625WL10
1 1.0 17 0.625
LOAD COMB 217 DL+0.625WL11
1 1.0 18 0.625
LOAD COMB 218 DL+0.625WL12
1 1.0 19 0.625
LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 8 0.75
LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 9 0.75
LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 10 0.75
LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 11 0.75
LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 12 0.75
LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 13 0.75
LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 14 0.75
LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 15 0.75
LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 16 0.75
LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 17 0.75
LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 18 0.75
LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 19 0.75
LOAD COMB 231 0.6DL+0.625WL1
1 0.6 8 0.625
LOAD COMB 232 0.6DL+0.625WL2
1 0.6 9 0.625
LOAD COMB 233 0.6DL+0.625WL3
1 0.6 10 0.625
LOAD COMB 234 0.6DL+0.625WL4
1 0.6 11 0.625
LOAD COMB 235 0.6DL+0.625WL5
1 0.6 12 0.625
LOAD COMB 236 0.6DL+0.625WL6
1 0.6 13 0.625
LOAD COMB 237 0.6DL+0.625WL7
1 0.6 14 0.625
LOAD COMB 238 0.6DL+0.625WL8
1 0.6 15 0.625
LOAD COMB 239 0.6DL+0.625WL9
1 0.6 16 0.625
LOAD COMB 240 0.6DL+0.625WL10
1 0.6 17 0.625
LOAD COMB 241 0.6DL+0.625WL11
1 0.6 18 0.625
LOAD COMB 242 0.6DL+0.625WL12
1 0.6 19 0.625
LOAD COMB 243 0.6DL+0.7SL1
1 0.6 20 0.7
LOAD COMB 244 0.6DL+0.7SL2
1 0.6 21 0.7
LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
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LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
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LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
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LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
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LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
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LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
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LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
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LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
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LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
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LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
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LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
************************************************************************************************
PERFORM ANALYSIS
FINISH
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FINISH

ATTACHMENT 4 STAAD INPUT – FOUNDATION F3

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28/05/2020
JOB NAME FDN. MKD. F3
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
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35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175;
128 0 913.4 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438;
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1090 1113 1104 983 990; 1091 1114 1110 1104 1113;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091
_RAFT_MOM 338 341 350 351 360 361 363 TO 402 410 411 420 421 430 431 -
433 TO 476 478 TO 485 487 TO 528 530 TO 537 539 TO 794 809 811 813 TO 826 -
841 843 845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091
_RAFT_SHEAR 539 TO 588 629 TO 794 809 811 813 TO 826 841 843 845 TO 901 903 -
905 907 909 TO 965 967 969 971 TO 1091
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 THICKNESS 1
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 2 ZD 1.5
MEMBER PROPERTY AMERICAN
1092 PRIS YD 0.7 ZD 0.7
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1090 -
1091 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
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1091 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
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LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 PR GY -40
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 PR GY -40
JOINT LOAD
128 FX 11.9 FY -391.7 MZ -88.6
1115 FX 7.7 FY -79.57
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FX 13.5 FY -144.5 MZ -162
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX 18.2 FY -325.7 MZ -120
1115 FX 22.9 FY -198.75
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE CRANE LOAD  ( CR1 )
JOINT LOAD
128 FX 14.7 FY -140.5 MZ -214.9
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE CRANE LOAD  ( CR2 )
JOINT LOAD
128 FX 14.8 FY -38.6 MZ -254.2
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE CRANE LOAD  ( CR3 )
JOINT LOAD
128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE CRANE LOAD  ( CR4 )
JOINT LOAD
128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FX 313.7 FY 142.4 MZ -3180.7
1115 FX 67.7 FY 38.4
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FX 188.5 FY 57.5 MZ -2560.2
1115 FX 67.7 FY 38.4
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FX 344.4 FY 21.3 MZ -3618.7
1115 FX 67.7 FY 38.4
*********************************************************************************************
LOAD 11 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FX 216.7 FY -60.9 MZ -2951.2
1115 FX 67.7 FY 38.4
*********************************************************************************************
LOAD 12 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FX -352.7 FY 276.8 MZ 3895.1
1115 FX -67.7 FY -38.4
*********************************************************************************************
LOAD 13 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FX -480.5 FY 195.6 MZ 4591.3
1115 FX -67.7 FY -38.4
*********************************************************************************************
LOAD 14 LOADTYPE None  TITLE WIND LOAD  ( WL7 )
JOINT LOAD
128 FX -331.7 FY 129.5 MZ 3703
1115 FX -67.7 FY -38.4
*********************************************************************************************
LOAD 15 LOADTYPE None  TITLE WIND LOAD  ( WL8 )
JOINT LOAD
128 FX -459.4 FY 48.4 MZ 4399.4
1115 FX -67.7 FY -38.4
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1115 FX -67.7 FY -38.4
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LOAD 16 LOADTYPE None  TITLE WIND LOAD  ( WL9 )
JOINT LOAD
128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
1115 FX 6.4 FY 0.5 FZ -2
*********************************************************************************************
LOAD 17 LOADTYPE None  TITLE WIND LOAD  ( WL10 )
JOINT LOAD
128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
1115 FX 6.4 FY 0.5 FZ -2
*********************************************************************************************
LOAD 18 LOADTYPE None  TITLE WIND LOAD  ( WL11 )
JOINT LOAD
128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
1115 FX 6.4 FY -0.5 FZ 2
*********************************************************************************************
LOAD 19 LOADTYPE None  TITLE WIND LOAD  ( WL12 )
JOINT LOAD
128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
1115 FX 6.4 FY -0.5 FZ 2
*********************************************************************************************
LOAD 20 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
*********************************************************************************************
LOAD 21 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
*********************************************************************************************
LOAD 22 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FX 1.1 FY 0 MZ -23.9
*********************************************************************************************
LOAD 23 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FX -1.1 FY 0 MZ 23.9
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L+TL1
1 1.0 3 1.0 22 1.0
LOAD COMB 203 DL+L+TL2
1 1.0 3 1.0 23 1.0
LOAD COMB 204 DL+LR
1 1.0 2 1.0
LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
1 1.0 2 0.75 3 0.75 22 0.75
LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
1 1.0 2 0.75 3 0.75 23 0.75
LOAD COMB 207 DL+0.625WL1
1 1.0 8 0.625
LOAD COMB 208 DL+0.625WL2
1 1.0 9 0.625
LOAD COMB 209 DL+0.625WL3
1 1.0 10 0.625
LOAD COMB 210 DL+0.625WL4
1 1.0 11 0.625
LOAD COMB 211 DL+0.625WL5
1 1.0 12 0.625
LOAD COMB 212 DL+0.625WL6
1 1.0 13 0.625
LOAD COMB 213 DL+0.625WL7
1 1.0 14 0.625
LOAD COMB 214 DL+0.625WL8
1 1.0 15 0.625
LOAD COMB 215 DL+0.625WL9
1 1.0 16 0.625
LOAD COMB 216 DL+0.625WL10
1 1.0 17 0.625
LOAD COMB 217 DL+0.625WL11
1 1.0 18 0.625
LOAD COMB 218 DL+0.625WL12
1 1.0 19 0.625
LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 8 0.75
LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 9 0.75
```



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1 1.0 2 0.75 3 0.75 9 0.75
LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 10 0.75
LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 11 0.75
LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 12 0.75
LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 13 0.75
LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 14 0.75
LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 15 0.75
LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 16 0.75
LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 17 0.75
LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 18 0.75
LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 19 0.75
LOAD COMB 231 0.6DL+0.625WL1
1 0.6 8 0.625
LOAD COMB 232 0.6DL+0.625WL2
1 0.6 9 0.625
LOAD COMB 233 0.6DL+0.625WL3
1 0.6 10 0.625
LOAD COMB 234 0.6DL+0.625WL4
1 0.6 11 0.625
LOAD COMB 235 0.6DL+0.625WL5
1 0.6 12 0.625
LOAD COMB 236 0.6DL+0.625WL6
1 0.6 13 0.625
LOAD COMB 237 0.6DL+0.625WL7
1 0.6 14 0.625
LOAD COMB 238 0.6DL+0.625WL8
1 0.6 15 0.625
LOAD COMB 239 0.6DL+0.625WL9
1 0.6 16 0.625
LOAD COMB 240 0.6DL+0.625WL10
1 0.6 17 0.625
LOAD COMB 241 0.6DL+0.625WL11
1 0.6 18 0.625
LOAD COMB 242 0.6DL+0.625WL12
1 0.6 19 0.625
LOAD COMB 243 0.6DL+0.7SL1
1 0.6 20 0.7
LOAD COMB 244 0.6DL+0.7SL2
1 0.6 21 0.7
LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
```

```
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
LOAD COMB 275 DL+0.75WL7+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
LOAD COMB 287 DL+0.75WL7+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.4TL1+1.7L+0.5LR
1 1.4 22 1.4 3 1.7 2 0.5
LOAD COMB 103 1.4DL+1.4TL2+1.7L+0.5LR
1 1.4 23 1.4 3 1.7 2 0.5
LOAD COMB 104 1.2DL+1.0LR+1.0L
1 1.2 2 1.0 3 1.0
LOAD COMB 105 1.2DL+1.0WL1+1.0L+0.5LR
1 1.2 8 1.0 2 0.5 3 1.0
```



```
1 1.2 8 1.0 2 0.5 3 1.0
LOAD COMB 106 1.2DL+1.0WL2+0.5L+0.5LR
1 1.2 9 1.0 2 0.5 3 1.0
LOAD COMB 107 1.2DL+1.0WL3+0.5L+0.5LR
1 1.2 10 1.0 2 0.5 3 1.0
LOAD COMB 108 1.2DL+1.0WL4+0.5L+0.5LR
1 1.2 11 1.0 2 0.5 3 1.0
LOAD COMB 109 1.2DL+1.0WL5+0.5L+0.5LR
1 1.2 12 1.0 2 0.5 3 1.0
LOAD COMB 110 1.2DL+1.0WL6+0.5L+0.5LR
1 1.2 13 1.0 2 0.5 3 1.0
LOAD COMB 111 1.2DL+1.0WL7+0.5L+0.5LR
1 1.2 14 1.0 2 0.5 3 1.0
LOAD COMB 112 1.2DL+1.0WL8+0.5L+0.5LR
1 1.2 15 1.0 2 0.5 3 1.0
LOAD COMB 113 1.2DL+1.0WL9+0.5L+0.5LR
1 1.2 16 1.0 2 0.5 3 1.0
LOAD COMB 114 1.2DL+1.0WL10+0.5L+0.5LR
1 1.2 17 1.0 2 0.5 3 1.0
LOAD COMB 115 1.2DL+1.0WL11+0.5L+0.5LR
1 1.2 18 1.0 2 0.5 3 1.0
LOAD COMB 116 1.2DL+1.0WL12+0.5L+0.5LR
1 1.2 19 1.0 2 0.5 3 1.0
LOAD COMB 117 1.2DL+1.0SL1+1.0L
1 1.2 20 1.0 3 1.0
LOAD COMB 118 1.2DL+1.0SL2+1.0L
1 1.2 21 1.0 3 1.0
LOAD COMB 119 0.9DL+1.0WL1
1 0.9 8 1.0
LOAD COMB 120 0.9DL+1.0WL2
1 0.9 9 1.0
LOAD COMB 121 0.9DL+1.0WL3
1 0.9 10 1.0
LOAD COMB 122 0.9DL+1.0WL4
1 0.9 11 1.0
LOAD COMB 123 0.9DL+1.0WL5
1 0.9 12 1.0
LOAD COMB 124 0.9DL+1.0WL6
1 0.9 13 1.0
LOAD COMB 125 0.9DL+1.0WL7
1 0.9 14 1.0
LOAD COMB 126 0.9DL+1.0WL8
1 0.9 15 1.0
LOAD COMB 127 0.9DL+1.0WL9
1 0.9 16 1.0
LOAD COMB 128 0.9DL+1.0WL10
1 0.9 17 1.0
LOAD COMB 129 0.9DL+1.0WL11
1 0.9 18 1.0
LOAD COMB 130 0.9DL+1.0WL12
1 0.9 19 1.0
LOAD COMB 131 0.9DL+1.0SL1
1 0.9 20 1.0
LOAD COMB 132 0.9DL+1.0SL2
1 0.9 21 1.0
LOAD COMB 133 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR1
1 1.2 8 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 134 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR1
1 1.2 9 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 135 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
1 1.2 10 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 136 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR1
1 1.2 11 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 137 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR1
1 1.2 12 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 138 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR1
1 1.2 13 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 139 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR1
1 1.2 14 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 140 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR1
1 1.2 15 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 141 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR1
1 1.2 16 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 142 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR1
1 1.2 17 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 143 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR1
1 1.2 18 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 144 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR1
1 1.2 19 1.0 2 0.5 3 1.0 4 1.0
```

```
1 1.2 19 1.0 2 0.5 3 1.0 4 1.0
LOAD COMB 145 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR2
1 1.2 8 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 146 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR2
1 1.2 9 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 147 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR2
1 1.2 10 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 148 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR2
1 1.2 11 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 149 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR2
1 1.2 12 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 150 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR2
1 1.2 13 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 151 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR2
1 1.2 14 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 152 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR2
1 1.2 15 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 153 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR2
1 1.2 16 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 154 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR2
1 1.2 17 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 155 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR2
1 1.2 18 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 156 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR2
1 1.2 19 1.0 2 0.5 3 1.0 5 1.0
LOAD COMB 157 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR3
1 1.2 8 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 158 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR3
1 1.2 9 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 159 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR3
1 1.2 10 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 160 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR3
1 1.2 11 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 161 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR3
1 1.2 12 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 162 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR3
1 1.2 13 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 163 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR3
1 1.2 14 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 164 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR3
1 1.2 15 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 165 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR3
1 1.2 16 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 166 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR3
1 1.2 17 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 167 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR3
1 1.2 18 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 168 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR3
1 1.2 19 1.0 2 0.5 3 1.0 6 1.0
LOAD COMB 169 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR4
1 1.2 8 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 170 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR4
1 1.2 9 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 171 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR4
1 1.2 10 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 172 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR4
1 1.2 11 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 173 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR4
1 1.2 12 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 174 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
1 1.2 13 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 175 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR4
1 1.2 14 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 176 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
1 1.2 15 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 177 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR4
1 1.2 16 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 178 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR4
1 1.2 17 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 179 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR4
1 1.2 18 1.0 2 0.5 3 1.0 7 1.0
LOAD COMB 180 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR4
1 1.2 19 1.0 2 0.5 3 1.0 7 1.0
********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 180
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
```



```
TRACK 2 ALL
FC 35000 MEMB 185 1092
CLB 0.05 MEMB 185 1092
CLS 0.05 ALL
CLT 0.05 MEMB 185 1092
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185 1092
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28/05/2020
JOB NAME FDN F3_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 913.4 3.175; 469 1.8 909.9 3.175; 1115 1.8 913.6 3.175;
MEMBER INCIDENCES
185 2 128; 1092 469 1115;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 2 ZD 1.5
MEMBER PROPERTY AMERICAN
1092 PRIS YD 0.7 ZD 0.7
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 469 FIXED
********************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FX 11.9 FY -391.7 MZ -88.6
1115 FX 7.7 FY -79.57
********************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FX 13.5 FY -144.5 MZ -162
********************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX 18.2 FY -325.7 MZ -120
1115 FX 22.9 FY -198.75
********************************************************************************
LOAD 4 LOADTYPE None  TITLE CRANE LOAD  ( CR1 )
JOINT LOAD
128 FX 14.7 FY -140.5 MZ -214.9
********************************************************************************
LOAD 5 LOADTYPE None  TITLE CRANE LOAD  ( CR2 )
JOINT LOAD
128 FX 14.8 FY -38.6 MZ -254.2
********************************************************************************
LOAD 6 LOADTYPE None  TITLE CRANE LOAD  ( CR3 )
JOINT LOAD
128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
********************************************************************************
LOAD 7 LOADTYPE None  TITLE CRANE LOAD  ( CR4 )
JOINT LOAD
128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
********************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FX 313.7 FY 142.4 MZ -3180.7
1115 FX 67.7 FY 38.4
********************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FX 188.5 FY 57.5 MZ -2560.2
1115 FX 67.7 FY 38.4
********************************************************************************
LOAD 10 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FX 344.4 FY 21.3 MZ -3618.7
1115 FX 67.7 FY 38.4
```



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1115 FX 67.7 FY 38.4
********************************************************************************
LOAD 11 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FX 216.7 FY -60.9 MZ -2951.2
1115 FX 67.7 FY 38.4
********************************************************************************
LOAD 12 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FX -352.7 FY 276.8 MZ 3895.1
1115 FX -67.7 FY -38.4
********************************************************************************
LOAD 13 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FX -480.5 FY 195.6 MZ 4591.3
1115 FX -67.7 FY -38.4
********************************************************************************
LOAD 14 LOADTYPE None  TITLE WIND LOAD  ( WL7 )
JOINT LOAD
128 FX -331.7 FY 129.5 MZ 3703
1115 FX -67.7 FY -38.4
********************************************************************************
LOAD 15 LOADTYPE None  TITLE WIND LOAD  ( WL8 )
JOINT LOAD
128 FX -459.4 FY 48.4 MZ 4399.4
1115 FX -67.7 FY -38.4
********************************************************************************
LOAD 16 LOADTYPE None  TITLE WIND LOAD  ( WL9 )
JOINT LOAD
128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
1115 FX 6.4 FY 0.5 FZ -2
********************************************************************************
LOAD 17 LOADTYPE None  TITLE WIND LOAD  ( WL10 )
JOINT LOAD
128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
1115 FX 6.4 FY 0.5 FZ -2
********************************************************************************
LOAD 18 LOADTYPE None  TITLE WIND LOAD  ( WL11 )
JOINT LOAD
128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
1115 FX 6.4 FY -0.5 FZ 2
********************************************************************************
LOAD 19 LOADTYPE None  TITLE WIND LOAD  ( WL12 )
JOINT LOAD
128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
1115 FX 6.4 FY -0.5 FZ 2
********************************************************************************
LOAD 20 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
********************************************************************************
LOAD 21 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
********************************************************************************
LOAD 22 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FX 1.1 FY 0 MZ -23.9
********************************************************************************
LOAD 23 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FX -1.1 FY 0 MZ 23.9
********************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L+TL1
1 1.0 3 1.0 22 1.0
LOAD COMB 203 DL+L+TL2
1 1.0 3 1.0 23 1.0
LOAD COMB 204 DL+LR
1 1.0 2 1.0
LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
1 1.0 2 0.75 3 0.75 22 0.75
LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
1 1.0 2 0.75 3 0.75 23 0.75
LOAD COMB 207 DL+0.625WL1
1 1.0 8 0.625
LOAD COMB 208 DL+0.625WL2
```

```
LOAD COMB 208 DL+0.625WL2
1 1.0 9 0.625
LOAD COMB 209 DL+0.625WL3
1 1.0 10 0.625
LOAD COMB 210 DL+0.625WL4
1 1.0 11 0.625
LOAD COMB 211 DL+0.625WL5
1 1.0 12 0.625
LOAD COMB 212 DL+0.625WL6
1 1.0 13 0.625
LOAD COMB 213 DL+0.625WL7
1 1.0 14 0.625
LOAD COMB 214 DL+0.625WL8
1 1.0 15 0.625
LOAD COMB 215 DL+0.625WL9
1 1.0 16 0.625
LOAD COMB 216 DL+0.625WL10
1 1.0 17 0.625
LOAD COMB 217 DL+0.625WL11
1 1.0 18 0.625
LOAD COMB 218 DL+0.625WL12
1 1.0 19 0.625
LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 8 0.75
LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 9 0.75
LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 10 0.75
LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 11 0.75
LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 12 0.75
LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 13 0.75
LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 14 0.75
LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 15 0.75
LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 16 0.75
LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 17 0.75
LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 18 0.75
LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 19 0.75
LOAD COMB 231 0.6DL+0.625WL1
1 0.6 8 0.625
LOAD COMB 232 0.6DL+0.625WL2
1 0.6 9 0.625
LOAD COMB 233 0.6DL+0.625WL3
1 0.6 10 0.625
LOAD COMB 234 0.6DL+0.625WL4
1 0.6 11 0.625
LOAD COMB 235 0.6DL+0.625WL5
1 0.6 12 0.625
LOAD COMB 236 0.6DL+0.625WL6
1 0.6 13 0.625
LOAD COMB 237 0.6DL+0.625WL7
1 0.6 14 0.625
LOAD COMB 238 0.6DL+0.625WL8
1 0.6 15 0.625
LOAD COMB 239 0.6DL+0.625WL9
1 0.6 16 0.625
LOAD COMB 240 0.6DL+0.625WL10
1 0.6 17 0.625
LOAD COMB 241 0.6DL+0.625WL11
1 0.6 18 0.625
LOAD COMB 242 0.6DL+0.625WL12
1 0.6 19 0.625
LOAD COMB 243 0.6DL+0.7SL1
1 0.6 20 0.7
LOAD COMB 244 0.6DL+0.7SL2
1 0.6 21 0.7
LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
```



```
LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
LOAD COMB 275 DL+0.75WL7+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
```

```
LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
LOAD COMB 287 DL+0.75WL7+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 5 STAAD OUTPUT – FOUNDATION F1

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    JUL 13, 2020                   *
*           Time=    11: 7:28                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
*****************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F1 - SBC 304.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 28/05/2020
    4. JOB NAME FDN. MKD. F1
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 10 -0.75 909.9 3.175; 34 -0.5 909.9 3.175
   14. 35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175
   15. 128 0 913.4 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438
   16. 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438; 306 0.25 909.9 3.42438
   17. 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438; 309 1 909.9 3.175
   18. 311 1.25 909.9 3.175; 313 1 909.9 3.42438; 314 1.25 909.9 3.42438
   19. 315 -1.25 909.9 3.175; 316 -1 909.9 3.175; 319 -1.25 909.9 3.42438
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   23. 329 1.25 909.9 3.67438; 330 -1.25 909.9 3.67438; 331 -1 909.9 3.67438
   24. 332 -0.75 909.9 3.92438; 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438
   25. 335 0 909.9 3.92438; 336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438
   26. 338 0.75 909.9 3.92438; 339 1 909.9 3.92438; 340 1.25 909.9 3.92438
   27. 341 -1.25 909.9 3.92438; 342 -1 909.9 3.92438; 343 -0.75 909.9 4.17438
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   29. 347 0.25 909.9 4.17438; 348 0.5 909.9 4.17438; 349 0.75 909.9 4.17438
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   36. 368 -0.75 909.9 1.925; 369 -0.25 909.9 2.17438; 370 -0.25 909.9 1.925
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   42. 386 -1.25 909.9 1.925; 387 -0.75 909.9 2.42438; 388 -0.5 909.9 2.42438
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   48. 404 0.75 909.9 2.67438; 405 1 909.9 2.67438; 406 1.25 909.9 2.67438
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   53. 419 -1 909.9 2.92438; 420 1.5 909.9 3.42438; 421 1.5 909.9 3.175
   54. 422 1.5 909.9 3.67438; 423 1.5 909.9 3.92438; 424 1.5 909.9 4.17438
   55. 425 1.5 909.9 4.42438; 426 1.5 909.9 2.17438; 427 1.5 909.9 1.925
   56. 428 1.5 909.9 2.42438; 429 1.5 909.9 2.67438; 430 1.5 909.9 2.92438
   57. 431 -1.5 909.9 3.42438; 432 -1.5 909.9 3.175; 433 -1.5 909.9 3.67438
   58. 434 -1.5 909.9 3.92438; 435 -1.5 909.9 4.17438; 436 -1.5 909.9 4.42438
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   60. 440 -1.5 909.9 2.67438; 441 -1.5 909.9 2.92438; 442 -0.75 909.9 4.67438
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   62. 446 0.25 909.9 4.67438; 447 0.5 909.9 4.67438; 448 0.75 909.9 4.67438
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   73. 479 1.75 909.9 4.67438; 480 1.75 909.9 1.675; 481 -1.75 909.9 3.42438
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   92. 536 2 909.9 1.675; 537 2 909.9 1.425; 538 2 909.9 4.92438
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   94. 542 -2 909.9 3.92438; 543 -2 909.9 4.17438; 544 -2 909.9 4.42438
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108. 584 1.75 909.9 5.17438; 585 -1.75 909.9 5.17438; 586 2 909.9 5.17438
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236. 1009 -0.75 909.9 0.175; 1010 -0.25 909.9 0.175; 1011 0 909.9 0.175
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272. MEMBER INCIDENCES
273. 185 2 128
274. ELEMENT INCIDENCES SHELL
275. 326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2
276. 333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 338 313 314 311 309
277. 341 319 320 316 315; 342 320 303 10 316; 343 321 322 302 303
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279. 347 325 326 307 306; 348 326 327 308 307; 349 327 328 313 308
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515. ELEMENT
516. _RAFT 326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
517. 845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091
518. _RAFT_MOM 338 341 350 351 360 361 363 TO 402 410 411 420 421 430 431 -
519. 433 TO 794 809 811 813 TO 826 841 843 845 TO 901 903 905 907 909 TO 965 967 -
520. 969 971 TO 1091
521. _RAFT_SHEAR 529 TO 794 809 811 813 TO 826 841 843 845 TO 901 903 905 907 909 -
522. 910 TO 965 967 969 971 TO 1091
523. MEMBER
524. END GROUP DEFINITION
525. ELEMENT PROPERTY
526. 326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
527. 845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 THICKNESS 1
528. DEFINE MATERIAL START
529. ISOTROPIC CONCRETE
530. E 2.17185E+007
531. POISSON 0.17
532. DENSITY 25
533. ALPHA 1E-005
534. DAMP 0.05
535. TYPE CONCRETE
536. STRENGTH FCU 27579
537. END DEFINE MATERIAL
538. MEMBER PROPERTY AMERICAN
539. 185 PRIS YD 2 ZD 1.5
540. CONSTANTS
541. MATERIAL CONCRETE ALL
542. SUPPORTS
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543. 326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
544. 845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1090 -
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STAAD SPACE -- PAGE NO. 21

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STAAD SPACE -- PAGE NO. 22

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547. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
548. SELFWEIGHT Y -1
549. ELEMENT LOAD
550. * LOAD DUE TO BACKFILL
551. 326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
552. 845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 PR GY -40
553. * LOAD DUE TO SURCHARGE
554. 326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
555. 845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 PR GY -40
556. JOINT LOAD
557. 128 FX 11.9 FY -391.7 MZ -88.6
558. ******************************************************************************
559. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
560. JOINT LOAD
561. 128 FX 13.5 FY -144.5 MZ -162
562. ******************************************************************************
563. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
564. JOINT LOAD
565. 128 FX 18.2 FY -325.7 MZ -120
566. ******************************************************************************
567. LOAD 4 LOADTYPE NONE  TITLE CRANE LOAD  ( CR1 )
568. JOINT LOAD
569. 128 FX 14.7 FY -140.5 MZ -214.9
570. ******************************************************************************
571. LOAD 5 LOADTYPE NONE  TITLE CRANE LOAD  ( CR2 )
572. JOINT LOAD
573. 128 FX 14.8 FY -38.6 MZ -254.2
574. ******************************************************************************
575. LOAD 6 LOADTYPE NONE  TITLE CRANE LOAD  ( CR3 )
576. JOINT LOAD
577. 128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
578. ******************************************************************************
579. LOAD 7 LOADTYPE NONE  TITLE CRANE LOAD  ( CR4 )
580. JOINT LOAD
581. 128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
582. ******************************************************************************
583. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
584. JOINT LOAD
585. 128 FX 313.7 FY 142.4 MZ -3180.7
586. ******************************************************************************
587. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
588. JOINT LOAD
589. 128 FX 188.5 FY 57.5 MZ -2560.2
590. ******************************************************************************
591. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
592. JOINT LOAD
593. 128 FX 344.4 FY 21.3 MZ -3618.7
594. ******************************************************************************
595. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
596. JOINT LOAD
597. 128 FX 216.7 FY -60.9 MZ -2951.2
598. ******************************************************************************
599. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
600. JOINT LOAD
601. 128 FX -352.7 FY 276.8 MZ 3895.1
602. ******************************************************************************
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603. LOAD 13 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
604. JOINT LOAD
605. 128 FX -480.5 FY 195.6 MZ 4591.3
606. ******************************************************************************
607. LOAD 14 LOADTYPE NONE  TITLE WIND LOAD  ( WL7 )
608. JOINT LOAD
609. 128 FX -331.7 FY 129.5 MZ 3703
610. ******************************************************************************
611. LOAD 15 LOADTYPE NONE  TITLE WIND LOAD  ( WL8 )
612. JOINT LOAD
613. 128 FX -459.4 FY 48.4 MZ 4399.4
614. ******************************************************************************
615. LOAD 16 LOADTYPE NONE  TITLE WIND LOAD  ( WL9 )
616. JOINT LOAD
617. 128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
618. ******************************************************************************
619. LOAD 17 LOADTYPE NONE  TITLE WIND LOAD  ( WL10 )
620. JOINT LOAD
621. 128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
622. ******************************************************************************
623. LOAD 18 LOADTYPE NONE  TITLE WIND LOAD  ( WL11 )
624. JOINT LOAD
625. 128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
626. ******************************************************************************
627. LOAD 19 LOADTYPE NONE  TITLE WIND LOAD  ( WL12 )
628. JOINT LOAD
629. 128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
630. ******************************************************************************
631. LOAD 20 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
632. JOINT LOAD
633. 128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
634. ******************************************************************************
635. LOAD 21 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
636. JOINT LOAD
637. 128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
638. ******************************************************************************
639. LOAD 22 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
640. JOINT LOAD
641. 128 FX 1.1 FY 0 MZ -23.9
642. ******************************************************************************
643. LOAD 23 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
644. JOINT LOAD
645. 128 FX -1.1 FY 0 MZ 23.9
646. ******************************************************************************
647. * UNFACTORED LOAD COMBINATIONS
648. LOAD COMB 201 DL
649. 1 1.0
650. LOAD COMB 202 DL+L+TL1
651. 1 1.0 3 1.0 22 1.0
652. LOAD COMB 203 DL+L+TL2
653. 1 1.0 3 1.0 23 1.0
654. LOAD COMB 204 DL+LR
655. 1 1.0 2 1.0
656. LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
657. 1 1.0 2 0.75 3 0.75 22 0.75
658. LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
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659. 1 1.0 2 0.75 3 0.75 23 0.75
660. LOAD COMB 207 DL+0.625WL1
661. 1 1.0 8 0.625
662. LOAD COMB 208 DL+0.625WL2
663. 1 1.0 9 0.625
664. LOAD COMB 209 DL+0.625WL3
665. 1 1.0 10 0.625
666. LOAD COMB 210 DL+0.625WL4
667. 1 1.0 11 0.625
668. LOAD COMB 211 DL+0.625WL5
669. 1 1.0 12 0.625
670. LOAD COMB 212 DL+0.625WL6
671. 1 1.0 13 0.625
672. LOAD COMB 213 DL+0.625WL7
673. 1 1.0 14 0.625
674. LOAD COMB 214 DL+0.625WL8
675. 1 1.0 15 0.625
676. LOAD COMB 215 DL+0.625WL9
677. 1 1.0 16 0.625
678. LOAD COMB 216 DL+0.625WL10
679. 1 1.0 17 0.625
680. LOAD COMB 217 DL+0.625WL11
681. 1 1.0 18 0.625
682. LOAD COMB 218 DL+0.625WL12
683. 1 1.0 19 0.625
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685. 1 1.0 2 0.75 3 0.75 8 0.75
686. LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
687. 1 1.0 2 0.75 3 0.75 9 0.75
688. LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
689. 1 1.0 2 0.75 3 0.75 10 0.75
690. LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
691. 1 1.0 2 0.75 3 0.75 11 0.75
692. LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
693. 1 1.0 2 0.75 3 0.75 12 0.75
694. LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
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696. LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
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698. LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
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700. LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
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702. LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
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704. LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
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706. LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
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714. LOAD COMB 234 0.6DL+0.625WL4

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730. LOAD COMB 242 0.6DL+0.625WL12
731. 1 0.6 19 0.625
732. LOAD COMB 243 0.6DL+0.7SL1
733. 1 0.6 20 0.7
734. LOAD COMB 244 0.6DL+0.7SL2
735. 1 0.6 21 0.7
736. LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
737. 1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
738. LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
739. 1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
740. LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
741. 1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
742. LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
743. 1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
744. LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
745. 1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
746. LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
747. 1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
748. LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
749. 1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
750. LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
751. 1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
752. LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
753. 1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
754. LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
755. 1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
756. LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
757. 1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
758. LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
759. 1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
760. LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
761. 1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
762. LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
763. 1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
764. LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
765. 1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
766. LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
767. 1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
768. LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
769. 1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
770. LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2

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771. 1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
772. LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
773. 1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
774. LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
775. 1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
776. LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
777. 1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
778. LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
779. 1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
780. LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
781. 1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
782. LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
783. 1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
784. LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
785. 1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
786. LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
787. 1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
788. LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
789. 1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
790. LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
791. 1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
792. LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
793. 1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
794. LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
795. 1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
796. LOAD COMB 275 DL+0.75WL7+0.75L+0.75LR+CR3
797. 1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
798. LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
799. 1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
800. LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
801. 1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
802. LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
803. 1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
804. LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
805. 1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
806. LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
807. 1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
808. LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
809. 1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
810. LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
811. 1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
812. LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
813. 1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
814. LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
815. 1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
816. LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
817. 1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
818. LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
819. 1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
820. LOAD COMB 287 DL+0.75WL7+0.75L+0.75LR+CR4
821. 1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
822. LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
823. 1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
824. LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
825. 1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
826. LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
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827. 1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
828. LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
829. 1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
830. LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
831. 1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
832. ********************************************************************************
833. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
834. LOAD COMB 101 1.4DL
835. 1 1.4
836. LOAD COMB 102 1.4DL+1.4TL1+1.7L+0.5LR
837. 1 1.4 22 1.4 3 1.7 2 0.5
838. LOAD COMB 103 1.4DL+1.4TL2+1.7L+0.5LR
839. 1 1.4 23 1.4 3 1.7 2 0.5
840. LOAD COMB 104 1.2DL+1.0LR+1.0L
841. 1 1.2 2 1.0 3 1.0
842. LOAD COMB 105 1.2DL+1.0WL1+1.0L+0.5LR
843. 1 1.2 8 1.0 2 0.5 3 1.0
844. LOAD COMB 106 1.2DL+1.0WL2+0.5L+0.5LR
845. 1 1.2 9 1.0 2 0.5 3 1.0
846. LOAD COMB 107 1.2DL+1.0WL3+0.5L+0.5LR
847. 1 1.2 10 1.0 2 0.5 3 1.0
848. LOAD COMB 108 1.2DL+1.0WL4+0.5L+0.5LR
849. 1 1.2 11 1.0 2 0.5 3 1.0
850. LOAD COMB 109 1.2DL+1.0WL5+0.5L+0.5LR
851. 1 1.2 12 1.0 2 0.5 3 1.0
852. LOAD COMB 110 1.2DL+1.0WL6+0.5L+0.5LR
853. 1 1.2 13 1.0 2 0.5 3 1.0
854. LOAD COMB 111 1.2DL+1.0WL7+0.5L+0.5LR
855. 1 1.2 14 1.0 2 0.5 3 1.0
856. LOAD COMB 112 1.2DL+1.0WL8+0.5L+0.5LR
857. 1 1.2 15 1.0 2 0.5 3 1.0
858. LOAD COMB 113 1.2DL+1.0WL9+0.5L+0.5LR
859. 1 1.2 16 1.0 2 0.5 3 1.0
860. LOAD COMB 114 1.2DL+1.0WL10+0.5L+0.5LR
861. 1 1.2 17 1.0 2 0.5 3 1.0
862. LOAD COMB 115 1.2DL+1.0WL11+0.5L+0.5LR
863. 1 1.2 18 1.0 2 0.5 3 1.0
864. LOAD COMB 116 1.2DL+1.0WL12+0.5L+0.5LR
865. 1 1.2 19 1.0 2 0.5 3 1.0
866. LOAD COMB 117 1.2DL+1.0SL1+1.0L
867. 1 1.2 20 1.0 3 1.0
868. LOAD COMB 118 1.2DL+1.0SL2+1.0L
869. 1 1.2 21 1.0 3 1.0
870. LOAD COMB 119 0.9DL+1.0WL1
871. 1 0.9 8 1.0
872. LOAD COMB 120 0.9DL+1.0WL2
873. 1 0.9 9 1.0
874. LOAD COMB 121 0.9DL+1.0WL3
875. 1 0.9 10 1.0
876. LOAD COMB 122 0.9DL+1.0WL4
877. 1 0.9 11 1.0
878. LOAD COMB 123 0.9DL+1.0WL5
879. 1 0.9 12 1.0
880. LOAD COMB 124 0.9DL+1.0WL6
881. 1 0.9 13 1.0
882. LOAD COMB 125 0.9DL+1.0WL7
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883. 1 0.9 14 1.0
884. LOAD COMB 126 0.9DL+1.0WL8
885. 1 0.9 15 1.0
886. LOAD COMB 127 0.9DL+1.0WL9
887. 1 0.9 16 1.0
888. LOAD COMB 128 0.9DL+1.0WL10
889. 1 0.9 17 1.0
890. LOAD COMB 129 0.9DL+1.0WL11
891. 1 0.9 18 1.0
892. LOAD COMB 130 0.9DL+1.0WL12
893. 1 0.9 19 1.0
894. LOAD COMB 131 0.9DL+1.0SL1
895. 1 0.9 20 1.0
896. LOAD COMB 132 0.9DL+1.0SL2
897. 1 0.9 21 1.0
898. LOAD COMB 133 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR1
899. 1 1.2 8 1.0 2 0.5 3 1.0 4 1.0
900. LOAD COMB 134 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR1
901. 1 1.2 9 1.0 2 0.5 3 1.0 4 1.0
902. LOAD COMB 135 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
903. 1 1.2 10 1.0 2 0.5 3 1.0 4 1.0
904. LOAD COMB 136 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR1
905. 1 1.2 11 1.0 2 0.5 3 1.0 4 1.0
906. LOAD COMB 137 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR1
907. 1 1.2 12 1.0 2 0.5 3 1.0 4 1.0
908. LOAD COMB 138 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR1
909. 1 1.2 13 1.0 2 0.5 3 1.0 4 1.0
910. LOAD COMB 139 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR1
911. 1 1.2 14 1.0 2 0.5 3 1.0 4 1.0
912. LOAD COMB 140 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR1
913. 1 1.2 15 1.0 2 0.5 3 1.0 4 1.0
914. LOAD COMB 141 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR1
915. 1 1.2 16 1.0 2 0.5 3 1.0 4 1.0
916. LOAD COMB 142 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR1
917. 1 1.2 17 1.0 2 0.5 3 1.0 4 1.0
918. LOAD COMB 143 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR1
919. 1 1.2 18 1.0 2 0.5 3 1.0 4 1.0
920. LOAD COMB 144 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR1
921. 1 1.2 19 1.0 2 0.5 3 1.0 4 1.0
922. LOAD COMB 145 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR2
923. 1 1.2 8 1.0 2 0.5 3 1.0 5 1.0
924. LOAD COMB 146 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR2
925. 1 1.2 9 1.0 2 0.5 3 1.0 5 1.0
926. LOAD COMB 147 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR2
927. 1 1.2 10 1.0 2 0.5 3 1.0 5 1.0
928. LOAD COMB 148 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR2
929. 1 1.2 11 1.0 2 0.5 3 1.0 5 1.0
930. LOAD COMB 149 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR2
931. 1 1.2 12 1.0 2 0.5 3 1.0 5 1.0
932. LOAD COMB 150 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR2
933. 1 1.2 13 1.0 2 0.5 3 1.0 5 1.0
934. LOAD COMB 151 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR2
935. 1 1.2 14 1.0 2 0.5 3 1.0 5 1.0
936. LOAD COMB 152 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR2
937. 1 1.2 15 1.0 2 0.5 3 1.0 5 1.0
938. LOAD COMB 153 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR2
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939. 1 1.2 16 1.0 2 0.5 3 1.0 5 1.0
940. LOAD COMB 154 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR2
941. 1 1.2 17 1.0 2 0.5 3 1.0 5 1.0
942. LOAD COMB 155 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR2
943. 1 1.2 18 1.0 2 0.5 3 1.0 5 1.0
944. LOAD COMB 156 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR2
945. 1 1.2 19 1.0 2 0.5 3 1.0 5 1.0
946. LOAD COMB 157 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR3
947. 1 1.2 8 1.0 2 0.5 3 1.0 6 1.0
948. LOAD COMB 158 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR3
949. 1 1.2 9 1.0 2 0.5 3 1.0 6 1.0
950. LOAD COMB 159 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR3
951. 1 1.2 10 1.0 2 0.5 3 1.0 6 1.0
952. LOAD COMB 160 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR3
953. 1 1.2 11 1.0 2 0.5 3 1.0 6 1.0
954. LOAD COMB 161 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR3
955. 1 1.2 12 1.0 2 0.5 3 1.0 6 1.0
956. LOAD COMB 162 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR3
957. 1 1.2 13 1.0 2 0.5 3 1.0 6 1.0
958. LOAD COMB 163 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR3
959. 1 1.2 14 1.0 2 0.5 3 1.0 6 1.0
960. LOAD COMB 164 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR3
961. 1 1.2 15 1.0 2 0.5 3 1.0 6 1.0
962. LOAD COMB 165 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR3
963. 1 1.2 16 1.0 2 0.5 3 1.0 6 1.0
964. LOAD COMB 166 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR3
965. 1 1.2 17 1.0 2 0.5 3 1.0 6 1.0
966. LOAD COMB 167 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR3
967. 1 1.2 18 1.0 2 0.5 3 1.0 6 1.0
968. LOAD COMB 168 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR3
969. 1 1.2 19 1.0 2 0.5 3 1.0 6 1.0
970. LOAD COMB 169 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR4
971. 1 1.2 8 1.0 2 0.5 3 1.0 7 1.0
972. LOAD COMB 170 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR4
973. 1 1.2 9 1.0 2 0.5 3 1.0 7 1.0
974. LOAD COMB 171 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR4
975. 1 1.2 10 1.0 2 0.5 3 1.0 7 1.0
976. LOAD COMB 172 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR4
977. 1 1.2 11 1.0 2 0.5 3 1.0 7 1.0
978. LOAD COMB 173 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR4
979. 1 1.2 12 1.0 2 0.5 3 1.0 7 1.0
980. LOAD COMB 174 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
981. 1 1.2 13 1.0 2 0.5 3 1.0 7 1.0
982. LOAD COMB 175 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR4
983. 1 1.2 14 1.0 2 0.5 3 1.0 7 1.0
984. LOAD COMB 176 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
985. 1 1.2 15 1.0 2 0.5 3 1.0 7 1.0
986. LOAD COMB 177 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR4
987. 1 1.2 16 1.0 2 0.5 3 1.0 7 1.0
988. LOAD COMB 178 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR4
989. 1 1.2 17 1.0 2 0.5 3 1.0 7 1.0
990. LOAD COMB 179 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR4
991. 1 1.2 18 1.0 2 0.5 3 1.0 7 1.0
992. LOAD COMB 180 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR4
993. 1 1.2 19 1.0 2 0.5 3 1.0 7 1.0
994. ********************************************************************************
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995. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 776 NUMBER OF MEMBERS 1
NUMBER OF PLATES 720 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 775

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 347/ 43/ 135 DOF
TOTAL PRIMARY LOAD CASES = 23, TOTAL DEGREES OF FREEDOM = 2331
TOTAL LOAD COMBINATION CASES = 172 SO FAR.
SIZE OF STIFFNESS MATRIX = 315 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 19.8/ 145135.2 MB

996. LOAD LIST 101 TO 180
997. START CONCRETE DESIGN

998. CODE ACI
999. TRACK 2 ALL
1000. FC 35000 MEMB 185
1001. CLB 0.05 MEMB 185
1002. CLS 0.05 ALL
1003. CLT 0.05 ALL
1004. MAXMAIN 25 ALL
1005. MINMAIN 25 ALL
1006. MINSEC 10 ALL
1007. DESIGN COLUMN 185

==========

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE -1500.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 30000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
64 - 25 MM	1.047	101	END	0.650

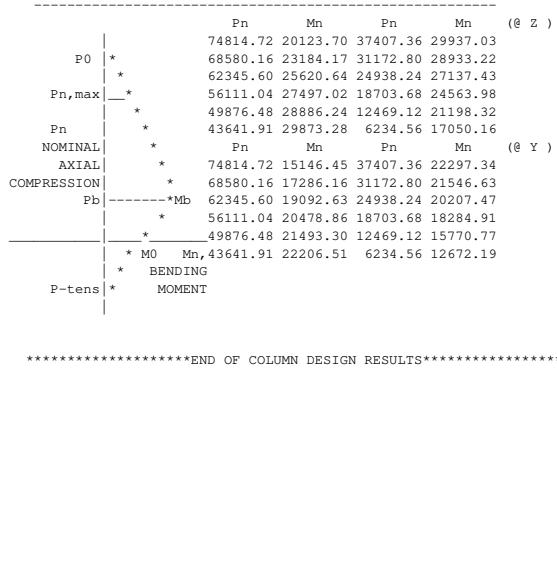
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
********	81049.27	41239.41	30170.30	731.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
12092.92	********	843.66	190.83	0.06463

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
********	81049.27	40737.83	22458.28	551.3
M0	P-tens.	Des.Pn	Des.Mn	e/h
8945.85	********	843.66	0.00	0.00000



**********END OF COLUMN DESIGN RESULTS**********

1008. END CONCRETE DESIGN
1009. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 13,2020 TIME= 11: 8:32 ****

For technical assistance on STAAD.Pro, please visit
http://selectservices.bentley.com/en-US/

Details about additional assistance from
Bentley and Partners can be found at program menu
Help->Technical Support

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http://www.bentley.com

****
* *
* STAAD.Pro V8i SELECTseries5 *
* Version 20.07.10.64 *
* Proprietary Program of *
* Bentley Systems, Inc. *
* Date= JUL 13, 2020 *
* Time= 12: 6:22 *
* *
* USER ID: LNT ECC DIV *
****

```
    1. STAAD SPACE
INPUT FILE: F1 - STABILITY_304.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 28/05/2020
    4. JOB NAME FDN. MKD. F1_STABILITY CHECK
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 128 0 913.4 3.175
   14. MEMBER INCIDENCES
   15. 185 2 128
   16. DEFINE MATERIAL START
   17. ISOTROPIC CONCRETE
   18. E 2.17185E+007
   19. POISSON 0.17
   20. DENSITY 25
   21. ALPHA 1E-005
   22. DAMP 0.05
   23. TYPE CONCRETE
   24. STRENGTH FCU 27579
   25. END DEFINE MATERIAL
   26. MEMBER PROPERTY AMERICAN
   27. 185 PRIS YD 2 ZD 1.5
   28. CONSTANTS
   29. MATERIAL CONCRETE ALL
   30. SUPPORTS
   31. 2 FIXED
   32. ******************************************************************************
   33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   34. SELFWEIGHT Y -1
   35. JOINT LOAD
   36. 128 FX 11.9 FY -391.7 MZ -88.6
   37. ******************************************************************************
   38. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
```



```
   39. JOINT LOAD
   40. 128 FX 13.5 FY -144.5 MZ -162
   41. ******************************************************************************
   42. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
   43. JOINT LOAD
   44. 128 FX 18.2 FY -325.7 MZ -120
   45. ******************************************************************************
   46. LOAD 4 LOADTYPE NONE  TITLE CRANE LOAD  ( CR1 )
   47. JOINT LOAD
   48. 128 FX 14.7 FY -140.5 MZ -214.9
   49. ******************************************************************************
   50. LOAD 5 LOADTYPE NONE  TITLE CRANE LOAD  ( CR2 )
   51. JOINT LOAD
   52. 128 FX 14.8 FY -38.6 MZ -254.2
   53. ******************************************************************************
   54. LOAD 6 LOADTYPE NONE  TITLE CRANE LOAD  ( CR3 )
   55. JOINT LOAD
   56. 128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
   57. ******************************************************************************
   58. LOAD 7 LOADTYPE NONE  TITLE CRANE LOAD  ( CR4 )
   59. JOINT LOAD
   60. 128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
   61. ******************************************************************************
   62. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
   63. JOINT LOAD
   64. 128 FX 313.7 FY 142.4 MZ -3180.7
   65. ******************************************************************************
   66. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
   67. JOINT LOAD
   68. 128 FX 188.5 FY 57.5 MZ -2560.2
   69. ******************************************************************************
   70. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
   71. JOINT LOAD
   72. 128 FX 344.4 FY 21.3 MZ -3618.7
   73. ******************************************************************************
   74. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
   75. JOINT LOAD
   76. 128 FX 216.7 FY -60.9 MZ -2951.2
   77. ******************************************************************************
   78. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
   79. JOINT LOAD
   80. 128 FX -352.7 FY 276.8 MZ 3895.1
   81. ******************************************************************************
   82. LOAD 13 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
   83. JOINT LOAD
   84. 128 FX -480.5 FY 195.6 MZ 4591.3
   85. ******************************************************************************
   86. LOAD 14 LOADTYPE NONE  TITLE WIND LOAD  ( WL7 )
   87. JOINT LOAD
   88. 128 FX -331.7 FY 129.5 MZ 3703
   89. ******************************************************************************
   90. LOAD 15 LOADTYPE NONE  TITLE WIND LOAD  ( WL8 )
   91. JOINT LOAD
   92. 128 FX -459.4 FY 48.4 MZ 4399.4
   93. ******************************************************************************
   94. LOAD 16 LOADTYPE NONE  TITLE WIND LOAD  ( WL9 )
```

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 95. JOINT LOAD
 96. 128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
 97. ********************************************************************************
 98. LOAD 17 LOADTYPE NONE  TITLE WIND LOAD  ( WL10 )
 99. JOINT LOAD
100. 128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
101. ********************************************************************************
102. LOAD 18 LOADTYPE NONE  TITLE WIND LOAD  ( WL11 )
103. JOINT LOAD
104. 128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
105. ********************************************************************************
106. LOAD 19 LOADTYPE NONE  TITLE WIND LOAD  ( WL12 )
107. JOINT LOAD
108. 128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
109. ********************************************************************************
110. LOAD 20 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
111. JOINT LOAD
112. 128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
113. ********************************************************************************
114. LOAD 21 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
115. JOINT LOAD
116. 128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
117. ********************************************************************************
118. LOAD 22 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
119. JOINT LOAD
120. 128 FX 1.1 FY 0 MZ -23.9
121. ********************************************************************************
122. LOAD 23 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
123. JOINT LOAD
124. 128 FX -1.1 FY 0 MZ 23.9
125. ********************************************************************************
126. * UNFACTORED LOAD COMBINATIONS
127. LOAD COMB 201 DL
128. 1 1.0
129. LOAD COMB 202 DL+L+TL1
130. 1 1.0 3 1.0 22 1.0
131. LOAD COMB 203 DL+L+TL2
132. 1 1.0 3 1.0 23 1.0
133. LOAD COMB 204 DL+LR
134. 1 1.0 2 1.0
135. LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
136. 1 1.0 2 0.75 3 0.75 22 0.75
137. LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
138. 1 1.0 2 0.75 3 0.75 23 0.75
139. LOAD COMB 207 DL+0.625WL1
140. 1 1.0 8 0.625
141. LOAD COMB 208 DL+0.625WL2
142. 1 1.0 9 0.625
143. LOAD COMB 209 DL+0.625WL3
144. 1 1.0 10 0.625
145. LOAD COMB 210 DL+0.625WL4
146. 1 1.0 11 0.625
147. LOAD COMB 211 DL+0.625WL5
148. 1 1.0 12 0.625
149. LOAD COMB 212 DL+0.625WL6
150. 1 1.0 13 0.625
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151. LOAD COMB 213 DL+0.625WL7
152. 1 1.0 14 0.625
153. LOAD COMB 214 DL+0.625WL8
154. 1 1.0 15 0.625
155. LOAD COMB 215 DL+0.625WL9
156. 1 1.0 16 0.625
157. LOAD COMB 216 DL+0.625WL10
158. 1 1.0 17 0.625
159. LOAD COMB 217 DL+0.625WL11
160. 1 1.0 18 0.625
161. LOAD COMB 218 DL+0.625WL12
162. 1 1.0 19 0.625
163. LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
164. 1 1.0 2 0.75 3 0.75 8 0.75
165. LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
166. 1 1.0 2 0.75 3 0.75 9 0.75
167. LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
168. 1 1.0 2 0.75 3 0.75 10 0.75
169. LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
170. 1 1.0 2 0.75 3 0.75 11 0.75
171. LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
172. 1 1.0 2 0.75 3 0.75 12 0.75
173. LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
174. 1 1.0 2 0.75 3 0.75 13 0.75
175. LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
176. 1 1.0 2 0.75 3 0.75 14 0.75
177. LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
178. 1 1.0 2 0.75 3 0.75 15 0.75
179. LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
180. 1 1.0 2 0.75 3 0.75 16 0.75
181. LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
182. 1 1.0 2 0.75 3 0.75 17 0.75
183. LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
184. 1 1.0 2 0.75 3 0.75 18 0.75
185. LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
186. 1 1.0 2 0.75 3 0.75 19 0.75
187. LOAD COMB 231 0.6DL+0.625WL1
188. 1 0.6 8 0.625
189. LOAD COMB 232 0.6DL+0.625WL2
190. 1 0.6 9 0.625
191. LOAD COMB 233 0.6DL+0.625WL3
192. 1 0.6 10 0.625
193. LOAD COMB 234 0.6DL+0.625WL4
194. 1 0.6 11 0.625
195. LOAD COMB 235 0.6DL+0.625WL5
196. 1 0.6 12 0.625
197. LOAD COMB 236 0.6DL+0.625WL6
198. 1 0.6 13 0.625
199. LOAD COMB 237 0.6DL+0.625WL7
200. 1 0.6 14 0.625
201. LOAD COMB 238 0.6DL+0.625WL8
202. 1 0.6 15 0.625
203. LOAD COMB 239 0.6DL+0.625WL9
204. 1 0.6 16 0.625
205. LOAD COMB 240 0.6DL+0.625WL10
206. 1 0.6 17 0.625
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207. LOAD COMB 241 0.6DL+0.625WL11
208. 1 0.6 18 0.625
209. LOAD COMB 242 0.6DL+0.625WL12
210. 1 0.6 19 0.625
211. LOAD COMB 243 0.6DL+0.7SL1
212. 1 0.6 20 0.7
213. LOAD COMB 244 0.6DL+0.7SL2
214. 1 0.6 21 0.7
215. LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
216. 1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
217. LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
218. 1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
219. LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
220. 1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
221. LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
222. 1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
223. LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
224. 1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
225. LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
226. 1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
227. LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
228. 1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
229. LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
230. 1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
231. LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
232. 1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
233. LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
234. 1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
235. LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
236. 1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
237. LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
238. 1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
239. LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
240. 1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
241. LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
242. 1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
243. LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
244. 1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
245. LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
246. 1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
247. LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
248. 1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
249. LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
250. 1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
251. LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
252. 1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
253. LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
254. 1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
255. LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
256. 1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
257. LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
258. 1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
259. LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
260. 1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
261. LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
262. 1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
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263. LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
264. 1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
265. LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
266. 1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
267. LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
268. 1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
269. LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
270. 1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
271. LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
272. 1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
273. LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
274. 1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
275. LOAD COMB 275 DL+0.75WL7+0.75L+0.75LR+CR3
276. 1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
277. LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
278. 1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
279. LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
280. 1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
281. LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
282. 1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
283. LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
284. 1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
285. LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
286. 1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
287. LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
288. 1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
289. LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
290. 1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
291. LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
292. 1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
293. LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
294. 1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
295. LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
296. 1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
297. LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
298. 1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
299. LOAD COMB 287 DL+0.75WL7+0.75L+0.75LR+CR4
300. 1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
301. LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
302. 1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
303. LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
304. 1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
305. LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
306. 1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
307. LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
308. 1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
309. LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
310. 1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
311. ********************************************************************************
312. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 2 NUMBER OF MEMBERS 1
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 1

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 23, TOTAL DEGREES OF FREEDOM = 6
TOTAL LOAD COMBINATION CASES = 92 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 145011.0 MB

313. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 13,2020 TIME= 12: 6:29 ****


```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*               http://www.bentley.com                      *
*************************************************************
```

ATTACHMENT 6 STAAD OUTPUT – FOUNDATION F2

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*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    JUL 13, 2020                   *
*           Time=    12: 8:51                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
*****************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F2 - SBC 304.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 10/06/2020
    4. JOB NAME FDN. MKD. F2
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 10 -0.75 909.9 3.175; 34 -0.5 909.9 3.175
   14. 35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175
   15. 128 0 913.4 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438
   16. 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438; 306 0.25 909.9 3.42438
   17. 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438; 309 1 909.9 3.175
   18. 311 1.25 909.9 3.175; 313 1 909.9 3.42438; 314 1.25 909.9 3.42438
   19. 315 -1.25 909.9 3.175; 316 -1 909.9 3.175; 319 -1.25 909.9 3.42438
   20. 320 -1 909.9 3.42438; 321 -0.75 909.9 3.67438; 322 -0.5 909.9 3.67438
   21. 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438; 325 0.25 909.9 3.67438
   22. 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438; 328 1 909.9 3.67438
   23. 329 1.25 909.9 3.67438; 330 -1.25 909.9 3.67438; 331 -1 909.9 3.67438
   24. 332 -0.75 909.9 3.92438; 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438
   25. 335 0 909.9 3.92438; 336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438
   26. 338 0.75 909.9 3.92438; 339 1 909.9 3.92438; 340 1.25 909.9 3.92438
   27. 341 -1.25 909.9 3.92438; 342 -1 909.9 3.92438; 343 -0.75 909.9 4.17438
   28. 344 -0.5 909.9 4.17438; 345 -0.25 909.9 4.17438; 346 0 909.9 4.17438
   29. 347 0.25 909.9 4.17438; 348 0.5 909.9 4.17438; 349 0.75 909.9 4.17438
   30. 350 1 909.9 4.17438; 351 1.25 909.9 4.17438; 352 -1.25 909.9 4.17438
   31. 353 -1 909.9 4.17438; 354 -0.75 909.9 4.42438; 355 -0.5 909.9 4.42438
   32. 356 -0.25 909.9 4.42438; 357 0 909.9 4.42438; 358 0.25 909.9 4.42438
   33. 359 0.5 909.9 4.42438; 360 0.75 909.9 4.42438; 361 1 909.9 4.42438
   34. 362 1.25 909.9 4.42438; 363 -1.25 909.9 4.42438; 364 -1 909.9 4.42438
   35. 365 -0.75 909.9 2.17438; 366 -0.5 909.9 2.17438; 367 -0.5 909.9 1.925
   36. 368 -0.75 909.9 1.925; 369 -0.25 909.9 2.17438; 370 -0.25 909.9 1.925
   37. 371 0 909.9 2.17438; 372 0 909.9 1.925; 373 0.25 909.9 2.17438
   38. 374 0.25 909.9 1.925; 375 0.5 909.9 2.17438; 376 0.5 909.9 1.925
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   39. 377 0.75 909.9 2.17438; 378 0.75 909.9 1.925; 379 1 909.9 2.17438
   40. 380 1 909.9 1.925; 381 1.25 909.9 2.17438; 382 1.25 909.9 1.925
   41. 383 -1.25 909.9 2.17438; 384 -1 909.9 2.17438; 385 -1 909.9 1.925
   42. 386 -1.25 909.9 1.925; 387 -0.75 909.9 2.42438; 388 -0.5 909.9 2.42438
   43. 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438; 391 0.25 909.9 2.42438
   44. 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438; 394 1 909.9 2.42438
   45. 395 1.25 909.9 2.42438; 396 -1.25 909.9 2.42438; 397 -1 909.9 2.42438
   46. 398 -0.75 909.9 2.67438; 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438
   47. 401 0 909.9 2.67438; 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438
   48. 404 0.75 909.9 2.67438; 405 1 909.9 2.67438; 406 1.25 909.9 2.67438
   49. 407 -1.25 909.9 2.67438; 408 -1 909.9 2.67438; 409 -0.75 909.9 2.92438
   50. 410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438
   51. 413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438
   52. 416 1 909.9 2.92438; 417 1.25 909.9 2.92438; 418 -1.25 909.9 2.92438
   53. 419 -1 909.9 2.92438; 420 1.5 909.9 3.42438; 421 1.5 909.9 3.175
   54. 422 1.5 909.9 3.67438; 423 1.5 909.9 3.92438; 424 1.5 909.9 4.17438
   55. 425 1.5 909.9 4.42438; 426 1.5 909.9 2.17438; 427 1.5 909.9 1.925
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498. ELEMENT
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504. 1200 1201 1204 TO 1229
505. _RAFT_MOM 338 341 350 351 360 361 363 TO 402 410 411 420 421 430 431 -
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516. MEMBER
517. END GROUP DEFINITION
518. ELEMENT PROPERTY
519. 326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 TO 731 745 TO 747 -
520. 750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 845 TO 891 895 TO 897 -
521. 901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 TO 1024 1028 1029 -
522. 1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 1147 1150 -
523. 1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 1200 1201 -
524. 1204 TO 1229 THICKNESS 1.2
525. DEFINE MATERIAL START
526. ISOTROPIC CONCRETE
527. E 2.17185E+007
528. POISSON 0.17
529. DENSITY 25
530. ALPHA 1E-005
531. DAMP 0.05
532. TYPE CONCRETE
533. STRENGTH FCU 27579
534. END DEFINE MATERIAL
535. MEMBER PROPERTY AMERICAN
536. 185 PRIS YD 2 ZD 1.5
537. CONSTANTS
538. MATERIAL CONCRETE ALL
539. SUPPORTS
540. 326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 TO 731 745 TO 747 -
541. 750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 845 TO 891 895 TO 897 -
542. 901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 TO 1024 1028 1029 -
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543. 1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 1147 1150 -
544. 1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 1200 1201 -
545. 1204 TO 1229 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT

JOINT	INFLUENCE AREA
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TOTAL	43.495438

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548. SELFWEIGHT Y -1
549. ELEMENT LOAD
550. * LOAD DUE TO BACKFILL
551. 326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 TO 731 745 TO 747 -
552. 750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 845 TO 891 895 TO 897 -
553. 901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 TO 1024 1028 1029 -
554. 1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 1147 1150 -
555. 1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 1200 1201 -
556. 1204 TO 1229 PR GY -36
557. * LOAD DUE TO SURCHARGE
558. 326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 TO 731 745 TO 747 -
559. 750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 845 TO 891 895 TO 897 -
560. 901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 TO 1024 1028 1029 -
561. 1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 1130 1146 1147 1150 -
562. 1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 1194 1195 1200 1201 -
563. 1204 TO 1229 PR GY -40
564. JOINT LOAD
565. 128 FX 11.9 FY -391.7 MZ -88.6
566. ********************************************************************************
567. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
568. JOINT LOAD
569. 128 FX 13.5 FY -144.5 MZ -162
570. ********************************************************************************
571. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
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572. JOINT LOAD
573. 128 FX 18.2 FY -325.7 MZ -120
574. ********************************************************************************
575. LOAD 4 LOADTYPE NONE  TITLE CRANE LOAD  ( CR1 )
576. JOINT LOAD
577. 128 FX 14.7 FY -140.5 MZ -214.9
578. ********************************************************************************
579. LOAD 5 LOADTYPE NONE  TITLE CRANE LOAD  ( CR2 )
580. JOINT LOAD
581. 128 FX 14.8 FY -38.6 MZ -254.2
582. ********************************************************************************
583. LOAD 6 LOADTYPE NONE  TITLE CRANE LOAD  ( CR3 )
584. JOINT LOAD
585. 128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
586. ********************************************************************************
587. LOAD 7 LOADTYPE NONE  TITLE CRANE LOAD  ( CR4 )
588. JOINT LOAD
589. 128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
590. ********************************************************************************
591. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
592. JOINT LOAD
593. 128 FX 313.7 FY 142.4 MZ -3180.7
594. ********************************************************************************
595. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
596. JOINT LOAD
597. 128 FX 188.5 FY 57.5 MZ -2560.2
598. ********************************************************************************
599. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
600. JOINT LOAD
601. 128 FX 344.4 FY 21.3 MZ -3618.7
602. ********************************************************************************
603. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
604. JOINT LOAD
605. 128 FX 216.7 FY -60.9 MZ -2951.2
606. ********************************************************************************
607. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
608. JOINT LOAD
609. 128 FX -352.7 FY 276.8 MZ 3895.1
610. ********************************************************************************
611. LOAD 13 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
612. JOINT LOAD
613. 128 FX -480.5 FY 195.6 MZ 4591.3
614. ********************************************************************************
615. LOAD 14 LOADTYPE NONE  TITLE WIND LOAD  ( WL7 )
616. JOINT LOAD
617. 128 FX -331.7 FY 129.5 MZ 3703
618. ********************************************************************************
619. LOAD 15 LOADTYPE NONE  TITLE WIND LOAD  ( WL8 )
620. JOINT LOAD
621. 128 FX -459.4 FY 48.4 MZ 4399.4
622. ********************************************************************************
623. LOAD 16 LOADTYPE NONE  TITLE WIND LOAD  ( WL9 )
624. JOINT LOAD
625. 128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
626. ********************************************************************************
627. LOAD 17 LOADTYPE NONE  TITLE WIND LOAD  ( WL10 )
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628. JOINT LOAD
629. 128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
630. ********************************************************************************
631. LOAD 18 LOADTYPE NONE  TITLE WIND LOAD  ( WL11 )
632. JOINT LOAD
633. 128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
634. ********************************************************************************
635. LOAD 19 LOADTYPE NONE  TITLE WIND LOAD  ( WL12 )
636. JOINT LOAD
637. 128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
638. ********************************************************************************
639. LOAD 20 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
640. JOINT LOAD
641. 128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
642. ********************************************************************************
643. LOAD 21 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
644. JOINT LOAD
645. 128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
646. ********************************************************************************
647. LOAD 22 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
648. JOINT LOAD
649. 128 FX 1.1 FY 0 MZ -23.9
650. ********************************************************************************
651. LOAD 23 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
652. JOINT LOAD
653. 128 FX -1.1 FY 0 MZ 23.9
654. ********************************************************************************
655. * UNFACTORED LOAD COMBINATIONS
656. LOAD COMB 201 DL
657. 1 1.0
658. LOAD COMB 202 DL+L+TL1
659. 1 1.0 3 1.0 22 1.0
660. LOAD COMB 203 DL+L+TL2
661. 1 1.0 3 1.0 23 1.0
662. LOAD COMB 204 DL+LR
663. 1 1.0 2 1.0
664. LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
665. 1 1.0 2 0.75 3 0.75 22 0.75
666. LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
667. 1 1.0 2 0.75 3 0.75 23 0.75
668. LOAD COMB 207 DL+0.625WL1
669. 1 1.0 8 0.625
670. LOAD COMB 208 DL+0.625WL2
671. 1 1.0 9 0.625
672. LOAD COMB 209 DL+0.625WL3
673. 1 1.0 10 0.625
674. LOAD COMB 210 DL+0.625WL4
675. 1 1.0 11 0.625
676. LOAD COMB 211 DL+0.625WL5
677. 1 1.0 12 0.625
678. LOAD COMB 212 DL+0.625WL6
679. 1 1.0 13 0.625
680. LOAD COMB 213 DL+0.625WL7
681. 1 1.0 14 0.625
682. LOAD COMB 214 DL+0.625WL8
683. 1 1.0 15 0.625
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684. LOAD COMB 215 DL+0.625WL9
685. 1 1.0 16 0.625
686. LOAD COMB 216 DL+0.625WL10
687. 1 1.0 17 0.625
688. LOAD COMB 217 DL+0.625WL11
689. 1 1.0 18 0.625
690. LOAD COMB 218 DL+0.625WL12
691. 1 1.0 19 0.625
692. LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
693. 1 1.0 2 0.75 3 0.75 8 0.75
694. LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
695. 1 1.0 2 0.75 3 0.75 9 0.75
696. LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
697. 1 1.0 2 0.75 3 0.75 10 0.75
698. LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
699. 1 1.0 2 0.75 3 0.75 11 0.75
700. LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
701. 1 1.0 2 0.75 3 0.75 12 0.75
702. LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
703. 1 1.0 2 0.75 3 0.75 13 0.75
704. LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
705. 1 1.0 2 0.75 3 0.75 14 0.75
706. LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
707. 1 1.0 2 0.75 3 0.75 15 0.75
708. LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
709. 1 1.0 2 0.75 3 0.75 16 0.75
710. LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
711. 1 1.0 2 0.75 3 0.75 17 0.75
712. LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
713. 1 1.0 2 0.75 3 0.75 18 0.75
714. LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
715. 1 1.0 2 0.75 3 0.75 19 0.75
716. LOAD COMB 231 0.6DL+0.625WL1
717. 1 0.6 8 0.625
718. LOAD COMB 232 0.6DL+0.625WL2
719. 1 0.6 9 0.625
720. LOAD COMB 233 0.6DL+0.625WL3
721. 1 0.6 10 0.625
722. LOAD COMB 234 0.6DL+0.625WL4
723. 1 0.6 11 0.625
724. LOAD COMB 235 0.6DL+0.625WL5
725. 1 0.6 12 0.625
726. LOAD COMB 236 0.6DL+0.625WL6
727. 1 0.6 13 0.625
728. LOAD COMB 237 0.6DL+0.625WL7
729. 1 0.6 14 0.625
730. LOAD COMB 238 0.6DL+0.625WL8
731. 1 0.6 15 0.625
732. LOAD COMB 239 0.6DL+0.625WL9
733. 1 0.6 16 0.625
734. LOAD COMB 240 0.6DL+0.625WL10
735. 1 0.6 17 0.625
736. LOAD COMB 241 0.6DL+0.625WL11
737. 1 0.6 18 0.625
738. LOAD COMB 242 0.6DL+0.625WL12
739. 1 0.6 19 0.625
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740. LOAD COMB 243 0.6DL+0.7SL1
741. 1 0.6 20 0.7
742. LOAD COMB 244 0.6DL+0.7SL2
743. 1 0.6 21 0.7
744. LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
745. 1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
746. LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
747. 1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
748. LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
749. 1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
750. LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
751. 1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
752. LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
753. 1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
754. LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
755. 1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
756. LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
757. 1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
758. LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
759. 1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
760. LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
761. 1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
762. LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
763. 1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
764. LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
765. 1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
766. LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
767. 1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
768. LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
769. 1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
770. LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
771. 1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
772. LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
773. 1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
774. LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
775. 1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
776. LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
777. 1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
778. LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
779. 1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
780. LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
781. 1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
782. LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
783. 1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
784. LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
785. 1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
786. LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
787. 1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
788. LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
789. 1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
790. LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
791. 1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
792. LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
793. 1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
794. LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
795. 1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
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796. LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
797. 1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
798. LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
799. 1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
800. LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
801. 1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
802. LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
803. 1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
804. LOAD COMB 275 DL+0.75WL7+0.75L+0.75LR+CR3
805. 1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
806. LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
807. 1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
808. LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
809. 1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
810. LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
811. 1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
812. LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
813. 1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
814. LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
815. 1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
816. LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
817. 1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
818. LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
819. 1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
820. LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
821. 1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
822. LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
823. 1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
824. LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
825. 1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
826. LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
827. 1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
828. LOAD COMB 287 DL+0.75WL7+0.75L+0.75LR+CR4
829. 1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
830. LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
831. 1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
832. LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
833. 1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
834. LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
835. 1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
836. LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
837. 1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
838. LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
839. 1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
840. ********************************************************************************
841. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
842. LOAD COMB 101 1.4DL
843. 1 1.4
844. LOAD COMB 102 1.4DL+1.4TL1+1.7L+0.5LR
845. 1 1.4 22 1.4 3 1.7 2 0.5
846. LOAD COMB 103 1.4DL+1.4TL2+1.7L+0.5LR
847. 1 1.4 23 1.4 3 1.7 2 0.5
848. LOAD COMB 104 1.2DL+1.0LR+1.0L
849. 1 1.2 2 1.0 3 1.0
850. LOAD COMB 105 1.2DL+1.0WL1+1.0L+0.5LR
851. 1 1.2 8 1.0 2 0.5 3 1.0
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852. LOAD COMB 106 1.2DL+1.0WL2+0.5L+0.5LR
853. 1 1.2 9 1.0 2 0.5 3 1.0
854. LOAD COMB 107 1.2DL+1.0WL3+0.5L+0.5LR
855. 1 1.2 10 1.0 2 0.5 3 1.0
856. LOAD COMB 108 1.2DL+1.0WL4+0.5L+0.5LR
857. 1 1.2 11 1.0 2 0.5 3 1.0
858. LOAD COMB 109 1.2DL+1.0WL5+0.5L+0.5LR
859. 1 1.2 12 1.0 2 0.5 3 1.0
860. LOAD COMB 110 1.2DL+1.0WL6+0.5L+0.5LR
861. 1 1.2 13 1.0 2 0.5 3 1.0
862. LOAD COMB 111 1.2DL+1.0WL7+0.5L+0.5LR
863. 1 1.2 14 1.0 2 0.5 3 1.0
864. LOAD COMB 112 1.2DL+1.0WL8+0.5L+0.5LR
865. 1 1.2 15 1.0 2 0.5 3 1.0
866. LOAD COMB 113 1.2DL+1.0WL9+0.5L+0.5LR
867. 1 1.2 16 1.0 2 0.5 3 1.0
868. LOAD COMB 114 1.2DL+1.0WL10+0.5L+0.5LR
869. 1 1.2 17 1.0 2 0.5 3 1.0
870. LOAD COMB 115 1.2DL+1.0WL11+0.5L+0.5LR
871. 1 1.2 18 1.0 2 0.5 3 1.0
872. LOAD COMB 116 1.2DL+1.0WL12+0.5L+0.5LR
873. 1 1.2 19 1.0 2 0.5 3 1.0
874. LOAD COMB 117 1.2DL+1.0SL1+1.0L
875. 1 1.2 20 1.0 3 1.0
876. LOAD COMB 118 1.2DL+1.0SL2+1.0L
877. 1 1.2 21 1.0 3 1.0
878. LOAD COMB 119 0.9DL+1.0WL1
879. 1 0.9 8 1.0
880. LOAD COMB 120 0.9DL+1.0WL2
881. 1 0.9 9 1.0
882. LOAD COMB 121 0.9DL+1.0WL3
883. 1 0.9 10 1.0
884. LOAD COMB 122 0.9DL+1.0WL4
885. 1 0.9 11 1.0
886. LOAD COMB 123 0.9DL+1.0WL5
887. 1 0.9 12 1.0
888. LOAD COMB 124 0.9DL+1.0WL6
889. 1 0.9 13 1.0
890. LOAD COMB 125 0.9DL+1.0WL7
891. 1 0.9 14 1.0
892. LOAD COMB 126 0.9DL+1.0WL8
893. 1 0.9 15 1.0
894. LOAD COMB 127 0.9DL+1.0WL9
895. 1 0.9 16 1.0
896. LOAD COMB 128 0.9DL+1.0WL10
897. 1 0.9 17 1.0
898. LOAD COMB 129 0.9DL+1.0WL11
899. 1 0.9 18 1.0
900. LOAD COMB 130 0.9DL+1.0WL12
901. 1 0.9 19 1.0
902. LOAD COMB 131 0.9DL+1.0SL1
903. 1 0.9 20 1.0
904. LOAD COMB 132 0.9DL+1.0SL2
905. 1 0.9 21 1.0
906. LOAD COMB 133 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR1
907. 1 1.2 8 1.0 2 0.5 3 1.0 4 1.0
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 908. LOAD COMB 134 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR1
 909. 1 1.2 9 1.0 2 0.5 3 1.0 4 1.0
 910. LOAD COMB 135 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
 911. 1 1.2 10 1.0 2 0.5 3 1.0 4 1.0
 912. LOAD COMB 136 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR1
 913. 1 1.2 11 1.0 2 0.5 3 1.0 4 1.0
 914. LOAD COMB 137 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR1
 915. 1 1.2 12 1.0 2 0.5 3 1.0 4 1.0
 916. LOAD COMB 138 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR1
 917. 1 1.2 13 1.0 2 0.5 3 1.0 4 1.0
 918. LOAD COMB 139 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR1
 919. 1 1.2 14 1.0 2 0.5 3 1.0 4 1.0
 920. LOAD COMB 140 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR1
 921. 1 1.2 15 1.0 2 0.5 3 1.0 4 1.0
 922. LOAD COMB 141 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR1
 923. 1 1.2 16 1.0 2 0.5 3 1.0 4 1.0
 924. LOAD COMB 142 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR1
 925. 1 1.2 17 1.0 2 0.5 3 1.0 4 1.0
 926. LOAD COMB 143 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR1
 927. 1 1.2 18 1.0 2 0.5 3 1.0 4 1.0
 928. LOAD COMB 144 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR1
 929. 1 1.2 19 1.0 2 0.5 3 1.0 4 1.0
 930. LOAD COMB 145 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR2
 931. 1 1.2 8 1.0 2 0.5 3 1.0 5 1.0
 932. LOAD COMB 146 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR2
 933. 1 1.2 9 1.0 2 0.5 3 1.0 5 1.0
 934. LOAD COMB 147 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR2
 935. 1 1.2 10 1.0 2 0.5 3 1.0 5 1.0
 936. LOAD COMB 148 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR2
 937. 1 1.2 11 1.0 2 0.5 3 1.0 5 1.0
 938. LOAD COMB 149 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR2
 939. 1 1.2 12 1.0 2 0.5 3 1.0 5 1.0
 940. LOAD COMB 150 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR2
 941. 1 1.2 13 1.0 2 0.5 3 1.0 5 1.0
 942. LOAD COMB 151 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR2
 943. 1 1.2 14 1.0 2 0.5 3 1.0 5 1.0
 944. LOAD COMB 152 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR2
 945. 1 1.2 15 1.0 2 0.5 3 1.0 5 1.0
 946. LOAD COMB 153 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR2
 947. 1 1.2 16 1.0 2 0.5 3 1.0 5 1.0
 948. LOAD COMB 154 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR2
 949. 1 1.2 17 1.0 2 0.5 3 1.0 5 1.0
 950. LOAD COMB 155 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR2
 951. 1 1.2 18 1.0 2 0.5 3 1.0 5 1.0
 952. LOAD COMB 156 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR2
 953. 1 1.2 19 1.0 2 0.5 3 1.0 5 1.0
 954. LOAD COMB 157 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR3
 955. 1 1.2 8 1.0 2 0.5 3 1.0 6 1.0
 956. LOAD COMB 158 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR3
 957. 1 1.2 9 1.0 2 0.5 3 1.0 6 1.0
 958. LOAD COMB 159 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR3
 959. 1 1.2 10 1.0 2 0.5 3 1.0 6 1.0
 960. LOAD COMB 160 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR3
 961. 1 1.2 11 1.0 2 0.5 3 1.0 6 1.0
 962. LOAD COMB 161 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR3
 963. 1 1.2 12 1.0 2 0.5 3 1.0 6 1.0
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 964. LOAD COMB 162 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR3
 965. 1 1.2 13 1.0 2 0.5 3 1.0 6 1.0
 966. LOAD COMB 163 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR3
 967. 1 1.2 14 1.0 2 0.5 3 1.0 6 1.0
 968. LOAD COMB 164 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR3
 969. 1 1.2 15 1.0 2 0.5 3 1.0 6 1.0
 970. LOAD COMB 165 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR3
 971. 1 1.2 16 1.0 2 0.5 3 1.0 6 1.0
 972. LOAD COMB 166 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR3
 973. 1 1.2 17 1.0 2 0.5 3 1.0 6 1.0
 974. LOAD COMB 167 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR3
 975. 1 1.2 18 1.0 2 0.5 3 1.0 6 1.0
 976. LOAD COMB 168 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR3
 977. 1 1.2 19 1.0 2 0.5 3 1.0 6 1.0
 978. LOAD COMB 169 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR4
 979. 1 1.2 8 1.0 2 0.5 3 1.0 7 1.0
 980. LOAD COMB 170 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR4
 981. 1 1.2 9 1.0 2 0.5 3 1.0 7 1.0
 982. LOAD COMB 171 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR4
 983. 1 1.2 10 1.0 2 0.5 3 1.0 7 1.0
 984. LOAD COMB 172 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR4
 985. 1 1.2 11 1.0 2 0.5 3 1.0 7 1.0
 986. LOAD COMB 173 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR4
 987. 1 1.2 12 1.0 2 0.5 3 1.0 7 1.0
 988. LOAD COMB 174 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
 989. 1 1.2 13 1.0 2 0.5 3 1.0 7 1.0
 990. LOAD COMB 175 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR4
 991. 1 1.2 14 1.0 2 0.5 3 1.0 7 1.0
 992. LOAD COMB 176 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
 993. 1 1.2 15 1.0 2 0.5 3 1.0 7 1.0
 994. LOAD COMB 177 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR4
 995. 1 1.2 16 1.0 2 0.5 3 1.0 7 1.0
 996. LOAD COMB 178 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR4
 997. 1 1.2 17 1.0 2 0.5 3 1.0 7 1.0
 998. LOAD COMB 179 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR4
 999. 1 1.2 18 1.0 2 0.5 3 1.0 7 1.0
1000. LOAD COMB 180 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR4
1001. 1 1.2 19 1.0 2 0.5 3 1.0 7 1.0
1002. ******************************************************************************
1003. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**751**	**NUMBER OF MEMBERS**	**1**
NUMBER OF PLATES	**696**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**750**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 271/ 42/ 129 DOF
TOTAL PRIMARY LOAD CASES = 23, TOTAL DEGREES OF FREEDOM = 2256
TOTAL LOAD COMBINATION CASES = 172 SO FAR.
SIZE OF STIFFNESS MATRIX = 292 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 19.4/ 145010.6 MB

```
1004. LOAD LIST 101 TO 180
1005. START CONCRETE DESIGN
```



```
1006. CODE ACI
1007. TRACK 2 ALL
1008. FC 35000 MEMB 185
1009. CLB 0.05 MEMB 185
1010. CLS 0.05 ALL
1011. CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 338 341 TO 714 725 TO 727 729 -
1012. 730 TO 731 745 TO 747 750 TO 752 769 TO 771 775 TO 777 781 TO 794 809 811 -
1013. 845 TO 891 895 TO 897 901 903 909 TO 955 959 TO 961 965 967 972 TO 1018 1022 -
1014. 1023 TO 1024 1028 1029 1032 TO 1078 1082 TO 1084 1088 TO 1113 1126 1127 1129 -
1015. 1130 1146 1147 1150 1151 1170 1171 1174 1175 1180 1181 1184 1185 1190 1191 -
1016. 1194 1195 1200 1201 1204 1205
1017. MAXMAIN 25 ALL
1018. MINMAIN 25 ALL
1019. MINSEC 10 ALL
1020. DESIGN COLUMN 185
```

==========

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE -1500.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 30000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
64 - 25 MM	1.047	101	END	0.650

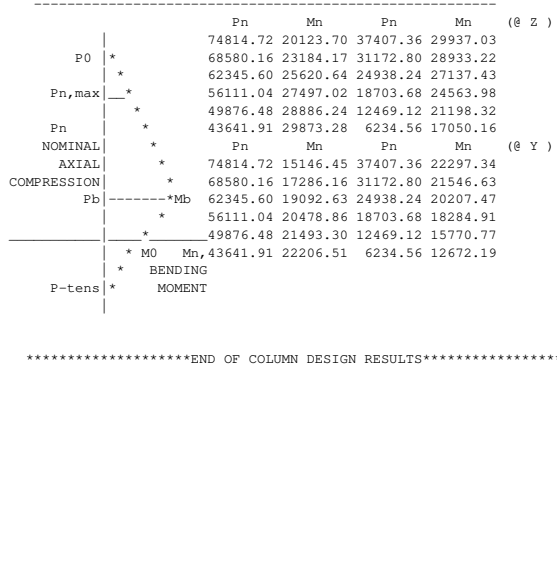
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
********	81049.27	41239.41	30170.30	731.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
12092.92	********	843.66	190.83	0.06463

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
********	81049.27	40737.83	22458.28	551.3
M0	P-tens.	Des.Pn	Des.Mn	e/h
8945.85	********	843.66	0.00	0.00000



**********END OF COLUMN DESIGN RESULTS**********

1021. END CONCRETE DESIGN
1022. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 13,2020 TIME= 12: 9:51 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc. *
* http://www.bentley.com *
****

```
            **************************************************
            *                                                *
            *           STAAD.Pro V8i SELECTseries5          *
            *           Version  20.07.10.64                 *
            *           Proprietary Program of               *
            *           Bentley Systems, Inc.                *
            *           Date=    JUL 13, 2020                *
            *           Time=    12:11:29                    *
            *                                                *
            *      USER ID: LNT ECC DIV                      *
            **************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F2 - STABILITY_304.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 10/06/2020
     4. JOB NAME FDN. MKD. F2_STABILITY CHECK
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. END JOB INFORMATION
    10. INPUT WIDTH 79
    11. UNIT METER KN
    12. JOINT COORDINATES
    13. 2 0 909.9 3.175; 128 0 913.4 3.175
    14. MEMBER INCIDENCES
    15. 185 2 128
    16. DEFINE MATERIAL START
    17. ISOTROPIC CONCRETE
    18. E 2.17185E+007
    19. POISSON 0.17
    20. DENSITY 25
    21. ALPHA 1E-005
    22. DAMP 0.05
    23. TYPE CONCRETE
    24. STRENGTH FCU 27579
    25. END DEFINE MATERIAL
    26. MEMBER PROPERTY AMERICAN
    27. 185 PRIS YD 2 ZD 1.5
    28. CONSTANTS
    29. MATERIAL CONCRETE ALL
    30. SUPPORTS
    31. 2 FIXED
    32. ******************************************************************************
    33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
    34. SELFWEIGHT Y -1
    35. JOINT LOAD
    36. 128 FX 11.9 FY -391.7 MZ -88.6
    37. ******************************************************************************
    38. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
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    39. JOINT LOAD
    40. 128 FX 13.5 FY -144.5 MZ -162
    41. ******************************************************************************
    42. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
    43. JOINT LOAD
    44. 128 FX 18.2 FY -325.7 MZ -120
    45. ******************************************************************************
    46. LOAD 4 LOADTYPE NONE  TITLE CRANE LOAD  ( CR1 )
    47. JOINT LOAD
    48. 128 FX 14.7 FY -140.5 MZ -214.9
    49. ******************************************************************************
    50. LOAD 5 LOADTYPE NONE  TITLE CRANE LOAD  ( CR2 )
    51. JOINT LOAD
    52. 128 FX 14.8 FY -38.6 MZ -254.2
    53. ******************************************************************************
    54. LOAD 6 LOADTYPE NONE  TITLE CRANE LOAD  ( CR3 )
    55. JOINT LOAD
    56. 128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
    57. ******************************************************************************
    58. LOAD 7 LOADTYPE NONE  TITLE CRANE LOAD  ( CR4 )
    59. JOINT LOAD
    60. 128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
    61. ******************************************************************************
    62. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
    63. JOINT LOAD
    64. 128 FX 313.7 FY 142.4 MZ -3180.7
    65. ******************************************************************************
    66. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
    67. JOINT LOAD
    68. 128 FX 188.5 FY 57.5 MZ -2560.2
    69. ******************************************************************************
    70. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
    71. JOINT LOAD
    72. 128 FX 344.4 FY 21.3 MZ -3618.7
    73. ******************************************************************************
    74. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
    75. JOINT LOAD
    76. 128 FX 216.7 FY -60.9 MZ -2951.2
    77. ******************************************************************************
    78. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
    79. JOINT LOAD
    80. 128 FX -352.7 FY 276.8 MZ 3895.1
    81. ******************************************************************************
    82. LOAD 13 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
    83. JOINT LOAD
    84. 128 FX -480.5 FY 195.6 MZ 4591.3
    85. ******************************************************************************
    86. LOAD 14 LOADTYPE NONE  TITLE WIND LOAD  ( WL7 )
    87. JOINT LOAD
    88. 128 FX -331.7 FY 129.5 MZ 3703
    89. ******************************************************************************
    90. LOAD 15 LOADTYPE NONE  TITLE WIND LOAD  ( WL8 )
    91. JOINT LOAD
    92. 128 FX -459.4 FY 48.4 MZ 4399.4
    93. ******************************************************************************
    94. LOAD 16 LOADTYPE NONE  TITLE WIND LOAD  ( WL9 )
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 95. JOINT LOAD
 96. 128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
 97. ********************************************************************************
 98. LOAD 17 LOADTYPE NONE  TITLE WIND LOAD  ( WL10 )
 99. JOINT LOAD
100. 128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
101. ********************************************************************************
102. LOAD 18 LOADTYPE NONE  TITLE WIND LOAD  ( WL11 )
103. JOINT LOAD
104. 128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
105. ********************************************************************************
106. LOAD 19 LOADTYPE NONE  TITLE WIND LOAD  ( WL12 )
107. JOINT LOAD
108. 128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
109. ********************************************************************************
110. LOAD 20 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
111. JOINT LOAD
112. 128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
113. ********************************************************************************
114. LOAD 21 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
115. JOINT LOAD
116. 128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
117. ********************************************************************************
118. LOAD 22 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
119. JOINT LOAD
120. 128 FX 1.1 FY 0 MZ -23.9
121. ********************************************************************************
122. LOAD 23 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
123. JOINT LOAD
124. 128 FX -1.1 FY 0 MZ 23.9
125. ********************************************************************************
126. * UNFACTORED LOAD COMBINATIONS
127. LOAD COMB 201 DL
128. 1 1.0
129. LOAD COMB 202 DL+L+TL1
130. 1 1.0 3 1.0 22 1.0
131. LOAD COMB 203 DL+L+TL2
132. 1 1.0 3 1.0 23 1.0
133. LOAD COMB 204 DL+LR
134. 1 1.0 2 1.0
135. LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
136. 1 1.0 2 0.75 3 0.75 22 0.75
137. LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
138. 1 1.0 2 0.75 3 0.75 23 0.75
139. LOAD COMB 207 DL+0.625WL1
140. 1 1.0 8 0.625
141. LOAD COMB 208 DL+0.625WL2
142. 1 1.0 9 0.625
143. LOAD COMB 209 DL+0.625WL3
144. 1 1.0 10 0.625
145. LOAD COMB 210 DL+0.625WL4
146. 1 1.0 11 0.625
147. LOAD COMB 211 DL+0.625WL5
148. 1 1.0 12 0.625
149. LOAD COMB 212 DL+0.625WL6
150. 1 1.0 13 0.625
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151. LOAD COMB 213 DL+0.625WL7
152. 1 1.0 14 0.625
153. LOAD COMB 214 DL+0.625WL8
154. 1 1.0 15 0.625
155. LOAD COMB 215 DL+0.625WL9
156. 1 1.0 16 0.625
157. LOAD COMB 216 DL+0.625WL10
158. 1 1.0 17 0.625
159. LOAD COMB 217 DL+0.625WL11
160. 1 1.0 18 0.625
161. LOAD COMB 218 DL+0.625WL12
162. 1 1.0 19 0.625
163. LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
164. 1 1.0 2 0.75 3 0.75 8 0.75
165. LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
166. 1 1.0 2 0.75 3 0.75 9 0.75
167. LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
168. 1 1.0 2 0.75 3 0.75 10 0.75
169. LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
170. 1 1.0 2 0.75 3 0.75 11 0.75
171. LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
172. 1 1.0 2 0.75 3 0.75 12 0.75
173. LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
174. 1 1.0 2 0.75 3 0.75 13 0.75
175. LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
176. 1 1.0 2 0.75 3 0.75 14 0.75
177. LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
178. 1 1.0 2 0.75 3 0.75 15 0.75
179. LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
180. 1 1.0 2 0.75 3 0.75 16 0.75
181. LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
182. 1 1.0 2 0.75 3 0.75 17 0.75
183. LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
184. 1 1.0 2 0.75 3 0.75 18 0.75
185. LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
186. 1 1.0 2 0.75 3 0.75 19 0.75
187. LOAD COMB 231 0.6DL+0.625WL1
188. 1 0.6 8 0.625
189. LOAD COMB 232 0.6DL+0.625WL2
190. 1 0.6 9 0.625
191. LOAD COMB 233 0.6DL+0.625WL3
192. 1 0.6 10 0.625
193. LOAD COMB 234 0.6DL+0.625WL4
194. 1 0.6 11 0.625
195. LOAD COMB 235 0.6DL+0.625WL5
196. 1 0.6 12 0.625
197. LOAD COMB 236 0.6DL+0.625WL6
198. 1 0.6 13 0.625
199. LOAD COMB 237 0.6DL+0.625WL7
200. 1 0.6 14 0.625
201. LOAD COMB 238 0.6DL+0.625WL8
202. 1 0.6 15 0.625
203. LOAD COMB 239 0.6DL+0.625WL9
204. 1 0.6 16 0.625
205. LOAD COMB 240 0.6DL+0.625WL10
206. 1 0.6 17 0.625
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207. LOAD COMB 241 0.6DL+0.625WL11
208. 1 0.6 18 0.625
209. LOAD COMB 242 0.6DL+0.625WL12
210. 1 0.6 19 0.625
211. LOAD COMB 243 0.6DL+0.7SL1
212. 1 0.6 20 0.7
213. LOAD COMB 244 0.6DL+0.7SL2
214. 1 0.6 21 0.7
215. LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
216. 1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
217. LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
218. 1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
219. LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
220. 1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
221. LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
222. 1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
223. LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
224. 1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
225. LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
226. 1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
227. LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
228. 1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
229. LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
230. 1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
231. LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
232. 1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
233. LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
234. 1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
235. LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
236. 1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
237. LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
238. 1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
239. LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
240. 1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
241. LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
242. 1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
243. LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
244. 1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
245. LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
246. 1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
247. LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
248. 1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
249. LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
250. 1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
251. LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
252. 1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
253. LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
254. 1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
255. LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
256. 1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
257. LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
258. 1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
259. LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
260. 1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
261. LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
262. 1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
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263. LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
264. 1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
265. LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
266. 1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
267. LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
268. 1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
269. LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
270. 1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
271. LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
272. 1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
273. LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
274. 1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
275. LOAD COMB 275 DL+0.75WL7+0.75L+0.75LR+CR3
276. 1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
277. LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
278. 1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
279. LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
280. 1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
281. LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
282. 1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
283. LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
284. 1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
285. LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
286. 1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
287. LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
288. 1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
289. LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
290. 1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
291. LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
292. 1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
293. LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
294. 1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
295. LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
296. 1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
297. LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
298. 1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
299. LOAD COMB 287 DL+0.75WL7+0.75L+0.75LR+CR4
300. 1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
301. LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
302. 1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
303. LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
304. 1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
305. LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
306. 1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
307. LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
308. 1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
309. LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
310. 1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
311. ********************************************************************************
312. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 2 NUMBER OF MEMBERS 1
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 1

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 23, TOTAL DEGREES OF FREEDOM = 6
TOTAL LOAD COMBINATION CASES = 92 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 144951.5 MB

313. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 13,2020 TIME= 12:11:34 ****


```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*               http://www.bentley.com                      *
*************************************************************
```

ATTACHMENT 7 STAAD OUTPUT – FOUNDATION F3

PROPRIETARY & CONFIDENTIAL

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    JUL 13, 2020                 *
*          Time=    11:15:50                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F3 - SBC 304.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 28/05/2020
    4. JOB NAME FDN. MKD. F3
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 10 -0.75 909.9 3.175; 34 -0.5 909.9 3.175
   14. 35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175
   15. 128 0 913.4 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438
   16. 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438; 306 0.25 909.9 3.42438
   17. 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438; 309 1 909.9 3.175
   18. 311 1.25 909.9 3.175; 313 1 909.9 3.42438; 314 1.25 909.9 3.42438
   19. 315 -1.25 909.9 3.175; 316 -1 909.9 3.175; 319 -1.25 909.9 3.42438
   20. 320 -1 909.9 3.42438; 321 -0.75 909.9 3.67438; 322 -0.5 909.9 3.67438
   21. 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438; 325 0.25 909.9 3.67438
   22. 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438; 328 1 909.9 3.67438
   23. 329 1.25 909.9 3.67438; 330 -1.25 909.9 3.67438; 331 -1 909.9 3.67438
   24. 332 -0.75 909.9 3.92438; 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438
   25. 335 0 909.9 3.92438; 336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438
   26. 338 0.75 909.9 3.92438; 339 1 909.9 3.92438; 340 1.25 909.9 3.92438
   27. 341 -1.25 909.9 3.92438; 342 -1 909.9 3.92438; 343 -0.75 909.9 4.17438
   28. 344 -0.5 909.9 4.17438; 345 -0.25 909.9 4.17438; 346 0 909.9 4.17438
   29. 347 0.25 909.9 4.17438; 348 0.5 909.9 4.17438; 349 0.75 909.9 4.17438
   30. 350 1 909.9 4.17438; 351 1.25 909.9 4.17438; 352 -1.25 909.9 4.17438
   31. 353 -1 909.9 4.17438; 354 -0.75 909.9 4.42438; 355 -0.5 909.9 4.42438
   32. 356 -0.25 909.9 4.42438; 357 0 909.9 4.42438; 358 0.25 909.9 4.42438
   33. 359 0.5 909.9 4.42438; 360 0.75 909.9 4.42438; 361 1 909.9 4.42438
   34. 362 1.25 909.9 4.42438; 363 -1.25 909.9 4.42438; 364 -1 909.9 4.42438
   35. 365 -0.75 909.9 2.17438; 366 -0.5 909.9 2.17438; 367 -0.5 909.9 1.925
   36. 368 -0.75 909.9 1.925; 369 -0.25 909.9 2.17438; 370 -0.25 909.9 1.925
   37. 371 0 909.9 2.17438; 372 0 909.9 1.925; 373 0.25 909.9 2.17438
   38. 374 0.25 909.9 1.925; 375 0.5 909.9 2.17438; 376 0.5 909.9 1.925
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   39. 377 0.75 909.9 2.17438; 378 0.75 909.9 1.925; 379 1 909.9 2.17438
   40. 380 1 909.9 1.925; 381 1.25 909.9 2.17438; 382 1.25 909.9 1.925
   41. 383 -1.25 909.9 2.17438; 384 -1 909.9 2.17438; 385 -1 909.9 1.925
   42. 386 -1.25 909.9 1.925; 387 -0.75 909.9 2.42438; 388 -0.5 909.9 2.42438
   43. 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438; 391 0.25 909.9 2.42438
   44. 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438; 394 1 909.9 2.42438
   45. 395 1.25 909.9 2.42438; 396 -1.25 909.9 2.42438; 397 -1 909.9 2.42438
   46. 398 -0.75 909.9 2.67438; 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438
   47. 401 0 909.9 2.67438; 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438
   48. 404 0.75 909.9 2.67438; 405 1 909.9 2.67438; 406 1.25 909.9 2.67438
   49. 407 -1.25 909.9 2.67438; 408 -1 909.9 2.67438; 409 -0.75 909.9 2.92438
   50. 410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438
   51. 413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438
   52. 416 1 909.9 2.92438; 417 1.25 909.9 2.92438; 418 -1.25 909.9 2.92438
   53. 419 -1 909.9 2.92438; 420 1.5 909.9 3.42438; 421 1.5 909.9 3.175
   54. 422 1.5 909.9 3.67438; 423 1.5 909.9 3.92438; 424 1.5 909.9 4.17438
   55. 425 1.5 909.9 4.42438; 426 1.5 909.9 2.17438; 427 1.5 909.9 1.925
   56. 428 1.5 909.9 2.42438; 429 1.5 909.9 2.67438; 430 1.5 909.9 2.92438
   57. 431 -1.5 909.9 3.42438; 432 -1.5 909.9 3.175; 433 -1.5 909.9 3.67438
   58. 434 -1.5 909.9 3.92438; 435 -1.5 909.9 4.17438; 436 -1.5 909.9 4.42438
   59. 437 -1.5 909.9 2.17438; 438 -1.5 909.9 1.925; 439 -1.5 909.9 2.42438
   60. 440 -1.5 909.9 2.67438; 441 -1.5 909.9 2.92438; 442 -0.75 909.9 4.67438
   61. 443 -0.5 909.9 4.67438; 444 -0.25 909.9 4.67438; 445 0 909.9 4.67438
   62. 446 0.25 909.9 4.67438; 447 0.5 909.9 4.67438; 448 0.75 909.9 4.67438
   63. 449 1 909.9 4.67438; 450 1.25 909.9 4.67438; 451 -1.25 909.9 4.67438
   64. 452 -1 909.9 4.67438; 453 1.5 909.9 4.67438; 454 -1.5 909.9 4.67438
   65. 455 -0.5 909.9 1.675; 456 -0.75 909.9 1.675; 457 -0.25 909.9 1.675
   66. 458 0 909.9 1.675; 459 0.25 909.9 1.675; 460 0.5 909.9 1.675
   67. 461 0.75 909.9 1.675; 462 1 909.9 1.675; 463 1.25 909.9 1.675
   68. 464 -1 909.9 1.675; 465 -1.25 909.9 1.675; 466 1.5 909.9 1.675
   69. 467 -1.5 909.9 1.675; 468 1.8 909.9 3.42438; 469 1.8 909.9 3.175
   70. 470 1.8 909.9 3.67438; 471 1.8 909.9 3.92438; 472 1.8 909.9 4.17438
   71. 473 1.8 909.9 4.42438; 474 1.8 909.9 2.17438; 475 1.8 909.9 1.925
   72. 476 1.8 909.9 2.42438; 477 1.8 909.9 2.67438; 478 1.8 909.9 2.92438
   73. 479 1.8 909.9 4.67438; 480 1.8 909.9 1.675; 481 -1.75 909.9 3.42438
   74. 482 -1.75 909.9 3.175; 483 -1.75 909.9 3.67438; 484 -1.75 909.9 3.92438
   75. 485 -1.75 909.9 4.17438; 486 -1.75 909.9 4.42438; 487 -1.75 909.9 2.17438
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   78. 494 -0.5 909.9 1.425; 495 -0.75 909.9 1.425; 496 -0.25 909.9 1.425
   79. 497 0 909.9 1.425; 498 0.25 909.9 1.425; 499 0.5 909.9 1.425
   80. 500 0.75 909.9 1.425; 501 1 909.9 1.425; 502 1.25 909.9 1.425
   81. 503 -1 909.9 1.425; 504 -1.25 909.9 1.425; 505 1.5 909.9 1.425
   82. 506 -1.5 909.9 1.425; 507 1.8 909.9 1.425; 508 -1.75 909.9 1.425
   83. 509 -0.75 909.9 4.92438; 510 -0.5 909.9 4.92438; 511 -0.25 909.9 4.92438
   84. 512 0 909.9 4.92438; 513 0.25 909.9 4.92438; 514 0.5 909.9 4.92438
   85. 515 0.75 909.9 4.92438; 516 1 909.9 4.92438; 517 1.25 909.9 4.92438
   86. 518 -1.25 909.9 4.92438; 519 -1 909.9 4.92438; 520 1.5 909.9 4.92438
   87. 521 -1.5 909.9 4.92438; 522 1.8 909.9 4.92438; 523 -1.75 909.9 4.92438
   88. 524 2.05 909.9 3.42438; 525 2.05 909.9 3.175; 526 2.05 909.9 3.67438
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   90. 530 2.05 909.9 2.17438; 531 2.05 909.9 1.925; 532 2.05 909.9 2.42438
   91. 533 2.05 909.9 2.67438; 534 2.05 909.9 2.92438; 535 2.05 909.9 4.67438
   92. 536 2.05 909.9 1.675; 537 2.05 909.9 1.425; 538 2.05 909.9 4.92438
   93. 539 -2 909.9 3.42438; 540 -2 909.9 3.175; 541 -2 909.9 3.67438
   94. 542 -2 909.9 3.92438; 543 -2 909.9 4.17438; 544 -2 909.9 4.42438
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96. 548 -2 909.9 2.67438; 549 -2 909.9 2.92438; 550 -2 909.9 4.67438
97. 551 -2 909.9 1.675; 552 -2 909.9 1.425; 553 -2 909.9 4.92438
98. 554 -0.5 909.9 1.175; 555 -0.75 909.9 1.175; 556 -0.25 909.9 1.175
99. 557 0 909.9 1.175; 558 0.25 909.9 1.175; 559 0.5 909.9 1.175
100. 560 0.75 909.9 1.175; 561 1 909.9 1.175; 562 1.25 909.9 1.175
101. 563 -1 909.9 1.175; 564 -1.25 909.9 1.175; 565 1.5 909.9 1.175
102. 566 -1.5 909.9 1.175; 567 1.8 909.9 1.175; 568 -1.75 909.9 1.175
103. 569 2.05 909.9 1.175; 570 -2 909.9 1.175; 571 -0.75 909.9 5.17438
104. 572 -0.5 909.9 5.17438; 573 -0.25 909.9 5.17438; 574 0 909.9 5.17438
105. 575 0.25 909.9 5.17438; 576 0.5 909.9 5.17438; 577 0.75 909.9 5.17438
106. 578 1 909.9 5.17438; 579 1.25 909.9 5.17438; 580 -1.25 909.9 5.17438
107. 581 -1 909.9 5.17438; 582 1.5 909.9 5.17438; 583 -1.5 909.9 5.17438
108. 584 1.8 909.9 5.17438; 585 -1.75 909.9 5.17438; 586 2.05 909.9 5.17438
109. 587 -2 909.9 5.17438; 588 2.3 909.9 3.42438; 589 2.3 909.9 3.175
110. 590 2.55 909.9 3.42438; 591 2.55 909.9 3.175; 592 2.8 909.9 3.42438
111. 593 2.8 909.9 3.175; 594 2.3 909.9 3.67438; 595 2.55 909.9 3.67438
112. 596 2.8 909.9 3.67438; 597 2.3 909.9 3.92438; 598 2.55 909.9 3.92438
113. 599 2.8 909.9 3.92438; 600 2.3 909.9 4.17438; 601 2.55 909.9 4.17438
114. 602 2.8 909.9 4.17438; 603 2.3 909.9 4.42438; 604 2.55 909.9 4.42438
115. 605 2.8 909.9 4.42438; 606 2.3 909.9 2.17438; 607 2.3 909.9 1.925
116. 608 2.55 909.9 2.17438; 609 2.55 909.9 1.925; 610 2.8 909.9 2.17438
117. 611 2.8 909.9 1.925; 612 2.3 909.9 2.42438; 613 2.55 909.9 2.42438
118. 614 2.8 909.9 2.42438; 615 2.3 909.9 2.67438; 616 2.55 909.9 2.67438
119. 617 2.8 909.9 2.67438; 618 2.3 909.9 2.92438; 619 2.55 909.9 2.92438
120. 620 2.8 909.9 2.92438; 621 3.05 909.9 3.42438; 622 3.05 909.9 3.175
121. 623 3.05 909.9 3.67438; 624 3.05 909.9 3.92438; 625 3.05 909.9 4.17438
122. 626 3.05 909.9 4.42438; 627 3.05 909.9 2.17438; 628 3.05 909.9 1.925
123. 629 3.05 909.9 2.42438; 630 3.05 909.9 2.67438; 631 3.05 909.9 2.92438
124. 632 2.3 909.9 4.67438; 633 2.55 909.9 4.67438; 634 2.8 909.9 4.67438
125. 635 3.05 909.9 4.67438; 636 2.3 909.9 1.675; 637 2.55 909.9 1.675
126. 638 2.8 909.9 1.675; 639 3.05 909.9 1.675; 640 3.3 909.9 3.42438
127. 641 3.3 909.9 3.175; 642 3.3 909.9 3.67438; 643 3.3 909.9 3.92438
128. 644 3.3 909.9 4.17438; 645 3.3 909.9 4.42438; 646 3.3 909.9 2.17438
129. 647 3.3 909.9 1.925; 648 3.3 909.9 2.42438; 649 3.3 909.9 2.67438
130. 650 3.3 909.9 2.92438; 651 3.3 909.9 4.67438; 652 3.3 909.9 1.675
131. 653 2.3 909.9 1.425; 654 2.55 909.9 1.425; 655 2.8 909.9 1.425
132. 656 3.05 909.9 1.425; 657 3.3 909.9 1.425; 658 2.3 909.9 4.92438
133. 659 2.55 909.9 4.92438; 660 2.8 909.9 4.92438; 661 3.05 909.9 4.92438
134. 662 3.3 909.9 4.92438; 663 3.55 909.9 3.42438; 664 3.55 909.9 3.175
135. 665 3.55 909.9 3.67438; 666 3.55 909.9 3.92438; 667 3.55 909.9 4.17438
136. 668 3.55 909.9 4.42438; 669 3.55 909.9 2.17438; 670 3.55 909.9 1.925
137. 671 3.55 909.9 2.42438; 672 3.55 909.9 2.67438; 673 3.55 909.9 2.92438
138. 674 3.55 909.9 4.67438; 675 3.55 909.9 1.675; 676 3.55 909.9 1.425
139. 677 3.55 909.9 4.92438; 678 2.3 909.9 1.175; 679 2.55 909.9 1.175
140. 680 2.8 909.9 1.175; 681 3.05 909.9 1.175; 682 3.3 909.9 1.175
141. 683 3.55 909.9 1.175; 684 2.3 909.9 5.17438; 685 2.55 909.9 5.17438
142. 686 2.8 909.9 5.17438; 687 3.05 909.9 5.17438; 688 3.3 909.9 5.17438
143. 689 3.55 909.9 5.17438; 690 -2.25 909.9 3.42438; 691 -2.25 909.9 3.175
144. 692 -2.75 909.9 3.42438; 693 -2.5 909.9 3.42438; 694 -2.5 909.9 3.175
145. 695 -2.75 909.9 3.175; 696 -2.25 909.9 3.67438; 697 -2.75 909.9 3.67438
146. 698 -2.5 909.9 3.67438; 699 -2.25 909.9 3.92438; 700 -2.75 909.9 3.92438
147. 701 -2.5 909.9 3.92438; 702 -2.25 909.9 4.17438; 703 -2.75 909.9 4.17438
148. 704 -2.5 909.9 4.17438; 705 -2.25 909.9 4.42438; 706 -2.75 909.9 4.42438
149. 707 -2.5 909.9 4.42438; 708 -2.25 909.9 2.17438; 709 -2.25 909.9 1.925
150. 710 -2.75 909.9 2.17438; 711 -2.5 909.9 2.17438; 712 -2.5 909.9 1.925

151. 713 -2.75 909.9 1.925; 714 -2.25 909.9 2.42438; 715 -2.75 909.9 2.42438
152. 716 -2.5 909.9 2.42438; 717 -2.25 909.9 2.67438; 718 -2.75 909.9 2.67438
153. 719 -2.5 909.9 2.67438; 720 -2.25 909.9 2.92438; 721 -2.75 909.9 2.92438
154. 722 -2.5 909.9 2.92438; 723 -3 909.9 3.42438; 724 -3 909.9 3.175
155. 725 -3 909.9 3.67438; 726 -3 909.9 3.92438; 727 -3 909.9 4.17438
156. 728 -3 909.9 4.42438; 729 -3 909.9 2.17438; 730 -3 909.9 1.925
157. 731 -3 909.9 2.42438; 732 -3 909.9 2.67438; 733 -3 909.9 2.92438
158. 734 -2.25 909.9 4.67438; 735 -2.75 909.9 4.67438; 736 -2.5 909.9 4.67438
159. 737 -3 909.9 4.67438; 738 -2.25 909.9 1.675; 739 -2.5 909.9 1.675
160. 740 -2.75 909.9 1.675; 741 -3 909.9 1.675; 742 -3.25 909.9 3.42438
161. 743 -3.25 909.9 3.175; 744 -3.25 909.9 3.67438; 745 -3.25 909.9 3.92438
162. 746 -3.25 909.9 4.17438; 747 -3.25 909.9 4.42438; 748 -3.25 909.9 2.17438
163. 749 -3.25 909.9 1.925; 750 -3.25 909.9 2.42438; 751 -3.25 909.9 2.67438
164. 752 -3.25 909.9 2.92438; 753 -3.25 909.9 4.67438; 754 -3.25 909.9 1.675
165. 755 -2.25 909.9 1.425; 756 -2.5 909.9 1.425; 757 -2.75 909.9 1.425
166. 758 -3 909.9 1.425; 759 -3.25 909.9 1.425; 760 -2.25 909.9 4.92438
167. 761 -2.75 909.9 4.92438; 762 -2.5 909.9 4.92438; 763 -3 909.9 4.92438
168. 764 -3.25 909.9 4.92438; 765 -3.5 909.9 3.42438; 766 -3.5 909.9 3.175
169. 767 -3.5 909.9 3.67438; 768 -3.5 909.9 3.92438; 769 -3.5 909.9 4.17438
170. 770 -3.5 909.9 4.42438; 771 -3.5 909.9 2.17438; 772 -3.5 909.9 1.925
171. 773 -3.5 909.9 2.42438; 774 -3.5 909.9 2.67438; 775 -3.5 909.9 2.92438
172. 776 -3.5 909.9 4.67438; 777 -3.5 909.9 1.675; 778 -3.5 909.9 1.425
173. 779 -3.5 909.9 4.92438; 780 -2.25 909.9 1.175; 781 -2.5 909.9 1.175
174. 782 -2.75 909.9 1.175; 783 -3 909.9 1.175; 784 -3.25 909.9 1.175
175. 785 -3.5 909.9 1.175; 786 -2.25 909.9 5.17438; 787 -2.75 909.9 5.17438
176. 788 -2.5 909.9 5.17438; 789 -3 909.9 5.17438; 790 -3.25 909.9 5.17438
177. 791 -3.5 909.9 5.17438; 792 3.75 909.9 3.42438; 793 3.75 909.9 3.175
178. 794 3.75 909.9 3.67438; 795 3.75 909.9 3.92438; 796 3.75 909.9 4.17438
179. 797 3.75 909.9 4.42438; 798 3.75 909.9 2.17438; 799 3.75 909.9 1.925
180. 800 3.75 909.9 2.42438; 801 3.75 909.9 2.67438; 802 3.75 909.9 2.92438
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883	0.062500

STAAD SPACE -- PAGE NO. 21

884	0.062500
885	0.062500
886	0.062500
887	0.062500
888	0.068750
889	0.062500
890	0.068750
891	0.062500
892	0.062500
893	0.062500
894	0.062500
895	0.062500
896	0.062500
897	0.062500
898	0.062500
899	0.056250
900	0.062500
901	0.062500
902	0.062500
903	0.062500
904	0.062500
905	0.056250
906	0.062500
907	0.062500
908	0.062500
909	0.062500
910	0.062500
911	0.062500
912	0.062500
913	0.062500
914	0.062500
915	0.062500
916	0.062500
917	0.062500
918	0.025000
920	0.025000
922	0.031250
924	0.031250
926	0.062500
927	0.062500
928	0.062500
929	0.062500
930	0.062500
931	0.062500
932	0.062500
933	0.062500
934	0.062500
935	0.062500
936	0.062500
937	0.068750
938	0.062500
939	0.068750
940	0.062500
941	0.062500
942	0.062500
943	0.062500

STAAD SPACE -- PAGE NO. 22

944	0.062500
945	0.062500
946	0.062500
947	0.062500
948	0.062500
949	0.062500
950	0.062500
951	0.062500
952	0.062500
953	0.062500
954	0.068750
955	0.062500
956	0.068750
957	0.062500
958	0.062500
959	0.062500
960	0.062500
961	0.062500
962	0.062500
963	0.062500
964	0.062500
965	0.056250
966	0.062500
967	0.062500
968	0.062500
969	0.062500
970	0.062500
971	0.056250
972	0.062500
973	0.062500
974	0.062500
975	0.062500
976	0.062500
977	0.062500
978	0.062500
979	0.062500
980	0.062500
981	0.062500
982	0.062500
983	0.062500
984	0.025000
986	0.025000
988	0.031250
990	0.031250
991	0.062500
992	0.062500
993	0.062500
994	0.062500
995	0.062500
996	0.062500
997	0.062500
998	0.062500
999	0.062500
1000	0.062500
1001	0.062500
1002	0.068750

1003	0.062500
1004	0.068750
1005	0.062500
1006	0.062500
1007	0.062500
1008	0.031250
1009	0.031250
1010	0.031250
1011	0.031250
1012	0.031250
1013	0.031250
1014	0.031250
1015	0.031250
1016	0.031250
1017	0.031250
1018	0.031250
1019	0.034375
1020	0.031250
1021	0.034375
1022	0.031250
1023	0.031250
1024	0.031250
1025	0.062500
1026	0.062500
1027	0.062500
1028	0.062500
1029	0.062500
1030	0.056250
1031	0.031250
1032	0.031250
1033	0.031250
1034	0.031250
1035	0.031250
1036	0.028125
1037	0.062500
1038	0.062500
1039	0.062500
1040	0.062500
1041	0.062500
1042	0.062500
1043	0.031250
1044	0.031250
1045	0.031250
1046	0.031250
1047	0.031250
1048	0.031250
1049	0.025000
1050	0.012500
1051	0.031250
1052	0.015625
1053	0.062500
1054	0.062500
1055	0.062500
1056	0.062500
1057	0.062500
1058	0.062500

1059	0.062500
1060	0.062500
1061	0.062500
1062	0.062500
1063	0.062500
1064	0.068750
1065	0.062500
1066	0.068750
1067	0.062500
1068	0.062500
1069	0.062500
1070	0.031250
1071	0.031250
1072	0.031250
1073	0.031250
1074	0.031250
1075	0.031250
1076	0.031250
1077	0.031250
1078	0.031250
1079	0.031250
1080	0.031250
1081	0.034375
1082	0.031250
1083	0.034375
1084	0.031250
1085	0.031250
1086	0.031250
1087	0.062500
1088	0.062500
1089	0.062500
1090	0.062500
1091	0.062500
1092	0.056250
1093	0.031250
1094	0.031250
1095	0.031250
1096	0.031250
1097	0.031250
1098	0.028125
1099	0.062500
1100	0.062500
1101	0.062500
1102	0.062500
1103	0.062500
1104	0.062500
1105	0.031250
1106	0.031250
1107	0.031250
1108	0.031250
1109	0.031250
1110	0.031250
1111	0.025000
1112	0.012500
1113	0.031250
1114	0.015625

```
    TOTAL          44.995335
548. ********************************************************************************
549. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
550. SELFWEIGHT Y -1
551. ELEMENT LOAD
552. * LOAD DUE TO BACKFILL
553. 326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
554. 845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 PR GY -40
555. * LOAD DUE TO SURCHARGE
556. 326 TO 328 332 TO 334 337 338 341 TO 794 809 811 813 TO 826 841 843 -
557. 845 TO 901 903 905 907 909 TO 965 967 969 971 TO 1091 PR GY -40
558. JOINT LOAD
559. 128 FX 11.9 FY -391.7 MZ -88.6
560. 1115 FX 7.7 FY -79.57
561. ********************************************************************************
562. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
563. JOINT LOAD
564. 128 FX 13.5 FY -144.5 MZ -162
565. ********************************************************************************
566. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
567. JOINT LOAD
568. 128 FX 18.2 FY -325.7 MZ -120
569. 1115 FX 22.9 FY -198.75
570. ********************************************************************************
571. LOAD 4 LOADTYPE NONE  TITLE CRANE LOAD  ( CR1 )
572. JOINT LOAD
573. 128 FX 14.7 FY -140.5 MZ -214.9
574. ********************************************************************************
575. LOAD 5 LOADTYPE NONE  TITLE CRANE LOAD  ( CR2 )
576. JOINT LOAD
577. 128 FX 14.8 FY -38.6 MZ -254.2
578. ********************************************************************************
579. LOAD 6 LOADTYPE NONE  TITLE CRANE LOAD  ( CR3 )
580. JOINT LOAD
581. 128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
582. ********************************************************************************
583. LOAD 7 LOADTYPE NONE  TITLE CRANE LOAD  ( CR4 )
584. JOINT LOAD
585. 128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
586. ********************************************************************************
587. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
588. JOINT LOAD
589. 128 FX 313.7 FY 142.4 MZ -3180.7
590. 1115 FX 67.7 FY 38.4
591. ********************************************************************************
592. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
593. JOINT LOAD
594. 128 FX 188.5 FY 57.5 MZ -2560.2
595. 1115 FX 67.7 FY 38.4
596. ********************************************************************************
597. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
598. JOINT LOAD
599. 128 FX 344.4 FY 21.3 MZ -3618.7
600. 1115 FX 67.7 FY 38.4
601. ********************************************************************************
602. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
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603. JOINT LOAD
604. 128 FX 216.7 FY -60.9 MZ -2951.2
605. 1115 FX 67.7 FY 38.4
606. ********************************************************************************
607. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
608. JOINT LOAD
609. 128 FX -352.7 FY 276.8 MZ 3895.1
610. 1115 FX -67.7 FY -38.4
611. ********************************************************************************
612. LOAD 13 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
613. JOINT LOAD
614. 128 FX -480.5 FY 195.6 MZ 4591.3
615. 1115 FX -67.7 FY -38.4
616. ********************************************************************************
617. LOAD 14 LOADTYPE NONE  TITLE WIND LOAD  ( WL7 )
618. JOINT LOAD
619. 128 FX -331.7 FY 129.5 MZ 3703
620. 1115 FX -67.7 FY -38.4
621. ********************************************************************************
622. LOAD 15 LOADTYPE NONE  TITLE WIND LOAD  ( WL8 )
623. JOINT LOAD
624. 128 FX -459.4 FY 48.4 MZ 4399.4
625. 1115 FX -67.7 FY -38.4
626. ********************************************************************************
627. LOAD 16 LOADTYPE NONE  TITLE WIND LOAD  ( WL9 )
628. JOINT LOAD
629. 128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
630. 1115 FX 6.4 FY 0.5 FZ -2
631. ********************************************************************************
632. LOAD 17 LOADTYPE NONE  TITLE WIND LOAD  ( WL10 )
633. JOINT LOAD
634. 128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
635. 1115 FX 6.4 FY 0.5 FZ -2
636. ********************************************************************************
637. LOAD 18 LOADTYPE NONE  TITLE WIND LOAD  ( WL11 )
638. JOINT LOAD
639. 128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
640. 1115 FX 6.4 FY -0.5 FZ 2
641. ********************************************************************************
642. LOAD 19 LOADTYPE NONE  TITLE WIND LOAD  ( WL12 )
643. JOINT LOAD
644. 128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
645. 1115 FX 6.4 FY -0.5 FZ 2
646. ********************************************************************************
647. LOAD 20 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
648. JOINT LOAD
649. 128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
650. ********************************************************************************
651. LOAD 21 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
652. JOINT LOAD
653. 128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
654. ********************************************************************************
655. LOAD 22 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
656. JOINT LOAD
657. 128 FX 1.1 FY 0 MZ -23.9
658. ********************************************************************************
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659. LOAD 23 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
660. JOINT LOAD
661. 128 FX -1.1 FY 0 MZ 23.9
662. **************************************************************************
663. * UNFACTORED LOAD COMBINATIONS
664. LOAD COMB 201 DL
665. 1 1.0
666. LOAD COMB 202 DL+L+TL1
667. 1 1.0 3 1.0 22 1.0
668. LOAD COMB 203 DL+L+TL2
669. 1 1.0 3 1.0 23 1.0
670. LOAD COMB 204 DL+LR
671. 1 1.0 2 1.0
672. LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
673. 1 1.0 2 0.75 3 0.75 22 0.75
674. LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
675. 1 1.0 2 0.75 3 0.75 23 0.75
676. LOAD COMB 207 DL+0.625WL1
677. 1 1.0 8 0.625
678. LOAD COMB 208 DL+0.625WL2
679. 1 1.0 9 0.625
680. LOAD COMB 209 DL+0.625WL3
681. 1 1.0 10 0.625
682. LOAD COMB 210 DL+0.625WL4
683. 1 1.0 11 0.625
684. LOAD COMB 211 DL+0.625WL5
685. 1 1.0 12 0.625
686. LOAD COMB 212 DL+0.625WL6
687. 1 1.0 13 0.625
688. LOAD COMB 213 DL+0.625WL7
689. 1 1.0 14 0.625
690. LOAD COMB 214 DL+0.625WL8
691. 1 1.0 15 0.625
692. LOAD COMB 215 DL+0.625WL9
693. 1 1.0 16 0.625
694. LOAD COMB 216 DL+0.625WL10
695. 1 1.0 17 0.625
696. LOAD COMB 217 DL+0.625WL11
697. 1 1.0 18 0.625
698. LOAD COMB 218 DL+0.625WL12
699. 1 1.0 19 0.625
700. LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
701. 1 1.0 2 0.75 3 0.75 8 0.75
702. LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
703. 1 1.0 2 0.75 3 0.75 9 0.75
704. LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
705. 1 1.0 2 0.75 3 0.75 10 0.75
706. LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
707. 1 1.0 2 0.75 3 0.75 11 0.75
708. LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
709. 1 1.0 2 0.75 3 0.75 12 0.75
710. LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
711. 1 1.0 2 0.75 3 0.75 13 0.75
712. LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
713. 1 1.0 2 0.75 3 0.75 14 0.75
714. LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
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715. 1 1.0 2 0.75 3 0.75 15 0.75
716. LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
717. 1 1.0 2 0.75 3 0.75 16 0.75
718. LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
719. 1 1.0 2 0.75 3 0.75 17 0.75
720. LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
721. 1 1.0 2 0.75 3 0.75 18 0.75
722. LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
723. 1 1.0 2 0.75 3 0.75 19 0.75
724. LOAD COMB 231 0.6DL+0.625WL1
725. 1 0.6 8 0.625
726. LOAD COMB 232 0.6DL+0.625WL2
727. 1 0.6 9 0.625
728. LOAD COMB 233 0.6DL+0.625WL3
729. 1 0.6 10 0.625
730. LOAD COMB 234 0.6DL+0.625WL4
731. 1 0.6 11 0.625
732. LOAD COMB 235 0.6DL+0.625WL5
733. 1 0.6 12 0.625
734. LOAD COMB 236 0.6DL+0.625WL6
735. 1 0.6 13 0.625
736. LOAD COMB 237 0.6DL+0.625WL7
737. 1 0.6 14 0.625
738. LOAD COMB 238 0.6DL+0.625WL8
739. 1 0.6 15 0.625
740. LOAD COMB 239 0.6DL+0.625WL9
741. 1 0.6 16 0.625
742. LOAD COMB 240 0.6DL+0.625WL10
743. 1 0.6 17 0.625
744. LOAD COMB 241 0.6DL+0.625WL11
745. 1 0.6 18 0.625
746. LOAD COMB 242 0.6DL+0.625WL12
747. 1 0.6 19 0.625
748. LOAD COMB 243 0.6DL+0.7SL1
749. 1 0.6 20 0.7
750. LOAD COMB 244 0.6DL+0.7SL2
751. 1 0.6 21 0.7
752. LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
753. 1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
754. LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
755. 1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
756. LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
757. 1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
758. LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
759. 1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
760. LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
761. 1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
762. LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
763. 1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
764. LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
765. 1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
766. LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
767. 1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
768. LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
769. 1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
770. LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
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771. 1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
772. LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
773. 1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
774. LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
775. 1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
776. LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
777. 1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
778. LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
779. 1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
780. LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
781. 1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
782. LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
783. 1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
784. LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
785. 1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
786. LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
787. 1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
788. LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
789. 1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
790. LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
791. 1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
792. LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
793. 1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
794. LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
795. 1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
796. LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
797. 1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
798. LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
799. 1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
800. LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
801. 1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
802. LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
803. 1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
804. LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
805. 1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
806. LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
807. 1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
808. LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
809. 1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
810. LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
811. 1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
812. LOAD COMB 275 DL+0.75WL7+0.75L+0.75LR+CR3
813. 1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
814. LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
815. 1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
816. LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
817. 1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
818. LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
819. 1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
820. LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
821. 1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
822. LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
823. 1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
824. LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
825. 1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
826. LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
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827. 1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
828. LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
829. 1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
830. LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
831. 1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
832. LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
833. 1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
834. LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
835. 1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
836. LOAD COMB 287 DL+0.75WL7+0.75L+0.75LR+CR4
837. 1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
838. LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
839. 1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
840. LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
841. 1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
842. LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
843. 1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
844. LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
845. 1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
846. LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
847. 1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
848. ********************************************************************************
849. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
850. LOAD COMB 101 1.4DL
851. 1 1.4
852. LOAD COMB 102 1.4DL+1.4TL1+1.7L+0.5LR
853. 1 1.4 22 1.4 3 1.7 2 0.5
854. LOAD COMB 103 1.4DL+1.4TL2+1.7L+0.5LR
855. 1 1.4 23 1.4 3 1.7 2 0.5
856. LOAD COMB 104 1.2DL+1.0LR+1.0L
857. 1 1.2 2 1.0 3 1.0
858. LOAD COMB 105 1.2DL+1.0WL1+1.0L+0.5LR
859. 1 1.2 8 1.0 2 0.5 3 1.0
860. LOAD COMB 106 1.2DL+1.0WL2+0.5L+0.5LR
861. 1 1.2 9 1.0 2 0.5 3 1.0
862. LOAD COMB 107 1.2DL+1.0WL3+0.5L+0.5LR
863. 1 1.2 10 1.0 2 0.5 3 1.0
864. LOAD COMB 108 1.2DL+1.0WL4+0.5L+0.5LR
865. 1 1.2 11 1.0 2 0.5 3 1.0
866. LOAD COMB 109 1.2DL+1.0WL5+0.5L+0.5LR
867. 1 1.2 12 1.0 2 0.5 3 1.0
868. LOAD COMB 110 1.2DL+1.0WL6+0.5L+0.5LR
869. 1 1.2 13 1.0 2 0.5 3 1.0
870. LOAD COMB 111 1.2DL+1.0WL7+0.5L+0.5LR
871. 1 1.2 14 1.0 2 0.5 3 1.0
872. LOAD COMB 112 1.2DL+1.0WL8+0.5L+0.5LR
873. 1 1.2 15 1.0 2 0.5 3 1.0
874. LOAD COMB 113 1.2DL+1.0WL9+0.5L+0.5LR
875. 1 1.2 16 1.0 2 0.5 3 1.0
876. LOAD COMB 114 1.2DL+1.0WL10+0.5L+0.5LR
877. 1 1.2 17 1.0 2 0.5 3 1.0
878. LOAD COMB 115 1.2DL+1.0WL11+0.5L+0.5LR
879. 1 1.2 18 1.0 2 0.5 3 1.0
880. LOAD COMB 116 1.2DL+1.0WL12+0.5L+0.5LR
881. 1 1.2 19 1.0 2 0.5 3 1.0
882. LOAD COMB 117 1.2DL+1.0SL1+1.0L
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883. 1 1.2 20 1.0 3 1.0
884. LOAD COMB 118 1.2DL+1.0SL2+1.0L
885. 1 1.2 21 1.0 3 1.0
886. LOAD COMB 119 0.9DL+1.0WL1
887. 1 0.9 8 1.0
888. LOAD COMB 120 0.9DL+1.0WL2
889. 1 0.9 9 1.0
890. LOAD COMB 121 0.9DL+1.0WL3
891. 1 0.9 10 1.0
892. LOAD COMB 122 0.9DL+1.0WL4
893. 1 0.9 11 1.0
894. LOAD COMB 123 0.9DL+1.0WL5
895. 1 0.9 12 1.0
896. LOAD COMB 124 0.9DL+1.0WL6
897. 1 0.9 13 1.0
898. LOAD COMB 125 0.9DL+1.0WL7
899. 1 0.9 14 1.0
900. LOAD COMB 126 0.9DL+1.0WL8
901. 1 0.9 15 1.0
902. LOAD COMB 127 0.9DL+1.0WL9
903. 1 0.9 16 1.0
904. LOAD COMB 128 0.9DL+1.0WL10
905. 1 0.9 17 1.0
906. LOAD COMB 129 0.9DL+1.0WL11
907. 1 0.9 18 1.0
908. LOAD COMB 130 0.9DL+1.0WL12
909. 1 0.9 19 1.0
910. LOAD COMB 131 0.9DL+1.0SL1
911. 1 0.9 20 1.0
912. LOAD COMB 132 0.9DL+1.0SL2
913. 1 0.9 21 1.0
914. LOAD COMB 133 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR1
915. 1 1.2 8 1.0 2 0.5 3 1.0 4 1.0
916. LOAD COMB 134 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR1
917. 1 1.2 9 1.0 2 0.5 3 1.0 4 1.0
918. LOAD COMB 135 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR1
919. 1 1.2 10 1.0 2 0.5 3 1.0 4 1.0
920. LOAD COMB 136 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR1
921. 1 1.2 11 1.0 2 0.5 3 1.0 4 1.0
922. LOAD COMB 137 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR1
923. 1 1.2 12 1.0 2 0.5 3 1.0 4 1.0
924. LOAD COMB 138 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR1
925. 1 1.2 13 1.0 2 0.5 3 1.0 4 1.0
926. LOAD COMB 139 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR1
927. 1 1.2 14 1.0 2 0.5 3 1.0 4 1.0
928. LOAD COMB 140 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR1
929. 1 1.2 15 1.0 2 0.5 3 1.0 4 1.0
930. LOAD COMB 141 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR1
931. 1 1.2 16 1.0 2 0.5 3 1.0 4 1.0
932. LOAD COMB 142 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR1
933. 1 1.2 17 1.0 2 0.5 3 1.0 4 1.0
934. LOAD COMB 143 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR1
935. 1 1.2 18 1.0 2 0.5 3 1.0 4 1.0
936. LOAD COMB 144 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR1
937. 1 1.2 19 1.0 2 0.5 3 1.0 4 1.0
938. LOAD COMB 145 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR2
```



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939. 1 1.2 8 1.0 2 0.5 3 1.0 5 1.0
940. LOAD COMB 146 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR2
941. 1 1.2 9 1.0 2 0.5 3 1.0 5 1.0
942. LOAD COMB 147 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR2
943. 1 1.2 10 1.0 2 0.5 3 1.0 5 1.0
944. LOAD COMB 148 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR2
945. 1 1.2 11 1.0 2 0.5 3 1.0 5 1.0
946. LOAD COMB 149 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR2
947. 1 1.2 12 1.0 2 0.5 3 1.0 5 1.0
948. LOAD COMB 150 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR2
949. 1 1.2 13 1.0 2 0.5 3 1.0 5 1.0
950. LOAD COMB 151 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR2
951. 1 1.2 14 1.0 2 0.5 3 1.0 5 1.0
952. LOAD COMB 152 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR2
953. 1 1.2 15 1.0 2 0.5 3 1.0 5 1.0
954. LOAD COMB 153 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR2
955. 1 1.2 16 1.0 2 0.5 3 1.0 5 1.0
956. LOAD COMB 154 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR2
957. 1 1.2 17 1.0 2 0.5 3 1.0 5 1.0
958. LOAD COMB 155 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR2
959. 1 1.2 18 1.0 2 0.5 3 1.0 5 1.0
960. LOAD COMB 156 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR2
961. 1 1.2 19 1.0 2 0.5 3 1.0 5 1.0
962. LOAD COMB 157 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR3
963. 1 1.2 8 1.0 2 0.5 3 1.0 6 1.0
964. LOAD COMB 158 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR3
965. 1 1.2 9 1.0 2 0.5 3 1.0 6 1.0
966. LOAD COMB 159 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR3
967. 1 1.2 10 1.0 2 0.5 3 1.0 6 1.0
968. LOAD COMB 160 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR3
969. 1 1.2 11 1.0 2 0.5 3 1.0 6 1.0
970. LOAD COMB 161 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR3
971. 1 1.2 12 1.0 2 0.5 3 1.0 6 1.0
972. LOAD COMB 162 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR3
973. 1 1.2 13 1.0 2 0.5 3 1.0 6 1.0
974. LOAD COMB 163 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR3
975. 1 1.2 14 1.0 2 0.5 3 1.0 6 1.0
976. LOAD COMB 164 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR3
977. 1 1.2 15 1.0 2 0.5 3 1.0 6 1.0
978. LOAD COMB 165 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR3
979. 1 1.2 16 1.0 2 0.5 3 1.0 6 1.0
980. LOAD COMB 166 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR3
981. 1 1.2 17 1.0 2 0.5 3 1.0 6 1.0
982. LOAD COMB 167 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR3
983. 1 1.2 18 1.0 2 0.5 3 1.0 6 1.0
984. LOAD COMB 168 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR3
985. 1 1.2 19 1.0 2 0.5 3 1.0 6 1.0
986. LOAD COMB 169 1.2DL+1.0WL1+1.0L+0.5LR+1.0CR4
987. 1 1.2 8 1.0 2 0.5 3 1.0 7 1.0
988. LOAD COMB 170 1.2DL+1.0WL2+1.0L+0.5LR+1.0CR4
989. 1 1.2 9 1.0 2 0.5 3 1.0 7 1.0
990. LOAD COMB 171 1.2DL+1.0WL3+1.0L+0.5LR+1.0CR4
991. 1 1.2 10 1.0 2 0.5 3 1.0 7 1.0
992. LOAD COMB 172 1.2DL+1.0WL4+1.0L+0.5LR+1.0CR4
993. 1 1.2 11 1.0 2 0.5 3 1.0 7 1.0
994. LOAD COMB 173 1.2DL+1.0WL5+1.0L+0.5LR+1.0CR4
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 995. 1 1.2 12 1.0 2 0.5 3 1.0 7 1.0
 996. LOAD COMB 174 1.2DL+1.0WL6+1.0L+0.5LR+1.0CR4
 997. 1 1.2 13 1.0 2 0.5 3 1.0 7 1.0
 998. LOAD COMB 175 1.2DL+1.0WL7+1.0L+0.5LR+1.0CR4
 999. 1 1.2 14 1.0 2 0.5 3 1.0 7 1.0
1000. LOAD COMB 176 1.2DL+1.0WL8+1.0L+0.5LR+1.0CR4
1001. 1 1.2 15 1.0 2 0.5 3 1.0 7 1.0
1002. LOAD COMB 177 1.2DL+1.0WL9+1.0L+0.5LR+1.0CR4
1003. 1 1.2 16 1.0 2 0.5 3 1.0 7 1.0
1004. LOAD COMB 178 1.2DL+1.0WL10+1.0L+0.5LR+1.0CR4
1005. 1 1.2 17 1.0 2 0.5 3 1.0 7 1.0
1006. LOAD COMB 179 1.2DL+1.0WL11+1.0L+0.5LR+1.0CR4
1007. 1 1.2 18 1.0 2 0.5 3 1.0 7 1.0
1008. LOAD COMB 180 1.2DL+1.0WL12+1.0L+0.5LR+1.0CR4
1009. 1 1.2 19 1.0 2 0.5 3 1.0 7 1.0
1010. ********************************************************************************
1011. PERFORM ANALYSIS


         P R O B L E M   S T A T I S T I C S
         -----------------------------------

   NUMBER OF JOINTS        777  NUMBER OF MEMBERS       2
   NUMBER OF PLATES        720  NUMBER OF SOLIDS        0
   NUMBER OF SURFACES        0  NUMBER OF SUPPORTS    775


        SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

   ORIGINAL/FINAL BAND-WIDTH=   605/    43/    135 DOF
 TOTAL      PRIMARY LOAD CASES =     23, TOTAL DEGREES OF FREEDOM =    2337
 TOTAL LOAD COMBINATION  CASES =    172  SO FAR.
   SIZE OF STIFFNESS MATRIX =      316 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =     19.9/ 145074.0 MB



1012. LOAD LIST 101 TO 180
1013. START CONCRETE DESIGN
```



```
1014. CODE ACI
1015. TRACK 2 ALL
1016. FC 35000 MEMB 185 1092
1017. CLB 0.05 MEMB 185 1092
1018. CLS 0.05 ALL
1019. CLT 0.05 MEMB 185 1092
1020. MAXMAIN 25 ALL
1021. MINMAIN 25 ALL
1022. MINSEC 10 ALL
1023. DESIGN COLUMN 185 1092
```

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE -1500.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 30000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
64 - 25 MM	1.047	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
********	81049.27	41239.41	30170.30	731.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
12092.92	********	843.66	190.83	0.06463

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
********	81049.27	40737.83	22458.28	551.3
M0	P-tens.	Des.Pn	Des.Mn	e/h
8945.85	********	843.66	0.00	0.00000

	Pn	Mn	Pn	Mn	(@ Z)
	74814.72	20123.70	37407.36	29937.03	
P0	68580.16	23184.17	31172.80	28933.22	
	62345.60	25620.64	24938.24	27137.43	
Pn,max	56111.04	27497.02	18703.68	24563.98	
	49876.48	28886.24	12469.12	21198.32	
Pn	43641.91	29873.28	6234.56	17050.16	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	74814.72	15146.45	37407.36	22297.34	
COMPRESSION	68580.16	17286.16	31172.80	21546.63	
Pb	62345.60	19092.63	24938.24	20207.47	
	56111.04	20478.86	18703.68	18284.91	
	49876.48	21493.30	12469.12	15770.77	
M0 Mn,	43641.91	22206.51	6234.56	12672.19	
BENDING					
P-tens MOMENT					

COLUMN NO. 1092 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 25 MM	1.202	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
16839.05	13471.24	6253.35	1726.04	276.0
M0	P-tens.	Des.Pn	Des.Mn	e/h
741.07	-2436.79	171.38	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
16839.05	13471.24	6253.35	1726.04	276.0
M0	P-tens.	Des.Pn	Des.Mn	e/h
741.07	-2436.79	171.38	0.00	0.00000

	Pn	Mn	Pn	Mn	(@ Z)
	12434.99	1145.49	6217.49	1724.81	
P0	11398.74	1312.93	5181.25	1667.55	
	10362.49	1450.83	4145.00	1582.87	
Pn,max	9326.24	1556.76	3108.75	1440.00	
	8289.99	1634.97	2072.50	1252.81	
Pn	7253.74	1688.98	1036.25	1023.18	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	12434.99	1145.49	6217.49	1724.81	
COMPRESSION	11398.74	1312.93	5181.25	1667.55	
Pb	10362.49	1450.83	4145.00	1582.87	
	9326.24	1556.76	3108.75	1440.00	
	8289.99	1634.97	2072.50	1252.81	
M0 Mn,	7253.74	1688.98	1036.25	1023.18	
BENDING					
P-tens MOMENT					

**********END OF COLUMN DESIGN RESULTS**********

```
1024. END CONCRETE DESIGN
1025. FINISH


           ********** END OF THE STAAD.Pro RUN **********

             **** DATE= JUL 13,2020   TIME= 11:16:52 ****

       ************************************************************
       *   For technical assistance on STAAD.Pro, please visit    *
       *   http://selectservices.bentley.com/en-US/               *
       *                                                          *
       *   Details about additional assistance from               *
       *   Bentley and Partners can be found at program menu      *
       *   Help->Technical Support                                *
       *                                                          *
       *      Copyright (c) 1997-2014 Bentley Systems, Inc.
       *                http://www.bentley.com                    *
       ************************************************************
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            *****************************************************
            *                                                   *
            *           STAAD.Pro V8i SELECTseries5             *
            *           Version  20.07.10.64                    *
            *           Proprietary Program of                  *
            *           Bentley Systems, Inc.                   *
            *           Date=    JUL 13, 2020                   *
            *           Time=    12:14: 1                       *
            *                                                   *
            *      USER ID: LNT ECC DIV                         *
            *****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F3 _STABILITY - 304.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 28/05/2020
     4. JOB NAME FDN F3_STABILITY CHECK
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. END JOB INFORMATION
    10. INPUT WIDTH 79
    11. UNIT METER KN
    12. JOINT COORDINATES
    13. 2 0 909.9 3.175; 128 0 913.4 3.175; 469 1.8 909.9 3.175; 1115 1.8 913.6 3.175
    14. MEMBER INCIDENCES
    15. 185 2 128; 1092 469 1115
    16. DEFINE MATERIAL START
    17. ISOTROPIC CONCRETE
    18. E 2.17185E+007
    19. POISSON 0.17
    20. DENSITY 25
    21. ALPHA 1E-005
    22. DAMP 0.05
    23. TYPE CONCRETE
    24. STRENGTH FCU 27579
    25. END DEFINE MATERIAL
    26. MEMBER PROPERTY AMERICAN
    27. 185 PRIS YD 2 ZD 1.5
    28. MEMBER PROPERTY AMERICAN
    29. 1092 PRIS YD 0.7 ZD 0.7
    30. CONSTANTS
    31. MATERIAL CONCRETE ALL
    32. SUPPORTS
    33. 2 469 FIXED
    34. ******************************************************************************
    35. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
  **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
            MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    36. SELFWEIGHT Y -1
```



```
    37. JOINT LOAD
    38. 128 FX 11.9 FY -391.7 MZ -88.6
    39. 1115 FX 7.7 FY -79.57
    40. ******************************************************************************
    41. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
    42. JOINT LOAD
    43. 128 FX 13.5 FY -144.5 MZ -162
    44. ******************************************************************************
    45. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
    46. JOINT LOAD
    47. 128 FX 18.2 FY -325.7 MZ -120
    48. 1115 FX 22.9 FY -198.75
    49. ******************************************************************************
    50. LOAD 4 LOADTYPE NONE  TITLE CRANE LOAD  ( CR1 )
    51. JOINT LOAD
    52. 128 FX 14.7 FY -140.5 MZ -214.9
    53. ******************************************************************************
    54. LOAD 5 LOADTYPE NONE  TITLE CRANE LOAD  ( CR2 )
    55. JOINT LOAD
    56. 128 FX 14.8 FY -38.6 MZ -254.2
    57. ******************************************************************************
    58. LOAD 6 LOADTYPE NONE  TITLE CRANE LOAD  ( CR3 )
    59. JOINT LOAD
    60. 128 FX -8.4 FY 12.1 FZ -11 MZ 157.2
    61. ******************************************************************************
    62. LOAD 7 LOADTYPE NONE  TITLE CRANE LOAD  ( CR4 )
    63. JOINT LOAD
    64. 128 FX -8.4 FY -173.6 FZ 11 MZ 196.3
    65. ******************************************************************************
    66. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
    67. JOINT LOAD
    68. 128 FX 313.7 FY 142.4 MZ -3180.7
    69. 1115 FX 67.7 FY 38.4
    70. ******************************************************************************
    71. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
    72. JOINT LOAD
    73. 128 FX 188.5 FY 57.5 MZ -2560.2
    74. 1115 FX 67.7 FY 38.4
    75. ******************************************************************************
    76. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
    77. JOINT LOAD
    78. 128 FX 344.4 FY 21.3 MZ -3618.7
    79. 1115 FX 67.7 FY 38.4
    80. ******************************************************************************
    81. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
    82. JOINT LOAD
    83. 128 FX 216.7 FY -60.9 MZ -2951.2
    84. 1115 FX 67.7 FY 38.4
    85. ******************************************************************************
    86. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
    87. JOINT LOAD
    88. 128 FX -352.7 FY 276.8 MZ 3895.1
    89. 1115 FX -67.7 FY -38.4
    90. ******************************************************************************
    91. LOAD 13 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
    92. JOINT LOAD
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 93. 128 FX -480.5 FY 195.6 MZ 4591.3
 94. 1115 FX -67.7 FY -38.4
 95. ********************************************************************************
 96. LOAD 14 LOADTYPE NONE  TITLE WIND LOAD  ( WL7 )
 97. JOINT LOAD
 98. 128 FX -331.7 FY 129.5 MZ 3703
 99. 1115 FX -67.7 FY -38.4
100. ********************************************************************************
101. LOAD 15 LOADTYPE NONE  TITLE WIND LOAD  ( WL8 )
102. JOINT LOAD
103. 128 FX -459.4 FY 48.4 MZ 4399.4
104. 1115 FX -67.7 FY -38.4
105. ********************************************************************************
106. LOAD 16 LOADTYPE NONE  TITLE WIND LOAD  ( WL9 )
107. JOINT LOAD
108. 128 FX 280.3 FY 518.52 FZ -78.25 MZ -1528.9
109. 1115 FX 6.4 FY 0.5 FZ -2
110. ********************************************************************************
111. LOAD 17 LOADTYPE NONE  TITLE WIND LOAD  ( WL10 )
112. JOINT LOAD
113. 128 FX 152.6 FY 436.92 FZ -78.25 MZ -850
114. 1115 FX 6.4 FY 0.5 FZ -2
115. ********************************************************************************
116. LOAD 18 LOADTYPE NONE  TITLE WIND LOAD  ( WL11 )
117. JOINT LOAD
118. 128 FX 293.1 FY -296.92 FZ 78.25 MZ -1683.9
119. 1115 FX 6.4 FY -0.5 FZ 2
120. ********************************************************************************
121. LOAD 19 LOADTYPE NONE  TITLE WIND LOAD  ( WL12 )
122. JOINT LOAD
123. 128 FX 165.6 FY -378.52 FZ 78.25 MZ -1005.4
124. 1115 FX 6.4 FY -0.5 FZ 2
125. ********************************************************************************
126. LOAD 20 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
127. JOINT LOAD
128. 128 FX 24.9 FY 69.96 FZ -15.28 MZ -217.4
129. ********************************************************************************
130. LOAD 21 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
131. JOINT LOAD
132. 128 FX -24.9 FY -69.96 FZ 15.28 MZ 217.4
133. ********************************************************************************
134. LOAD 22 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
135. JOINT LOAD
136. 128 FX 1.1 FY 0 MZ -23.9
137. ********************************************************************************
138. LOAD 23 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
139. JOINT LOAD
140. 128 FX -1.1 FY 0 MZ 23.9
141. ********************************************************************************
142. * UNFACTORED LOAD COMBINATIONS
143. LOAD COMB 201 DL
144. 1 1.0
145. LOAD COMB 202 DL+L+TL1
146. 1 1.0 3 1.0 22 1.0
147. LOAD COMB 203 DL+L+TL2
148. 1 1.0 3 1.0 23 1.0
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149. LOAD COMB 204 DL+LR
150. 1 1.0 2 1.0
151. LOAD COMB 205 DL+0.75L+0.75TL1+0.75LR
152. 1 1.0 2 0.75 3 0.75 22 0.75
153. LOAD COMB 206 DL+0.75L+0.75TL2+0.75LR
154. 1 1.0 2 0.75 3 0.75 23 0.75
155. LOAD COMB 207 DL+0.625WL1
156. 1 1.0 8 0.625
157. LOAD COMB 208 DL+0.625WL2
158. 1 1.0 9 0.625
159. LOAD COMB 209 DL+0.625WL3
160. 1 1.0 10 0.625
161. LOAD COMB 210 DL+0.625WL4
162. 1 1.0 11 0.625
163. LOAD COMB 211 DL+0.625WL5
164. 1 1.0 12 0.625
165. LOAD COMB 212 DL+0.625WL6
166. 1 1.0 13 0.625
167. LOAD COMB 213 DL+0.625WL7
168. 1 1.0 14 0.625
169. LOAD COMB 214 DL+0.625WL8
170. 1 1.0 15 0.625
171. LOAD COMB 215 DL+0.625WL9
172. 1 1.0 16 0.625
173. LOAD COMB 216 DL+0.625WL10
174. 1 1.0 17 0.625
175. LOAD COMB 217 DL+0.625WL11
176. 1 1.0 18 0.625
177. LOAD COMB 218 DL+0.625WL12
178. 1 1.0 19 0.625
179. LOAD COMB 219 DL+0.75WL1+0.75L+0.75LR
180. 1 1.0 2 0.75 3 0.75 8 0.75
181. LOAD COMB 220 DL+0.75WL2+0.75L+0.75LR
182. 1 1.0 2 0.75 3 0.75 9 0.75
183. LOAD COMB 221 DL+0.75WL3+0.75L+0.75LR
184. 1 1.0 2 0.75 3 0.75 10 0.75
185. LOAD COMB 222 DL+0.75WL4+0.75L+0.75LR
186. 1 1.0 2 0.75 3 0.75 11 0.75
187. LOAD COMB 223 DL+0.75WL5+0.75L+0.75LR
188. 1 1.0 2 0.75 3 0.75 12 0.75
189. LOAD COMB 224 DL+0.75WL67+0.75L+0.75LR
190. 1 1.0 2 0.75 3 0.75 13 0.75
191. LOAD COMB 225 DL+0.75WL7+0.75L+0.75LR
192. 1 1.0 2 0.75 3 0.75 14 0.75
193. LOAD COMB 226 DL+0.75WL8+0.75L+0.75LR
194. 1 1.0 2 0.75 3 0.75 15 0.75
195. LOAD COMB 227 DL+0.75WL9+0.75L+0.75LR
196. 1 1.0 2 0.75 3 0.75 16 0.75
197. LOAD COMB 228 DL+0.75WL10+0.75L+0.75LR
198. 1 1.0 2 0.75 3 0.75 17 0.75
199. LOAD COMB 229 DL+0.75WL11+0.75L+0.75LR
200. 1 1.0 2 0.75 3 0.75 18 0.75
201. LOAD COMB 230 DL+0.75WL12+0.75L+0.75LR
202. 1 1.0 2 0.75 3 0.75 19 0.75
203. LOAD COMB 231 0.6DL+0.625WL1
204. 1 0.6 8 0.625
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205. LOAD COMB 232 0.6DL+0.625WL2
206. 1 0.6 9 0.625
207. LOAD COMB 233 0.6DL+0.625WL3
208. 1 0.6 10 0.625
209. LOAD COMB 234 0.6DL+0.625WL4
210. 1 0.6 11 0.625
211. LOAD COMB 235 0.6DL+0.625WL5
212. 1 0.6 12 0.625
213. LOAD COMB 236 0.6DL+0.625WL6
214. 1 0.6 13 0.625
215. LOAD COMB 237 0.6DL+0.625WL7
216. 1 0.6 14 0.625
217. LOAD COMB 238 0.6DL+0.625WL8
218. 1 0.6 15 0.625
219. LOAD COMB 239 0.6DL+0.625WL9
220. 1 0.6 16 0.625
221. LOAD COMB 240 0.6DL+0.625WL10
222. 1 0.6 17 0.625
223. LOAD COMB 241 0.6DL+0.625WL11
224. 1 0.6 18 0.625
225. LOAD COMB 242 0.6DL+0.625WL12
226. 1 0.6 19 0.625
227. LOAD COMB 243 0.6DL+0.7SL1
228. 1 0.6 20 0.7
229. LOAD COMB 244 0.6DL+0.7SL2
230. 1 0.6 21 0.7
231. LOAD COMB 245 DL+0.75WL1+0.75L+0.75LR+CR1
232. 1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
233. LOAD COMB 246 DL+0.75WL2+0.75L+0.75LR+CR1
234. 1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
235. LOAD COMB 247 DL+0.75WL3+0.75L+0.75LR+CR1
236. 1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
237. LOAD COMB 248 DL+0.75WL4+0.75L+0.75LR+CR1
238. 1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
239. LOAD COMB 249 DL+0.75WL5+0.75L+0.75LR+CR1
240. 1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
241. LOAD COMB 250 DL+0.75WL67+0.75L+0.75LR+CR1
242. 1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
243. LOAD COMB 251 DL+0.75WL7+0.75L+0.75LR+CR1
244. 1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
245. LOAD COMB 252 DL+0.75WL8+0.75L+0.75LR+CR1
246. 1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
247. LOAD COMB 253 DL+0.75WL9+0.75L+0.75LR+CR1
248. 1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
249. LOAD COMB 254 DL+0.75WL10+0.75L+0.75LR+CR1
250. 1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
251. LOAD COMB 255 DL+0.75WL11+0.75L+0.75LR+CR1
252. 1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
253. LOAD COMB 256 DL+0.75WL12+0.75L+0.75LR+CR1
254. 1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
255. LOAD COMB 257 DL+0.75WL1+0.75L+0.75LR+CR2
256. 1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
257. LOAD COMB 258 DL+0.75WL2+0.75L+0.75LR+CR2
258. 1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
259. LOAD COMB 259 DL+0.75WL3+0.75L+0.75LR+CR2
260. 1 1.0 2 0.75 3 0.75 10 0.75 5 1.0

261. LOAD COMB 260 DL+0.75WL4+0.75L+0.75LR+CR2
262. 1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
263. LOAD COMB 261 DL+0.75WL5+0.75L+0.75LR+CR2
264. 1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
265. LOAD COMB 262 DL+0.75WL67+0.75L+0.75LR+CR2
266. 1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
267. LOAD COMB 263 DL+0.75WL7+0.75L+0.75LR+CR2
268. 1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
269. LOAD COMB 264 DL+0.75WL8+0.75L+0.75LR+CR2
270. 1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
271. LOAD COMB 265 DL+0.75WL9+0.75L+0.75LR+CR2
272. 1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
273. LOAD COMB 266 DL+0.75WL10+0.75L+0.75LR+CR2
274. 1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
275. LOAD COMB 267 DL+0.75WL11+0.75L+0.75LR+CR2
276. 1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
277. LOAD COMB 268 DL+0.75WL12+0.75L+0.75LR+CR2
278. 1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
279. LOAD COMB 269 DL+0.75WL1+0.75L+0.75LR+CR3
280. 1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
281. LOAD COMB 270 DL+0.75WL2+0.75L+0.75LR+CR3
282. 1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
283. LOAD COMB 271 DL+0.75WL3+0.75L+0.75LR+CR3
284. 1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
285. LOAD COMB 272 DL+0.75WL4+0.75L+0.75LR+CR3
286. 1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
287. LOAD COMB 273 DL+0.75WL5+0.75L+0.75LR+CR3
288. 1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
289. LOAD COMB 274 DL+0.75WL67+0.75L+0.75LR+CR3
290. 1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
291. LOAD COMB 275 DL+0.75WL7+0.75L+0.75LR+CR3
292. 1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
293. LOAD COMB 276 DL+0.75WL8+0.75L+0.75LR+CR3
294. 1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
295. LOAD COMB 277 DL+0.75WL9+0.75L+0.75LR+CR3
296. 1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
297. LOAD COMB 278 DL+0.75WL10+0.75L+0.75LR+CR3
298. 1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
299. LOAD COMB 279 DL+0.75WL11+0.75L+0.75LR+CR3
300. 1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
301. LOAD COMB 280 DL+0.75WL12+0.75L+0.75LR+CR3
302. 1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
303. LOAD COMB 281 DL+0.75WL1+0.75L+0.75LR+CR4
304. 1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
305. LOAD COMB 282 DL+0.75WL2+0.75L+0.75LR+CR4
306. 1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
307. LOAD COMB 283 DL+0.75WL3+0.75L+0.75LR+CR4
308. 1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
309. LOAD COMB 284 DL+0.75WL4+0.75L+0.75LR+CR4
310. 1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
311. LOAD COMB 285 DL+0.75WL5+0.75L+0.75LR+CR4
312. 1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
313. LOAD COMB 286 DL+0.75WL67+0.75L+0.75LR+CR4
314. 1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
315. LOAD COMB 287 DL+0.75WL7+0.75L+0.75LR+CR4
316. 1 1.0 2 0.75 3 0.75 14 0.75 7 1.0

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317. LOAD COMB 288 DL+0.75WL8+0.75L+0.75LR+CR4
318. 1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
319. LOAD COMB 289 DL+0.75WL9+0.75L+0.75LR+CR4
320. 1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
321. LOAD COMB 290 DL+0.75WL10+0.75L+0.75LR+CR4
322. 1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
323. LOAD COMB 291 DL+0.75WL11+0.75L+0.75LR+CR4
324. 1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
325. LOAD COMB 292 DL+0.75WL12+0.75L+0.75LR+CR4
326. 1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
327. ********************************************************************************
328. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 4 NUMBER OF MEMBERS 2
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 2

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 23, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 92 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 144951.0 MB

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329. FINISH

          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= JUL 13,2020   TIME= 12:14: 8 ****
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